COMPATIBILITY AND LOCAL GRADING CHARACTERIZE THE GENERAL TWISTED P(Z) TENSOR-PRODUCT SPACE

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Allowed standard prerequisites (editorial).

- Vertex operator algebras, twisted modules and ordinary complex analytic continuation. Complete original bibliography and source-local definitions retained.

1. Twisted modules and twisted intertwining operators

We first recall in this section the notion of (generalized) twisted module from [15]. We then introduce a notion of twisted intertwining operators more general than the one in [17]. We also give some basic results on such twisted intertwining operators.

For $z \in \mathbb{C}^\times$ and $p \in \mathbb{Z}$, we shall use the notation $l_p(z) = \log|z| + \mathrm{i}\arg z + 2\pi p\mathrm{i}$, where $0 \le \arg z < 2\pi$. We shall also use the notation $\log z = l_0(z) = \log|z| + \mathrm{i}\arg z$. For a vector space U, $p \in \mathbb{Z}$ and a formal series

$$f(x) = \sum_{k=0}^{K} \sum_{n\in\mathbb{C}} a_{n,k} x^n (\log x)^k,$$

where $a_{n,k} \in U$, the series

$$f^p(z) = \sum_{k=0}^{K} \sum_{n\in\mathbb{C}} a_{n,k} \mathrm{e}^{n l_p(z)} (l_p(z))^k$$

is called the p-th analytic branch of $f(x)$. We also denote $f^0(z)$ simply by $f(z)$.

Let V be a vertex operator algebra and g an automorphism of V. We recall the definition of generalized g-twisted V-module first introduced in [15]. Note that as in [15, 17] in this paper, the vertex operator map for a generalized g-twisted V-module in general contains the logarithm of the variable and the operator $L(0)$ in general does not have to act semisimply.

Definition 1.1. *A generalized g-twisted V-module without a g-action is a $\mathbb{C}$-graded vector space $W = \coprod_{n\in\mathbb{C}} W_{[n]}$ (graded by weights) equipped with a linear map*

$$Y_W^g : \ V \otimes W \to W\{x\}[\log x],$$

$$v \otimes w \mapsto Y_W^g(v,x)w = \sum_{k=0}^{K} \sum_{n\in\mathbb{C}} (Y_W^g)_{n,k}(v) w x^{-n-1} (\log x)^k$$

satisfying the following conditions:

1. *The equivariance property: For $p \in \mathbb{Z}$, $z \in \mathbb{C}^\times$, $v \in V$ and $w \in W$, $\left(Y_W^g\right)^{p+1}(gv,z)w = \left(Y_W^g\right)^p(v,z)w$, where for $p \in \mathbb{Z}$, $\left(Y_W^g\right)^p(v,z)$ is the p-th analytic branch of $Y_W^g(v,x)$.*
2. *The identity property: For $w \in W$, $Y_W^g(\mathbf{1},x)w = w$.*

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(3) *The duality property: For any $u, v \in V$, $w \in W$ and $w' \in W'$, there exists a maximally-extended multivalued analytic function with preferred branch of the form*

$$f(z_1, z_2) = \sum_{i,j,k,l=0}^{N} a_{ijkl} z_1^{m_i} z_2^{n_j} (\log z_1)^k (\log z_2)^l (z_1 - z_2)^{-t}$$

for $N \in \mathbb{N}$, $m_1, \dots, m_N$, $n_1, \dots, n_N \in \mathbb{C}$ and $t \in \mathbb{Z}_+$, such that the series

$$\langle w', (Y_W^g)^p(u, z_1)(Y_W^g)^p(v, z_2)w\rangle = \sum_{n\in\mathbb{C}} \langle w', (Y_W^g)^p(u, z_1)\pi_n(Y_W^g)^p(v, z_2)w\rangle,$$

$$\langle w', (Y_W^g)^p(v, z_2)(Y_W^g)^p(u, z_1)w\rangle = \sum_{n\in\mathbb{C}} \langle w', (Y_W^g)^p(v, z_2)\pi_n(Y_W^g)^p(u, z_1)w\rangle,$$

$$\langle w', (Y_W^g)^p(Y_V(u, z_1 - z_2)v, z_2)w\rangle = \sum_{n\in\mathbb{C}} \langle w', (Y_W^g)^p(\pi_n Y_V(u, z_1 - z_2)v, z_2)w\rangle$$

are absolutely convergent on the regions $|z_1| > |z_2| > 0$, $|z_2| > |z_1| > 0$, $|z_2| > |z_1 - z_2| > 0$, respectively, and their sums are equal to the branch

$$f^{p,p}(z_1, z_2) = \sum_{i,j,k,l=0}^{N} a_{ijkl} \mathrm{e}^{m_i l_p(z_1)} \mathrm{e}^{n_j l_p(z_2)} l_p(z_1)^k l_p(z_2)^l (z_1 - z_2)^{-t}$$

of $f(z_1, z_2)$ on the region $|z_1| > |z_2| > 0$, the region $|z_2| > |z_1| > 0$, the region given by $|z_2| > |z_1 - z_2| > 0$ and $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$, respectively.

(4) *The $L(0)$-grading condition: Let $L_W^g(0) = \operatorname{Res}_x x Y_W^g(\omega, x)$. Then for $n \in \mathbb{C}$, $w \in W_{[n]}$, there exists $K \in \mathbb{Z}_+$ such that $\big(L_W^g(0) - n\big)^K w = 0$.*

(5) *The $L(-1)$-derivative property: For $v \in V$,*

$$\frac{\mathrm{d}}{\mathrm{d}x} Y_W^g(v, x) = Y_W^g(L_V(-1)v, x).$$

A lower-bounded generalized g-twisted V-module is a generalized g-twisted V-module W such that for each $n \in \mathbb{C}$, $W_{[n+l]} = 0$ for sufficiently negative real numbers l. A generalized g-twisted V-module W is said to be grading-restricted if it is lower bounded and for each $n \in \mathbb{C}$, $\dim W_{[n]} < \infty$. A generalized g-twisted V-module with a g-action is a $\mathbb{C} \times \mathbb{C}/\mathbb{Z}$-graded vector space $W = \coprod_{n\in\mathbb{C},\alpha\in\mathbb{C}/\mathbb{Z}} W_{[n]}^{[\alpha]}$ (graded by weights and g-weights) equipped with a linear map $Y_W^g \colon V \otimes W \to W\{x\}[\log x]$ as above and an action of g such that W equipped with Y_W^g is a generalized g-twisted V-module without a g-action satisfying in addition the following g-grading condition: There exists $\Lambda \in \mathbb{Z}_+$ such that for $\alpha \in \mathbb{C}/\mathbb{Z}$, $w \in W^{[\alpha]} = \coprod_{n\in\mathbb{N}} W_{[n]}^{[\alpha]}$, $\big(g - \mathrm{e}^{2\pi\alpha\mathrm{i}}\big)^\Lambda w = 0$. Moreover, $gY_W^g(u, x)w = Y_W^g(gu, x)gw$ for $u \in V$ and $w \in W$.

Module maps between g-twisted V-modules are defined in the obvious way. For two different automorphisms g_1 and g_2 of V, there might exist linear maps from a g_1-twisted V-module to a g_2-twisted V-module commuting with the twisted vertex operator maps when V is not simple. But we will not consider such maps in this paper. So in the categories of twisted V-modules considered in this paper, we always choose the space of module maps between a g_1-twisted V-module and a g_2-twisted V-module to be 0 if $g_1 \neq g_2$.

In this paper, generalized g-twisted V-modules with or without g-actions are all called *generalized g-twisted V-modules*. Only when the g-actions are specifically discussed, we shall add the words "with a g-action" or "without a g-action". Later, we shall also need generalized g-twisted V-modules with g-actions equipped with an h-action for another automorphism h of V. Also, for simplicity, we shall sometimes omit the subscript W to denote the twisted vertex operator map Y_W^g by Y^g.

Remark 1.2. Note that a lower-bounded generalized g-twisted V-module defined above also satisfies the lower-truncation property: For $v \in V$ and $w \in W$, $n \in \mathbb{C}$ and $k = 0, \dots, K$, $\big(Y_W^g\big)_{n+l,k}(w_1)w_2 = 0$ for $l \in \mathbb{N}$ sufficiently large. This in fact follows from the $L(0)$-commutator formula, which is a consequence of the duality property and the $L(0)$-grading condition.

Remark 1.3. In this paper, we need to study generalized g-twisted V-modules with and without g-actions in a different way because the skew-symmetry and contragredient isomorphisms in Section 2 are different in these two types of twisted modules. In fact, there is also a notion of generalized g-twisted module with a G action for a group G of automorphisms of V. See [22] for a definition and the construction $P(z)$-tensor product of such twisted modules.

In this paper, we shall consider only lower-bounded generalized g-twisted V-modules and grading-restricted generalized V-modules. For simplicity, we shall often call lower-bounded generalized g-twisted V-modules simply g-twisted V-modules and grading-restricted generalized V-modules simply grading-restricted g-twisted V-modules.

Let $\big(W, Y_W^g\big)$ be a g-twisted V-module without a g-action. Let h be an automorphism of V. We recall the hgh^{-1}-twisted V-module $\big(W, \phi_h\big(Y_W^g\big)\big)$ without a hgh^{-1}-action (see, for example, [17]). Let

$$\phi_h\big(Y_W^g\big)\colon\ V \times W \to W\{x\}[\log x], \qquad v \otimes w \mapsto \phi_h\big(Y_W^g\big)(v,x)w$$

be the linear map defined by

$$\phi_h\big(Y_W^g\big)(v,x)w = Y_W^g\big(h^{-1}v, x\big)w.$$

Then the pair $\big(W, \phi_h\big(Y_W^g\big)\big)$ is an hgh^{-1}-twisted V-module without a hgh^{-1}-action.

In the case that W is a g-twisted V-module with a g-action and also has an action of h in the sense that W is a module for the group generated by g and h, we define

$$\varphi_h\big(Y_W^g\big)\colon\ V \times W \to W\{x\}[\log x], \qquad v \otimes w \mapsto \varphi_h\big(Y_W^g\big)(v,x)w$$

by

$$\varphi_h\big(Y_W^g\big)(v,x)w = hY_W^g\big(h^{-1}v, x\big)h^{-1}w.$$

Then $\big(W, \varphi_h\big(Y_W^g\big)\big)$ is also an hgh^{-1}-twisted V-module without a hgh^{-1}-action. Since the conformal element ω is fixed under the action of g^{-1}, the coefficients

$$L_{\varphi_h(W)}(n) = \operatorname{Res}_x x^{n+1}\varphi_h\big(Y_W^g\big)(\omega,x) = h\operatorname{Res}_x x^{n+1}Y_W^g(\omega,x)h^{-1} = hL_W^g(n)h^{-1}$$

of the vertex operator $\varphi_h\big(Y_W^g\big)(\omega,x)$ for $n \in \mathbb{N}$ also give Virasoro operators on W with the same central charge. Since h is an isomorphism of the vector space W,

$$hW = \coprod_{n\in\mathbb{C},\alpha\in\mathbb{C}/\mathbb{Z}} hW_{[n]}^{[\alpha]}.$$

We denote the graded space hW by $\varphi_h(W)$ and $hW_{[n]}^{[\alpha]}$ by $\varphi_h(W)_{[n]}^{[\alpha]}$ $n \in \mathbb{C}, \alpha \in \mathbb{C}/\mathbb{Z}$. Then we have

$$\varphi_h(W) = \coprod_{n\in\mathbb{C},\alpha\in\mathbb{C}/\mathbb{Z}} \varphi_h(W)_{[n]}^{[\alpha]}.$$

Note that as a vector space $\varphi_h(W) = hW$ is the same as W, but as graded spaces, they are different. For $w \in \varphi_h(W)_{[n]} = \coprod_{\alpha\in\mathbb{C}/\mathbb{Z}} \varphi_h(W)_{[n]}^{[\alpha]}$, $h^{-1}w \in W_{[n]} = \coprod_{\alpha\in\mathbb{C}/\mathbb{Z}} W_{[n]}^{[\alpha]}$. So there exists $K \in \mathbb{Z}_+$ such that $\big(L_W^g - n\big)^K h^{-1}w = 0$. Then

$$(L_{\varphi_h(W)}(0) - n)^K w = h\big(L_W^g - n\big)^K h^{-1}w = 0.$$

For $w \in \varphi_h(W)^{[\alpha]} = \coprod_{n\in\mathbb{Z}} \varphi_h(W)^{[\alpha]}_{[n]}$, $h^{-1}w \in W^{[\alpha]} = \coprod_{n\in\mathbb{Z}} W^{[\alpha]}_{[n]}$. So there exists $\Lambda \in \mathbb{Z}_+$ such that $\big(g - \mathrm{e}^{2\pi\mathrm{i}\alpha}\big)^{\Lambda} h^{-1}w = 0$. Then $\big(hgh^{-1} - \mathrm{e}^{2\pi\mathrm{i}\alpha}\big)^{\Lambda} w = h\big(g - \mathrm{e}^{2\pi\mathrm{i}\alpha}\big)^{\Lambda} h^{-1}w = 0$. Also we have

$$\begin{aligned} hgh^{-1}\varphi_h\big(Y_W^g\big)(v,x)w &= hgY_W^g\big(h^{-1}v,x\big)h^{-1}w = hY_W^g\big(h^{-1}hgh^{-1}v,x\big)h^{-1}hgh^{-1}w \\ &= \varphi_h\big(Y_W^g\big)\big(hgh^{-1}v,x\big)hgh^{-1}w. \end{aligned}$$

So $(\varphi_h(W), \varphi_h\big(Y_W^g\big))$ is a hgh^{-1}-twisted V-module with an hgh^{-1}-action. We shall denote this hgh^{-1}-twisted V-module with an hgh^{-1}-action by $\varphi_h(W)$.

Note that when $h = g$, $\varphi_h(W) = W$ and $\phi_g(W)$ with the twisted vertex operator given by

$$\phi_g\big(Y_W^g\big)(v,x)w = Y_W^g\big(g^{-1}v,x\big)w$$

is equivalent to the original g-twisted module W with the equivalence $g^{-1}\colon W \to W$. By the equivariance property, we have $\big(Y_W^g\big)^p\big(g^{-1}v,x\big) = \big(Y_W^g\big)^{p+1}(v,x)$ and, if

$$Y_W^g(v,x)w = \sum_{k=0}^{K}\sum_{n\in\mathbb{C}} \big(Y_W^g\big)_{n,k}(v)x^{-n-1}(\log x)^k$$

for $v \in V$ and $w \in W$, we have

$$Y_W^g\big(g^{-1}v,x\big)w = \sum_{k=0}^{K}\sum_{n\in\mathbb{C}} \big(Y_W^g\big)_{n,k}(v)\mathrm{e}^{-2\pi\mathrm{i}n}x^{-n-1}(\log x + 2\pi\mathrm{i})^k.$$

Remark 1.4. In the definitions of $\big(\varphi_h(W), \varphi_h\big(Y_W^g\big)\big)$, we assume that there is an action of h on W. One immediate question is whether there exist such g-twisted V-modules with h-actions. In fact, the same construction in Section 4 of [18] can be used to construct such a g-twisted V-module with an h-action or even a g-twisted V-module with a G-action for a group G of automorphisms of V satisfying the additional condition $hY_W^g(v,x)w = Y_W^g(hv,x)hw$ for $h \in G$. In this case, though $\varphi_h(W) = W$ as vector spaces and $\varphi_h\big(Y_W^g\big) = Y_W^g$ as linear maps from $V \otimes W$ to $W\{x\}[\log x]$ when we view $\varphi_h(W)$ as a vector space equal to W, $\varphi_h(W)$ is graded by generalized eigenspaces of hgh^{-1} but $W$is graded by the generalized eigenspaces of g. In particular, $\big(\varphi_h(W), \varphi_h\big(Y_W^g\big)\big)$ and $\big(W, Y_W^g\big)$ are indeed different twisted V-modules.

We also recall contragredient twisted V-modules (see, for example, [17]). Let $\big(W, Y_W^g\big)$ be a g-twisted V-module relative to G. Let W' be the graded dual of W. Define a linear map

$$\big(Y_W^g\big)'\colon\ V \otimes W' \to W'\{x\}[\log x], \qquad v \otimes w' \mapsto \big(Y_W^g\big)'(v,x)w'$$

by

$$\big\langle \big(Y_W^g\big)'(v,x)w', w\big\rangle = \big\langle w', Y_W^g\big(\mathrm{e}^{xL(1)}\big(-x^{-2}\big)^{L(0)}v, x^{-1}\big)w\big\rangle$$

for $v \in V$, $w \in W$ and $w' \in W'$. Then the pair $\big(W', \big(Y_W^g\big)'\big)$ is a g^{-1}-twisted V-module.

Let $M^2 = \big\{(z_1,z_2) \in \mathbb{C}^2 \mid z_1 \neq 0,\ z_2 \neq 0,\ z_1 \neq z_2\big\}$. Let $f(z_1,z_2)$ be a maximally extended multivalued analytic function on M^2 with a preferred single-valued branch $f^e(z_1,z_2)$ on the simply-connected region M_0^2 given by cutting M^2 along the positive real lines in the z_1-, z_2- and $(z_1 - z_2)$-planes, that is, the sets

$$\big\{(z_1,z_2) \in M^2 \mid z_1 \in \mathbb{R}_+\big\}, \qquad \big\{(z_1,z_2) \in M^2 \mid z_2 \in \mathbb{R}_+\big\},$$
$$\big\{(z_1,z_2) \in M^2 \mid z_1 - z_2 \in \mathbb{R}_+\big\},$$

with these sets attached to the upper half z_1-, z_2- and $(z_1 - z_2)$-planes, that is, the points in these sets are viewed as limits in M_0^2 of sequences of points approaching them from the upper z_1-, z_2- and $(z_1 - z_2)$-planes. Note that given any point

$\left(z_1^0, z_2^0\right) \in M^2$ and any loop γ in M^2 based at $\left(z_1^0, z_2^0\right)$, we obtain from the preferred branch of $f(z_1, z_2)$ another single-valued branch by going around γ. The resulting single-valued branch depends only on the homotopy class of the loop and is independent of the choices of the base point. Thus we obtain a right action of the fundamental group of M^2 on the set of single-valued branches of $f(z_1, z_2)$. Note that M^2 is homotopically equivalent to the configuration space

$$F_3(\mathbb{C}) = \left\{(z_1, z_2, z_3) \in \mathbb{C}^3 \mid z_i \neq z_j,\ i \neq j\right\}.$$

So the fundamental group of M^2 is in fact the pure braid group PB_3 and has three generators b_{12}, b_{13} and b_{23}, which are given as follows: Choose the base point to be $(-3, -2)$. Then the generator b_{12} is the homotopy class of the loop given by letting z_1 go counterclockwise around the circle of radius 1 centered at -2 (see Figure 1). The generator b_{23} is the homotopy class of the loop given by letting z_2 go counterclockwise around the circle of radius 2 centered at 0 (see Figure 2). The generator b_{13} is the homotopy class of the loop given by letting z_1 go around first the lower half circle of radius 3 centered at 0 counterclockwise, then the upper half circle of radius 2 centered at 1 counterclockwise and finally the lower half circle of radius 1 centered at -2 clockwise (see Figure 3). We know that the pure braid group PB_3 is isomorphic to the group generated by b_{12}, b_{13}, b_{23} with the relations $b_{13}b_{12}b_{23} = b_{12}b_{23}b_{13} = b_{23}b_{13}b_{12}$. See [4] for more detailed discussions on M^2, its fundamental group, the pure braid groups and multivalued functions on M^2.

For $a \in \mathrm{PB}_3$, we denote the single-valued branch of $f(z_1, z_2)$ obtained by applying a to $f^e(z_1, z_2)$ by $f^a(z_1, z_2)$.

Let $\delta = (\delta_1, \ldots, \delta_n)$, where $\delta_i \in \{0, \infty\}$, $i = 1, \ldots, n$. Suppose $f(z_1, \ldots, z_n)$ is a multi-valued analytic function defined on an open region Ω of $\mathbb{C}^n$. We say $(z_1, \ldots, z_n) = \delta$ is a *component-isolated singularity of* $f(z_1, \ldots, z_n)$ if there exists $r = (r_1, \ldots, r_n) \in \mathbb{R}_+^n$ such that $\Delta^\times(\delta, r) \subset \Omega$, where $\Delta^\times(\delta, r) = \{(z_1, \ldots, z_n) \in \mathbb{C}^n \mid |z_i| > r_i \text{ if } \delta_i = \infty, 0 < |z_i| < r_i \text{ if } \delta_i = 0\}$. Let $A \in \mathrm{GL}(n, \mathbb{C})$ and $\beta \in \mathbb{C}^n$ (written as a row vector). Then $\zeta_1, \ldots, \zeta_n$ given by

$$(\zeta_1, \ldots, \zeta_n) = (z_1, \ldots, z_n)A - \beta$$

are also independent variables. Define

$$g(\zeta_1, \ldots, \zeta_n) = f\left((\zeta_1, \ldots, \zeta_n)A^{-1} + \beta A^{-1}\right). \tag{1.1}$$

For $\delta \in (\mathbb{C} \cup \{\infty\})^n$, we say that $(\zeta_1, \ldots, \zeta_n) = \delta$ is a *component-isolated singularity of the function* $f(z_1, \ldots, z_n)$ if $(\zeta_1, \ldots, \zeta_n) = \delta$ is a component-isolated singularity of $g(\zeta_1, \ldots, \zeta_n)$.

Remark 1.5. Notice that $(\zeta_1, \ldots, \zeta_n)(= (z_1, \ldots, z_n)A - \beta) = \delta$ being a component-isolated singularity of a function is not equivalent to $(z_1, \ldots, z_n) = \delta A^{-1} + \beta A^{-1}$ being a component-isolated singularity of the same function. This is because we have different sets of independent variables. For example, consider $f(z_1, z_2) = \frac{1}{z_1 - z_2}$. Since $(\zeta_1, \zeta_2) = (0, 0)$ is a component-isolated singularity of the function $g(\zeta_1, \zeta_2) = 1/\zeta_1$, we also say that $(z_1 - z_2, z_2) = (0, 0)$ is a component-isolated singularity of $f(z_1, z_2)$. In this case, $z_1 - z_2$ and z_2 are independent variables.

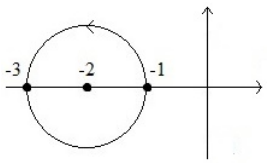


FIGURE 1. The loop for b_{12}.

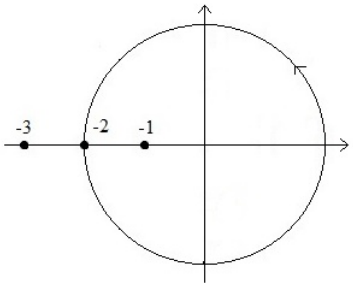


FIGURE 2. The loop for b_{23}.

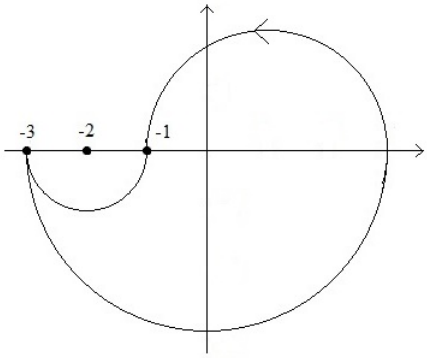


FIGURE 3. The loop for b_{13}.

However, $(z_1, z_2) = (0, 0)$ is clearly not a component-isolated singularity of $f(z_1, z_2)$. In this case, the independent variables are z_1 and z_2.

Definition 1.6. Let $f(z_1, \dots, z_n)$ be a multi-valued analytic function defined on an open region of $\mathbb{C}^n$. Let $\delta = (\delta_1, \dots, \delta_n) \in \{0, \infty\}^n$. Suppose $(z_1, \dots, z_n) = \delta$ is a component-isolated singularity of $f(z_1, \dots, z_n)$. Let $\{f^b(z_1, \dots, z_n)\}_{b\in B}$ be the set of all single valued branches of $f(z_1, \dots, z_n)$ near δ with cuts at $z_i \in \mathbb{R}_+$. Then for each $b \in B$, there exists $r_b \in \mathbb{R}_+^n$ such that $f^b(z_1, \dots, z_n)$ is analytic on $\Delta^\times(\delta, r_b)$. We say that $(z_1, \dots, z_n) = \delta$ is a *regular singularity* of $f(z_1, \dots, z_n)$ if there exists $K \in \mathbb{N}$, $D_i = \cup_{j=1}^{N_i} r_j^{(i)} + \mathbb{N}$ $\big($or $D_i = \cup_{j=1}^{N_i} r_j^{(i)} - \mathbb{N}\big)$ where $r_1^{(i)}, \dots, r_{N_i}^{(i)} \in \mathbb{C}$ for $\delta_i = 0$ (or $\delta_i = \infty$), and $\alpha^{(b)}_{a_1,j_1;\dots;a_n,j_n} \in \mathbb{C}$, such that on the region $\Delta^\times(\delta, r_b)$, the right-hand-side of the following equation is absolutely convergent, and

(1.2)
$$f^b(z_1, \dots, z_n) = \sum_{i=1}^{n} \sum_{a_i \in D_i} \sum_{j_1,\dots,j_n=0}^{K} \alpha^{(b)}_{a_1,j_2;\dots;a_n,j_n} z_1^{a_1} (\log z_1)^{j_1} \cdots z_n^{a_n} (\log z_n)^{j_n}.$$

If $K = 0$ and $D_i = -n^{(i)} + \mathbb{N}$ in the case $\delta_i = 0$ and $D_i = n^{(i)} - \mathbb{N}$ in the case $\delta_i = \infty$ for $i = 1, \dots, n$ in ((1.2)), where $n^{(i)} \in \mathbb{Z}_+$ for $i \in I \subset \{1, \dots, n\}$ and $n^{(i)} = 0$ for $i \neq I$, we say that $z_i = \delta_i$ for $i \in I$ are *poles of* $f(z_1, \dots, z_n)$. If $n^{(i)} \in \mathbb{Z}_+$ for $i \in I \subset \{1, \dots, n\}$, $\delta_i = 0$ are the smallest and such that (1.2) holds, we call n_i the orders of the poles $z_i = \delta_i$, respectively, for $i \in \mathbb{I}$. Let $A \in \mathrm{GL}(n, \mathbb{C})$ and $\beta \in \mathbb{C}^n$ be the same as above. We say that $(\zeta_1, \dots, \zeta_n) = \delta$ is a *regular singularity* of $f(z_1, \dots, z_n)$ if $(\zeta_1, \dots, \zeta_n) = \delta$ is a regular singularity of $g(\zeta_1, \dots, \zeta_n)$, where $g(\zeta_1, \dots, \zeta_n)$ is given by (1.1). We say that $\zeta_i = \delta_i$ for $i \in I$

are poles of $f(z_1,\dots,z_n)$ with orders n_i, respectively, if $\zeta_i=\delta_i$ for $i\in I$ are poles of $g(\zeta_1,\dots,\zeta_n)$ with orders n_i, respectively.

Remark 1.7. Let $f(z_1,\dots,z_n)$ and $g(z_1,\dots,z_n)$ be multivalued analytic functions with preferred branches defined on an open region. Then $\lambda f(z_1,\dots,z_n)+\mu g(z_1,\dots,z_n)$ for $\lambda,\mu\in\mathbb{C}$ and $f(z_1,\dots,z_n)g(z_1,\dots,z_n)$ are well defined using the preferred branches and are also multivalued analytic functions on the same region with preferred branches. If $(\zeta_1,\dots,\zeta_n)=((z_1,\dots,z_n)A-\beta=)\delta$ is a regular singular point of both $f(z_1,\dots,z_n)$ and $g(z_1,\dots,z_n)$, then it is also a regular singular point for $\lambda f(z_1,\dots,z_n)+\mu g(z_1,\dots,z_n)$ and $f(z_1,\dots,z_n)g(z_1,\dots,z_n)$. Therefore, the set of multivalued analytic functions on the same region with a preferred branch such that $(\zeta_1,\dots,\zeta_n)=\delta$ is a regular singular point form a commutative associative algebra over $\mathbb{C}$.

We also need the region $M^n=\{(z_1,\dots,z_n)\in\mathbb{C}^n \mid z_i\neq 0, z_i\neq z_j, i\neq j\}$ for $n\in\mathbb{Z}_+$.

Definition 1.8. *Let g_1, g_2, g_3 be automorphisms of V and let W_1, W_2 and W_3 be g_1-, g_2- and g_3-twisted V-modules (with or without actions of the corresponding automorphisms of V), respectively. A twisted intertwining operator of type $\binom{W_3}{W_1\ W_2}$ is a linear map*

$$\mathcal{Y}:\ W_1\otimes W_2\to W_3\{x\}[\log x],$$

$$w_1\otimes w_2\mapsto \mathcal{Y}(w_1,x)w_2=\sum_{k=0}^{K}\sum_{n\in\mathbb{C}}\mathcal{Y}_{n,k}(w_1)w_2x^{-n-1}(\log x)^k$$

satisfying the following conditions:

1. *The lower truncation property: For $w_1\in W_1$ and $w_2\in W_2$, $n\in\mathbb{C}$ and $k=0,\dots,K$, $\mathcal{Y}_{n+l,k}(w_1)w_2=0$ for $l\in\mathbb{N}$ sufficiently large.*
2. *The duality property: For $u\in V$, $w_1\in W_1$, $w_2\in W_2$ and $w_3'\in W_3'$, there exists a maximally extended multivalued analytic function $f(z_1,z_2;u,w_1,w_2,w_3')$ on M^2 with a preferred single-valued branch $f^e(z_1,z_2;u,w_1,w_2,w_3')$ on M_0^2 such that the series*

$$\big\langle w_3', Y_{W_3}^{g_3}(u,z_1)\mathcal{Y}(w_1,z_2)w_2\big\rangle=\sum_{n\in\mathbb{C}}\big\langle w_3', Y_{W_3}^{g_3}(u,z_1)\pi_n\mathcal{Y}(w_1,z_2)w_2\big\rangle, \tag{1.3}$$

$$\big\langle w_3', \mathcal{Y}(w_1,z_2)Y_{W_2}^{g_2}(u,z_1)w_2\big\rangle=\sum_{n\in\mathbb{C}}\big\langle w_3', \mathcal{Y}(w_1,z_2)\pi_nY_{W_2}^{g_2}(u,z_1)w_2\big\rangle, \tag{1.4}$$

(1.5)

$$\big\langle w_3', \mathcal{Y}\big(Y_{W_1}^{g_1}(u,z_1-z_2)w_1,z_2\big)w_2\big\rangle=\sum_{n\in\mathbb{C}}\big\langle w_3', \mathcal{Y}(\pi_nY_{W_1}^{g_1}(u,z_1-z_2)w_1,z_2)w_2\big\rangle$$

are absolutely convergent on the regions $|z_1|>|z_2|>0$, $|z_2|>|z_1|>0$, $|z_2|>|z_1-z_2|>0$, respectively. Moreover, their sums are equal to $f^e(z_1,z_2;u,w_1,w_2,w_3')$ on the region given by $|z_1|>|z_2|>0$ and $|\arg(z_1-z_2)-\arg z_1|<\frac{\pi}{2}$, the region given by $|z_2|>|z_1|>0$ and $-\frac{3\pi}{2}<\arg(z_1-z_2)-\arg z_2<-\frac{\pi}{2}$, the region given by $|z_2|>|z_1-z_2|>0$ and $|\arg z_1-\arg z_2|<\frac{\pi}{2}$, respectively.

3. *The convergence and analytic extension for products with more than one twisted vertex operators: For $k\in\mathbb{N}+3$, $u_1,\dots,u_{k-1}\in V$, $w_1\in W_1$, $w_2\in W_2$ and $w_3'\in W_3'$, the series*

$$\big\langle w_3', Y_{W_3}^{g_3}(u_1,z_1)\cdots Y_{W_3}^{g_3}(u_{k-1},z_{k-1})\mathcal{Y}(w_1,z_k)w_2\big\rangle$$

$$=\sum_{n_1,\dots,n_{k-1}\in\mathbb{C}}\big\langle w_3', Y_{W_3}^{g_3}(u_1,z_1)\pi_{n_1}\cdots\pi_{n_{k-2}}Y_{W_3}^{g_3}(u_{k-1},z_{k-1})\pi_{k-1}\mathcal{Y}(w_1,z_k)w_2\big\rangle$$

is absolutely convergent on the region $|z_1| > \cdots > |z_k| > 0$ *and can be maximally extended to a multivalued analytic function* $f(z_1, \dots, z_k)$ *on the region* M^k*. Moreover, for any fixed* $z_k \in \mathbb{C}^\times$ *and* $M_{ij} \in \mathbb{Z}_+$ *such that* $x^{M_{ij}} Y_V(u_i, x)u_j \in V[[x]]$ *for* $i, j = 1, \dots, k-1$*, any component-isolated singularity of the form* $(z_1, \dots, z_{k-1}) - \beta = \delta$ *for* $\beta \in \mathbb{C}^{k-1}$ *and* $\delta \in \{0, \infty\}^{k-1}$ *of the multivalued analytic function of* $z_1, \dots, z_{k-1}$ *obtained as a value at* $\zeta = z_k$ *of* $\prod_{1 \le i < j \le k-1}(z_i - z_j)^{M_{ij}} f(z_1, \dots, \zeta)$ *is regular.*

(4) *The* $L(-1)$*-derivative property* $\frac{\mathrm{d}}{\mathrm{d}x}\mathcal{Y}(w_1, x) = \mathcal{Y}(L(-1)w_1, x)$.

Remark 1.9. Note that the series in the definition of twisted intertwining operator $\mathcal{Y}$ above are all summed over unique expansion sets (see [29, Definition 7.5]). In fact, by the duality property and the fact that ω is fixed under g_1, g_2 and g_3, the $L(0)$-commutator formula for $\mathcal{Y}$ holds. Then for homogeneous $v \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $w_3' \in W_3'$, the series above are in fact summed over sets of the form $(\{r_1, \dots, r_m\} + \mathbb{N}) \times \{0, \dots, K\}$, which is a unique expansion set (see [20]).

Remark 1.10. For simplicity, in the duality property in Definition 1.8, we use only the preferred branches of the twisted intertwining operator and the twisted vertex operators. One can derive what the products and iterates of other branches of a twisted intertwining operator converge to using the actions of the elements of PB_3 on the single-valued branch $f^e(z_1, z_2; u, w_1, w_2, w_3')$ of the multivalued function $f(z_1, z_2; u, w_1, w_2, w_3')$ in the definition. Let $\mathcal{Y}$ be a twisted intertwining operator of type $\binom{W_3}{W_1 W_2}$. For any $p_1, p_2, p_{12} \in \mathbb{Z}$, the series

$$\big\langle w_3', \big(Y_{W_3}^{g_3}\big)^{p_1}(u, z_1)\mathcal{Y}^{p_2}(w_1, z_2)w_2\big\rangle = \sum_{n \in \mathbb{C}} \big\langle w_3', \big(Y_{W_3}^{g_3}\big)^{p_1}(u, z_1)\pi_n\mathcal{Y}^{p_2}(w_1, z_2)w_2\big\rangle,$$

$$\big\langle w_3', \mathcal{Y}^{p_2}(w_1, z_2)\big(Y_{W_2}^{g_2}\big)^{p_1}(u, z_1)w_2\big\rangle = \sum_{n \in \mathbb{C}} \big\langle w_3', \mathcal{Y}^{p_2}(w_1, z_2)\pi_n\big(Y_{W_2}^{g_2}\big)^{p_1}(u, z_1)w_2\big\rangle,$$

$$\big\langle w_3', \mathcal{Y}^{p_2}(\big(Y_{W_1}^{g_1}\big)^{p_{12}}(u, z_1 - z_2)w_1, z_2)w_2\big\rangle = \sum_{n \in \mathbb{C}} \big\langle w_3', \mathcal{Y}^{p_2}(\pi_n\big(Y_{W_1}^{g_1}\big)^{p_{12}}(u, z_1 - z_2)w_1, z_2)w_2\big\rangle$$

are absolutely convergent on the regions $|z_1| > |z_2| > 0$, $|z_2| > |z_1| > 0$, $|z_2| > |z_1 - z_2| > 0$, respectively. Moreover, their sums are equal to the branches

$$f^{(b_{13}b_{12})^{p_1}b_{23}^{p_2}}(z_1, z_2; u, w_1, w_2, w_3'), \qquad f^{(b_{12}b_{23})^{p_2}b_{13}^{p_1}}(z_1, z_2; u, w_1, w_2, w_3'),$$

$$f^{(b_{23}b_{13})^{p_2}b_{12}^{p_{12}}}(z_1, z_2; u, w_1, w_2, w_3'),$$

respectively, of $f(z_1, z_2; u, w_1, w_2, w_3')$ on the region given by $|z_1| > |z_2| > 0$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$, the region given by $|z_2| > |z_1| > 0$ and $-\frac{3\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$, the region given by $|z_2| > |z_1 - z_2| > 0$ and $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$, respectively. See [4] for more details.

Remark 1.11. From Definition 1.8, we see that a twisted intertwining operator of type $\binom{W_3}{W_1 W_2}$ is in fact an intertwining operator of type $\binom{W_3}{W_1 W_2}$ when W_1, W_2, and W_3 are viewed as (untwisted) modules for the fixed-point subalgebra of V under g_1 and g_2. In particular, all the usual properties for intertwining operators holds for the vertex operators associated to elements of the fixed-point subalgebra and the twisted intertwining operator. For example, the Jacobi identity and the commutator formula for these operators are satisfied.

Proposition 1.12. *Let* g_1*,* g_2*,* g_3 *be automorphisms of* V*,* W_1*,* W_2*,* W_3 g_1*-,* g_2*-,* g_3*-twisted generalized* V*-modules and* $\mathcal{Y}$ *a twisted intertwining operator of type* $\binom{W_3}{W_1 W_2}$*. Assume that the map* $u \mapsto Y_{W_3}^{g_3}(u, x)$ *is injective and* $\mathcal{Y}$ *is surjective in the sense that the coefficients of the series* $\mathcal{Y}(w_1, x)w_2$ *for* $w_1 \in W_1$*,* $w_2 \in W_2$ *span* W_3*. Then* $g_3 = g_1 g_2$.

Proof. By the definition of twisted intertwining operator, for $u \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $w_3' \in W_3'$, there exists a multivalued analytic function $f(z_1, z_2; u, w_1, w_2, w_3')$ on M^2 with a preferred single-valued branch $f^e(z_1, z_2; u, w_1, w_2, w_3')$ on M_0^2 such that (1.3), (1.4) and (1.5) are absolutely convergent to $f^e(z_1, z_2; u, w_1, w_2, w_3')$ on the corresponding regions in given in Definition 1.8. In particular, on the region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ the sum of the series $\langle w_3', Y_{W_3}^{g_3}(g_3 u, z_1)\mathcal{Y}(w_1, z_2)w_2\rangle$ is equal to $f^e(z_1, z_2; g_3 u, w_1, w_2, w_3')$. By the equivariance property for W_3,

$$\langle w_3', Y_{W_3}^{g_3}(g_3 u, z_1)\mathcal{Y}(w_1, z_2)w_2\rangle = \langle w_3', \left(Y_{W_3}^{g_3}\right)^{-1}(u, z_1)\mathcal{Y}(w_1, z_2)w_2\rangle, \tag{1.6}$$

where as above $\left(Y_{W_3}^{g_3}\right)^{-1}$ is the (-1)-th branch of the twisted vertex operator map $Y_{W_3}^{g_3}$. But the right-hand side of (1.6) can be obtained from $\langle w_3', Y_{W_3}^{g_3}(u, z_1)\mathcal{Y}(w_1, z_2)w_2\rangle$ on the region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ by letting z_1 go around clockwise a circle of radius larger than $|z_2|$. The homotopy class of such a circle is equal to $(b_{13}b_{12})^{-1}$. So (1.6) gives

$$f^e(z_1, z_2; g_3 u, w_1, w_2, w_3') = f^{(b_{13}b_{12})^{-1}}(z_1, z_2; u, w_1, w_2, w_3')$$

or equivalently

$$f^{b_{13}b_{12}}(z_1, z_2; g_3 u, w_1, w_2, w_3') = f^e(z_1, z_2; u, w_1, w_2, w_3'). \tag{1.7}$$

Similarly, on the region $|z_2| > |z_1| > 0$, $-\frac{3\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$, the sum of the series $\langle w_3', \mathcal{Y}(w_1, z_2)Y_{W_3}^{g_2}(g_2 u, z_1)w_2\rangle$ is equal to $f^e(z_1, z_2; g_2 u, w_1, w_2, w_3')$. By the equivariance property for W_2,

$$\langle w_3', \mathcal{Y}(w_1, z_2)Y_{W_3}^{g_2}(g_2 u, z_1)w_2\rangle = \langle w_3', \mathcal{Y}(w_1, z_2)\left(Y_{W_2}^{g_2}\right)^{-1}(u, z_1)w_2\rangle, \tag{1.8}$$

where $\left(Y_{W_2}^{g_2}\right)^{-1}$ is the (-1)-th branch of the twisted vertex operator map $Y_{W_2}^{g_2}$. The right-hand side of (1.8) can be obtained from $\langle w_3', \mathcal{Y}(w_1, z_2)Y_{W_3}^{g_2}(u, z_1)w_2\rangle$ on the region $|z_2| > |z_1| > 0$, $-\frac{3\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$ by letting z_1 go around clockwise a circle of radius less than $|z_2|$. Such a circle as a loop must have a base point in the region $|z_2| > |z_1| > 0$, $-\frac{3\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$. But there is a canonical isomorphism between the fundamental group of M^2 with such a base point and PB_3 which has a base point $(-3, -2)$. To see how the loop given by the circle above acts on the single-valued branches of $f(z_1, z_2; u, w_1, w_2, w_3')$, we need to find the element of PB_3 corresponding to this loop. We choose the following loop γ based at $(-3, -2)$: The first part γ_1 of γ is the lower half circle centered at -2 with radius 1 from -3 to -1 for z_1 and trivial for z_2 (meaning z_2 is always equal to -2). The second part γ_2 is the counterclockwise circle centered at 0 with radius 1 based at -1 for z_1 and trivial for z_2. The third part $\gamma_3 = \gamma_1^{-1}$ is also the lower half circle centered at -1 with radius 1 but from -1 to -3 for z_1 and trivial for z_2. It is clear that γ is homotopically equivalent to the loop given in Figure 3. Then we have $b_{13} = [\gamma] = [\gamma_1][\gamma_2][\gamma_1]^{-1}$, where $[\gamma]$ for a path γ means its homotopy class. Equivalently, we have $[\gamma_2] = [\gamma_1]^{-1}b_{13}[\gamma_1]$. When we let z_1 go from -1 to -3 along γ_1^{-1}, since γ_1^{-1} passes the cut along the positive real line in the $z_1 - z_2$-plane, the single-valued branch $f^e(z_1, z_2; u, w_1, w_2, w_3')$ is changed to the single-valued branch $f^{b_{12}^{-1}}(z_1, z_2; u, w_1, w_2, w_3')$. Similarly, when we let z_1 go from -1 to -3 along γ_1, $f^b(z_1, z_2; u, w_1, w_2, w_3')$ is changed to the single-valued branch $f^{bb_{12}}(z_1, z_2; u, w_1, w_2, w_3')$ for any $b \in \mathrm{PB}_3$. Thus when we let z_1 go around the loop γ_2, $f^e(z_1, z_2; u, w_1, w_2, w_3')$ is changed to $f^{b_{12}^{-1}b_{13}b_{12}}(z_1, z_2; u, w_1, w_2, w_3')$, that is,

$$f^{[\gamma_2]}(z_1, z_2; u, w_1, w_2, w_3') = f^{b_{12}^{-1}b_{13}b_{12}}(z_1, z_2; u, w_1, w_2, w_3').$$

Note that the circle γ_2^{-1} is exactly the circle we use to obtain the right-hand side of (1.8) from $\langle w_3', \mathcal{Y}(w_1, z_2)Y_{W_3}^{g_2}(u, z_1)w_2\rangle$ on the region $|z_2| > |z_1| > 0$, $-\frac{3\pi}{2} <$

$\arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$. So the right-hand side of (1.8) is equal to

$$f^{[\gamma_2^{-1}]}(z_1, z_2; u, w_1, w_2, w_3') = f^{b_{12}^{-1}b_{13}^{-1}b_{12}}(z_1, z_2; u, w_1, w_2, w_3').$$

But the sum of the left-hand side of (1.8) is equal to $f^e(z_1, z_2; g_2u, w_1, w_2, w_3')$. So we obtain

$$f^e(z_1, z_2; g_2u, w_1, w_2, w_3') = f^{b_{12}^{-1}b_{13}^{-1}b_{12}}(z_1, z_2; u, w_1, w_2, w_3')$$

or equivalently

$$f^{b_{12}^{-1}b_{13}b_{12}}(z_1, z_2; g_2u, w_1, w_2, w_3') = f^e(z_1, z_2; u, w_1, w_2, w_3'). \tag{1.9}$$

Similarly we also have

$$f^{b_{12}}(z_1, z_2; g_1u, w_1, w_2, w_3') = f^e(z_1, z_2; u, w_1, w_2, w_3'). \tag{1.10}$$

Using the right action of PB_3 on the set of single-valued branches of the multivalued analytic function $f(z_1, z_2; u, w_1, w_2, w_3')$, for $b \in \mathrm{PB}_3$, we obtain from (1.10)

$$f^{b_{12}b}(z_1, z_2; g_2u, w_1, w_2, w_3') = f^b(z_1, z_2; u, w_1, w_2, w_3'). \tag{1.11}$$

From (1.7), (1.9) and (1.11), we have

$$\begin{aligned} f^{b_{13}b_{12}}(z_1, z_2; g_3u, w_1, w_2, w_3') &= f^e(z_1, z_2; u, w_1, w_2, w_3') = f^{b_{12}^{-1}b_{13}b_{12}}(z_1, z_2; g_2u, w_1, w_2, w_3') \\ &= f^{b_{12}(b_{12}^{-1}b_{13}b_{12})}(z_1, z_2; g_1g_2u, w_1, w_2, w_3') \\ &= f^{b_{13}b_{12}}(z_1, z_2; g_1g_2u, w_1, w_2, w_3') \end{aligned} \tag{1.12}$$

When $|z_1| > |z_2| > 0$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$, the left- and right-hand sides of (1.12) are equal to the sum of $\big\langle w_3', \big(Y_{W_3}^{g_3}\big)^1(g_3u, z_1)\mathcal{Y}(w_1, z_2)w_2\big\rangle$ and $\big\langle w_3', \big(Y_{W_3}^{g_3}\big)^1(g_1g_2u, z_1)\mathcal{Y}(w_1, z_2)w_2\big\rangle$, respectively. Therefore, we obtain

$$\big\langle w_3', \big(Y_{W_3}^{g_3}\big)^1(g_1g_2u - g_3u, z_1)\mathcal{Y}(w_1, z_2)w_2\big\rangle = 0 \tag{1.13}$$

for $w_1 \in W_1$, $w_2 \in W_2$, $w_3 \in W_3'$. Since $\mathcal{Y}$ is surjective in the sense above and w_3' is arbitrary, (1.13) implies $\big(Y_{W_3}^{g_3}\big)^1(g_1g_2u - g_3u, z_1) = 0$. But the map given by $v \to Y_{W_3}^{g_3}(v, x)$ is injective, we obtain $g_1g_2u - g_3u = 0$ for $u \in V$. So we have $g_3 = g_1g_2$. □

2. Skew-symmetry and contragredient isomorphisms

Let g_1, g_2 be automorphisms of V, W_1, W_2 and W_3 g_1-, g_2- and g_1g_2-twisted V-modules without g_1-, g_2- and g_1g_2-actions, respectively and $\mathcal{Y}$ a twisted intertwining operator of type $\binom{W_3}{W_1\ W_2}$. We define linear maps $\Omega_\pm(\mathcal{Y})\colon W_2 \otimes W_1 \to W_3\{x\}[\log x]$, $w_2 \otimes w_1 \mapsto \Omega_\pm(\mathcal{Y})(w_2, x)w_1$ by

$$\Omega_\pm(\mathcal{Y})(w_2, x)w_1 = \mathrm{e}^{xL(-1)}\mathcal{Y}(w_1, y)w_2\big|_{y^n = \mathrm{e}^{\pm n\pi\mathrm{i}}x^n, \log y = \log x \pm \pi\mathrm{i}} \tag{2.1}$$

for $w_1 \in W_1$ and $w_2 \in W_2$. Note that we can also define Ω_p for $p \in \mathbb{Z}$ by changing $\pm$ in the right-hand side of (2.1) to $+(2p+1)$. But we will not discuss Ω_p in this paper.

From the definition (2.1), for $p \in \mathbb{Z}$, $w_1 \in W_1$, $w_2 \in W_2$ and $z \in \mathbb{C}^\times$,

$$\begin{aligned} \Omega_\pm(\mathcal{Y})^p(w_2, z)w_1 &= \Omega_\pm(\mathcal{Y})(w_2, x)w_1\big|_{x^n = \mathrm{e}^{nl_p(z)}, \log x = l_p(z)} \\ &= \big(\mathrm{e}^{xL(-1)}\mathcal{Y}(w_1, y)w_2\big|_{y^n = \mathrm{e}^{\pm n\pi\mathrm{i}}x^n, \log y = \log x \pm \pi\mathrm{i}}\big)\big|_{x^n = \mathrm{e}^{nl_p(z)}, \log x = l_p(z)} \\ &= \mathrm{e}^{zL(-1)}\mathcal{Y}(w_1, y)w_2\big|_{y^n = \mathrm{e}^{n(l_p(z) \pm \pi\mathrm{i})}, \log y = l_p(z) \pm \pi\mathrm{i}}. \end{aligned}$$

When $\arg z < \pi$ and $\arg z \geq \pi$, $\arg(-z) = \arg z + \pi$ and $\arg(-z) = \arg z - \pi$, respectively. Hence

$$\mathrm{e}^{zL(-1)}\mathcal{Y}(w_1, y)w_2\big|_{y^n = \mathrm{e}^{n(l_p(z) + \pi\mathrm{i})}, \log y = l_p(z) + \pi\mathrm{i}} = \mathrm{e}^{zL(-1)}\mathcal{Y}^p(w_1, -z)w_2$$

when $\arg z < \pi$ and

$$\mathrm{e}^{zL(-1)}\mathcal{Y}(w_1,y)w_2\big|_{y^n=\mathrm{e}^{n(l_p(z)-\pi\mathrm{i})},\log y=l_p(z)-\pi\mathrm{i}} = \mathrm{e}^{zL(-1)}\mathcal{Y}^p(w_1,-z)w_2$$

when $\arg z \geq \pi$. In particular, for $w_1 \in W_1$, $w_2 \in W_2$ and $z \in \mathbb{C}^\times$ satisfying $\arg z < \pi$ and $\arg z \geq \pi$, we have

$$\Omega_+(\mathcal{Y})^p(w_2,z)w_1 = \mathrm{e}^{zL(-1)}\mathcal{Y}^p(w_1,-z)w_2 \tag{2.2}$$

and

$$\Omega_-(\mathcal{Y})^p(w_2,z)w_1 = \mathrm{e}^{zL(-1)}\mathcal{Y}^p(w_1,-z)w_2, \tag{2.3}$$

respectively.

Now assume that W_1, W_2, and W_3 are g_1-, g_2-, and g_1g_2-twisted V-modules with g_1-, g_2- and g_1g_2-actions, respectively. In the case that W_1 also has an action of g_2^{-1}, we have the $g_2^{-1}g_1g_2$-twisted V-module $\varphi_{g_2^{-1}}(W_1)$. In this case, we define

$$\Omega_+^{g_2^{-1}}(\mathcal{Y})\colon\ W_2\otimes W_1 \to W_3\{x\}[\log x], \qquad w_2\otimes w_1 \mapsto \Omega_+^{g_2^{-1}}(\mathcal{Y})(w_2,x)w_1$$

by $\Omega_+^{g_2^{-1}}(\mathcal{Y})(w_2,x)w_1 = \mathrm{e}^{xL(-1)}\mathcal{Y}(g_2w_1,y)w_2\big|_{y^n=\mathrm{e}^{n\pi\mathrm{i}}x^n,\log y=\log x+\pi\mathrm{i}}$ for $w_1 \in W_1$ and $w_2 \in W_2$. In the case that W_2 has an action of g_1, we have the $g_1g_2g_1^{-1}$-twisted V-module $\varphi_{g_1}(W_2)$. In this case, we define

$$\Omega_-^{g_1}(\mathcal{Y})\colon\ W_2\otimes W_1 \to W_3\{x\}[\log x], \qquad w_2\otimes w_1 \mapsto \Omega_-^{g_1}(\mathcal{Y})(w_2,x)w_1$$

by $\Omega_-^{g_1}(\mathcal{Y})(w_2,x)w_1 = \mathrm{e}^{xL(-1)}\mathcal{Y}(w_1,y)g_1^{-1}w_2\big|_{y^n=\mathrm{e}^{-n\pi\mathrm{i}}x^n,\log y=\log x-\pi\mathrm{i}}$ for $w_1 \in W_1$ and $w_2 \in W_2$.

For $w_1 \in W_1$, $w_2 \in W_2$ and $z \in \mathbb{C}^\times$ satisfying $\arg z < \pi$ and $\arg z \geq \pi$, we have

$$\Omega_+^{g_2^{-1}}(\mathcal{Y})^p(w_2,z)w_1 = \mathrm{e}^{zL(-1)}g_2\mathcal{Y}^p(g_2w_1,-z)g_2^{-1}w_2 \tag{2.4}$$

and

$$\Omega_-^{g_1}(\mathcal{Y})^p(w_2,z)w_1 = \mathrm{e}^{zL(-1)}\mathcal{Y}^p(w_1,-z)g_1^{-1}w_2, \tag{2.5}$$

respectively.

Theorem 2.1. *The linear maps $\Omega_+(\mathcal{Y})$, $\Omega_-(\mathcal{Y})$, $\Omega_+^{g_2^{-1}}(\mathcal{Y})$ and $\Omega_-^{g_1}(\mathcal{Y})$ are twisted intertwining operators of types*

$$\binom{W_3}{W_2\phi_{g_2^{-1}}(W_1)}, \quad \binom{W_3}{\phi_{g_1}(W_2)W_1}, \quad \binom{W_3}{W_2\varphi_{g_2^{-1}}(W_1)}, \quad \binom{W_3}{\varphi_{g_1}(W_2)W_1},$$

respectively (recall the definition of ϕ_g and φ_g for an automorphism g of V in Section 1).

Proof. We need only prove the results for $\Omega_+(\mathcal{Y})$ and $\Omega_-(\mathcal{Y})$. Using the definitions of φ_{g_1} and $\varphi_{g_2^{-1}}$ and (2.4) and (2.5), we see that the proofs for $\Omega_+^{g_2^{-1}}(\mathcal{Y})$ and $\Omega_-^{g_1}(\mathcal{Y})$ are reduced to the proofs for $\Omega_+(\mathcal{Y})$ and $\Omega_-(\mathcal{Y})$.

The proofs of the lower-truncation property and the $L(-1)$-derivative property of $\Omega_+(\mathcal{Y})$ and $\Omega_-(\mathcal{Y})$ are easy to verify. As in the proof of [17, Theorem 5.1], we omit their proofs here. The main difference between the proof here and the proof of [17, Theorem 5.1] is that here we cannot use the explicit form of the correlation functions in [17] obtained from the products and iterates of twisted intertwining operators and twisted vertex operators. So our proof here is much more complicated and involves some technical convergence and analytic extension results, even though the idea is the same as in the proof of [17, Theorem 5.1].

Let $u \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $w_3' \in W_3'$. As in the proof of [17, Theorem 5.1], we use $f(z_1,z_2;u,w_1,w_2,w_3')$ to denote the multivalued analytic function in the duality property for the twisted intertwining operator $\mathcal{Y}$ with the preferred branch $f^e(z_1,z_2;u,w_1,w_2,w_3')$. Note that $f(z_1,z_2;u,w_1,w_2,w_3')$ in [17] is of the

particular form in the definition of twisted intertwining operator there. But in this proof, it is a maximally-extended multivalued analytic function on M^2 with regular singular points at $z_1, z_2 = 0$ and $z_1 - z_2 = 0$ and in general might not have the special form in [17].

Define

$$g_{\pm}(z_1, z_2; u, w_2, w_1, w_3') = f\big(z_1 - z_2, -z_2; u, w_1, w_2, \mathrm{e}^{z_2 L'(1)} w_3'\big)$$

and choose the preferred branch $g_{\pm}^e(z_1, z_2; u, w_2, w_1, w_3')$ of $g_{\pm}(z_1, z_2; u, w_2, w_1, w_3')$ as follows: On the subregion $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_2 < \pi$ (for Ω_+) or $\arg z_2 \geq \pi$ (for Ω_-) of M_0^2, let

$$g_{\pm}^e(z_1, z_2; u, w_2, w_1, w_3') = f^e\big(z_1 - z_2, -z_2; u, w_1, w_2, \mathrm{e}^{z_2 L'(1)} w_3'\big).$$

For general $(z_1, z_2) \in M_0^2$, we define $g_{\pm}^e(z_1, z_2; u, w_2, w_1, w_3')$ to be the unique analytic extension on M_0^2.

When $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_2 < \pi$ (for Ω_+) or $\arg z_2 \geq \pi$ (for Ω_-), from (2.2) and (2.3) and the $L(-1)$-derivative property for $Y_{W_3}^{g_3}$, we have

$$\begin{aligned}
&\big\langle w_3', Y_{W_3}^{g_3}(u, z_1)\Omega_{\pm}(\mathcal{Y})(w_2, z_2)w_1\big\rangle \\
&= \big\langle w_3', Y_{W_3}^{g_3}(u, x_1)\Omega_{\pm}(\mathcal{Y})(w_2, x_2)w_1\big\rangle\big|_{x_1^n = \mathrm{e}^{n\log z_1}, \log x_1 = \log z_1, x_2^n = \mathrm{e}^{n\log z_2}, \log x_2 = \log z_2} \\
&= \big\langle w_3', Y_{W_3}^{g_3}(u, x_1)\mathrm{e}^{-yL(-1)}\mathcal{Y}(w_1, y)w_2\big\rangle\big|_{x_1^n = \mathrm{e}^{n\log z_1}, \log x_1 = \log z_1, y^n = \mathrm{e}^{n\log(-z_2)}, \log y = \log(-z_2)} \\
&= \big\langle \mathrm{e}^{z_2 L'(1)} w_3', (Y_{W_3}^{g_3})(u, x_1 + y) \\
&\qquad \times \mathcal{Y}(w_1, y)w_2\big\rangle\big|_{x_1^n = \mathrm{e}^{n\log z_1}, \log x_1 = \log z_1, y^n = \mathrm{e}^{n\log(-z_2)}, \log y = \log(-z_2)}.
\end{aligned} \tag{2.6}$$

We first prove that the right-hand side of formula (2.6) is absolutely convergent on the region $|z_1| > |z_2| > 0$ and is convergent to $g_{\pm}^e(z_1, z_2; u, w_1, w_2, w_3')$ on the region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$. The proof is in fact the same as the proof that the right-hand side of [30, equation (9.170)] is absolutely convergent in the region $|z_2| > |z_0| > 0$. Here we give a slightly different proof.

We can always take $u \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $\mathrm{e}^{z_2 L'(1)} w_3' \in W_3'$ to be homogeneous. Let $\Delta = -\operatorname{wt} \mathrm{e}^{z_2 L'(1)} w_3' + \operatorname{wt} u + \operatorname{wt} w_1 + \operatorname{wt} w_2$. Let

$$D = \big\{n \in \mathbb{C} \mid \text{there exist } i, j \in \mathbb{N}, \big\langle \big(Y_{W_3}^{g_3}\big)_{\Delta - n - 2, j}(u)\mathcal{Y}_{n,i}(w_1)w_2\big\rangle \neq 0\big\}$$

and $M, N \in \mathbb{N}$ such that $\big(Y_{W_3}^{g_3}\big)_{m,j}(u) = 0$ for $m \in \mathbb{C}$, $j > M$ and $\mathcal{Y}_{n,i}(w_1) = 0$ for $i > N$. Then by the lower truncation property of $\mathcal{Y}$, the fact that u is a finite sum of generalized eigenvectors of g_3 and the equivariance property of the g_3-twisted module W_3, we know that there exist a finite subset A of $\mathbb{C}/\mathbb{Z}$ and $R_\mu \in \mu$ for each $\mu \in A$ such that

$$D \subset \bigcup_{\mu \in A} = (R_\mu - \mathbb{N}).$$

Let

$$a_{n,j,i} = \big\langle \mathrm{e}^{z_2 L'(1)} w_3', \big(Y_{W_3}^{g_3}\big)_{\Delta - n - 2, j}(u)(\mathcal{Y})_{n,i}(w_1)w_2\big\rangle \in \mathbb{C}$$

for $n \in D$, $0 \leq j \leq M$ and $0 \leq n \leq N$. Then, by the convergence of (1.3), the $L(-1)$-derivative properties for $Y_{W_3}^{g_3}$ and $\mathcal{Y}$, and [29, Proposition 7.9], we know that the triple series

$$\sum_{n \in D} \sum_{j=0}^{M} \sum_{i=0}^{N} a_{n,j,i} \mathrm{e}^{(-\Delta + n + 1)\log z_1} (\log z_1)^j \mathrm{e}^{(-n-1)\log z_2} (\log z_2)^i \tag{2.7}$$

is absolutely convergent on the region given by $|z_1| > |z_2| > 0$ and is convergent on the region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ to

$$f^e\big(z_1, z_2; u, w_1, w_2, \mathrm{e}^{z_2 L'(1)} w_3'\big) = \big\langle \mathrm{e}^{z_2 L'(1)} w_3', Y_{W_3}^{g_3}(u, z_1)\mathcal{Y}(w_1, z_2)w_2\big\rangle.$$

For $n \in D$, $j = 0, \dots, M$, $k \in \mathbb{Z}_{\geq 0}$, $s = 0, \dots, j$, let $b_{n,j,k,s} \in \mathbb{C}$ be the numbers defined in (5.4). Then

$$\sum_{m \in D - \mathbb{N}} \sum_{s=0}^{M} \sum_{i=0}^{N} \left(\sum_{j=s}^{M} \sum_{\substack{n-k=m \\ n \in D,\, k \in \mathbb{N}}} a_{n,j,i} b_{n,j,k,s} \right) \mathrm{e}^{(-\Delta + m + 1)\log z_1} (\log z_1)^s \times \mathrm{e}^{(-m-1)\log(-z_2)} (\log(-z_2))^i$$

is equal to the right-hand side of (2.6) and, by Lemma 5.1, is absolutely convergent on the region $|z_1| > |z_2| > 0$ and is convergent to $f^e\big(z_1 - z_2, -z_2; u, w_1, w_2, \mathrm{e}^{z_2 L'(1)} w_3'\big)$ on the region $|z_1| > |z_2| > 0$, $|\arg z_1 - \arg(z_1 - z_2)| < \frac{\pi}{2}$.

Now it is easy to see that the left-hand side of (2.6) is absolutely convergent on the region $|z_1| > |z_2| > 0$ and its sum is equal to $g_{\pm}^e(z_1, z_2; u, w_1, w_2, w_3')$ on the region $|z_1| > |z_2| > 0$, $|\arg z_1 - \arg(z_1 - z_2)| < \frac{\pi}{2}$. In fact, we know that the left-hand side of (2.6) as a series is equal to the right-hand side of (2.6) on the region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_2 < \pi$ (for Ω_+) or $\arg z_2 \geq \pi$ (for Ω_-). But we have just proved that the right-hand side of (2.6) is absolutely convergent on the larger region $|z_1| > |z_2| > 0$. The left-hand side of (2.6) is also a series of the same form as (2.7). In particular, its absolute convergence on the smaller region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_2 < \pi$ (for Ω_+) or $\arg z_2 \geq \pi$ (for Ω_-) implies its absolute convergence on the larger region $|z_1| > |z_2| > 0$.

On the region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_2 < \pi$ (for Ω_+) or $\arg z_2 \geq \pi$ (for Ω_-), by (2.6) and the discussion above, the left-hand side of (2.6) is convergent to $f^e\big(z_1 - z_2, -z_2; u, w_1, w_2, \mathrm{e}^{z_2 L'(1)} w_3'\big)$, which in turn is by definition equal to $g_{\pm}^e(z_1, z_2; u, w_1, w_2, w_3')$ on the same region. We know that the left-hand side of (2.6) on the region given by $|z_1| > |z_2| > 0$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ with cuts along the positive real lines on the z_1- and z_2-planes is convergent to the analytic extension of the sum of the left-hand side of (2.6) on the smaller region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_2 < \pi$ (for Ω_+) or $\arg z_2 \geq \pi$ (for Ω_-). Also, by definition, $g_{\pm}^e(z_1, z_2; u, w_1, w_2, w_3')$ on M_0^2 is obtained by analytically extending $g_{\pm}^e(z_1, z_2; u, w_1, w_2, w_3')$ on the smaller region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_2 < \pi$ (for Ω_+) or $\arg z_2 \geq \pi$ (for Ω_-). Thus the left-hand side of (2.6) on the region given by $|z_1| > |z_2| > 0$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ is absolutely convergent to $g_{\pm}^e(z_1, z_2; u, w_1, w_2, w_3')$ on M_0^2.

Generalizing the proof of the convergence of (2.6) above, we can prove that

$$\big\langle w_3', Y_{W_3}^{g_3}(u_1, z_1) \cdots Y_{W_3}^{g_3}(u_{k-1}, z_{k-1})\Omega_{\pm}(\mathcal{Y})(w_2, z_k)w_1\big\rangle \tag{2.8}$$

is absolutely convergent on the region $|z_1| > \cdots > |z_k| > 0$ and can be maximally extended to a multivalued analytic function on the region M^k for $k \in \mathbb{N} + 3$, $u_1, \dots, u_{k-1} \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $w_3' \in W_3'$. In fact, generalizing (2.6), we see that (2.8) is equal to

$$\begin{aligned} &\big\langle \mathrm{e}^{z_k L'(1)} w_3', \big(Y_{W_3}^{g_3}\big)(u_1, x_1 + y) \cdots \big(Y_{W_3}^{g_3}\big)(u_{k-1}, x_{k-1} + y) \\ &\times \mathcal{Y}(w_1, y)w_2\big\rangle\big|_{x_i^n = \mathrm{e}^{n \log z_i},\, \log x_i = \log z_i,\, i = 1, \dots, k-1,\, y^n = \mathrm{e}^{n \log(-z_k)},\, \log y = \log(-z_k)}. \end{aligned} \tag{2.9}$$

From Definition 1.1, we see that the convergence and analytic extension of (2.9) on the region $|z_1| > \cdots > |z_k| > 0$ is equivalent to the convergence and analytic

extension of

$$\prod_{1\le i<j\le k-1}(z_i-z_j)^{M_{ij}}\left\langle \mathrm{e}^{z_k L'(1)}w_3', \left(Y_{W_3}^{g_3}\right)(u_1,x_1+y)\cdots\left(Y_{W_3}^{g_3}\right)(u_{k-1},x_{k-1}+y)\right.$$

$$\left.\times\mathcal{Y}(w_1,y)w_2\right\rangle\Big|_{x_i^n=\mathrm{e}^{n\log z_i},\log x_i=\log z_i,i=1,\dots,k-1,y^n=\mathrm{e}^{n\log(-z_k)},\log y=\log(-z_k)} \tag{2.10}$$

on the region $|z_i|>|z_k|>0$ for $i=1,\dots,k-1$, $z_i\ne z_j$ for $i\ne j$, where $M_{ij}\in\mathbb{Z}_+$ for $i\ne j$ satisfy $x^{M_{ij}}Y_V(u_i,x)u_j\in V[[x]]$. From Remark 5.3, we know that Lemma 5.1 can be generalized to the case of more than two variables for a series of the form (2.10). It is important to note that the polynomial $\prod_{1\le i<j\le k-1}(z_i-z_j)^{M_{ij}}$ in (2.10) is multiplied to cancel the poles at $z_i=z_j$ for $1\le i<j\le k-1$ so that the series (2.10) is a different iterated sum of a convergent iterated series of the form (5.22). Using the convergence of products with more than one twisted vertex operator for the twisted intertwining operator $\mathcal{Y}$ and this generalization of Lemma 5.1, we see that (2.10) is absolutely convergent on the region $|z_i|>|z_k|>0$ for $i=1,\dots,k-1$, $z_i\ne z_j$ for $i\ne j$ and its sum has analytic extension on the region M^k. Thus (2.8) is absolutely convergent on the region $|z_1|>\cdots>|z_k|>0$ and its sum has maximal analytic extension on the region M^k. Moreover, from the discussion above, we have

$$g(z_1,\dots,z_k)=\sum_{j=0}^{J}\frac{z_k^j}{j!}f_j(z_1-z_k,\dots,z_{k-1}-z_k,-z_k),$$

where $g(z_1,\dots,z_k)$ is the maximal analytic extension of the sum of (2.8) on the region M^k, $J\in\mathbb{Z}_+$ such that $L'(1)^{J+1}w_3'=0$, and $f_j(z_1,\dots,z_k)$ for $j=0,\dots,J$ are the maximal analytic extensions of the sum of

$$\left\langle L'(1)^j w_3', Y_{W_3}^{g_3}(u_1,z_1)\cdots Y_{W_3}^{g_3}(u_{k-1},z_{k-1})\mathcal{Y}(w_1,z_k)w_2\right\rangle.$$

For fixed $z_k\in\mathbb{C}^\times$, let $(z_1,\dots,z_{k-1})-\beta=\delta$ for $\beta=\mathbb{C}^{k-1}$ and $\delta\in\{0,\infty\}^{k-1}$ be a component-isolated singularity of the multivalued analytic function of of $z_1,\dots,z_{k-1}$ obtained as a value $\prod_{1\le i<j\le k-1}(z_i-z_j)^{M_{ij}}g(z_1,\dots,-z_k)$ at $\zeta=-z_k$ of $\prod_{1\le i<j\le k-1}(z_i-z_j)^{M_{ij}}g(z_1,\dots,\zeta)$. Then this singularity, which can also be written as $(z_1-z_k,\dots,z_{k-1}-z_k)-(\beta+(z_k,\dots,z_k))=\delta$, is also a component-isolated singularity of the corresponding values

$$\prod_{1\le i<j\le k-1}(z_i-z_j)^{M_{ij}}f_j(z_1-z_k,\dots,z_{k-1}-z_k,-z_k)$$

at $\zeta=-z_k$ of

$$\prod_{1\le i<j\le k-1}(z_i-z_j)^{M_{ij}}f_j(z_1-z_k,\dots,z_{k-1}-z_k,\zeta)$$

for $j=0,\dots,J$. Thus this singularity of

$$\prod_{1\le i<j\le k-1}(z_i-z_j)^{M_{ij}}f_j(z_1-z_k,\dots,z_{k-1}-z_k,-z_k)$$

corresponds to the component-isolated singularity $(z_1,\dots,z_{k-1})-(\beta+(z_k,\dots,z_k))=\delta$ of $\prod_{1\le i<j\le k-1}(z_i-z_j)^{M_{ij}}f_j(z_1,\dots,z_k)$. But we know that a component-isolated singularity of $\prod_{1\le i<j\le k-1}(z_i-z_j)^{M_{ij}}f_j(z_1,\dots,z_{k-1},-z_k)$ of such a form is regular. So this component-isolated singularity $(z_1,\dots,z_{k-1})-(\beta+(z_k,\dots,z_k))=\delta$ is regular. Equivalently, the component-isolated singularity $(z_1,\dots,z_{k-1})-\beta=\delta$ of

$$\prod_{1\le i<j\le k-1}(z_i-z_j)^{M_{ij}}f_j(z_1-z_k,\dots,z_{k-1}-z_k,-z_k)$$

for $j = 0, \dots, J$ is regular and thus the component-isolated singularity of $\prod_{1\le i<j\le k-1}(z_i - z_j)^{M_{ij}} g(z_1, \dots, z_{k-1}, -z_k)$ is regular. But z_k is arbitrary, $-z_k$ is also arbitrary. We see that for any fixed $z_k \in \mathbb{C}^\times$, $(z_1, \dots, z_{k-1}) - \beta = \delta$ is a regular singularity of any value at $\zeta = z_k$ of $\prod_{1\le i<j\le k-1}(z_i - z_j)^{M_{ij}} g(z_1, \dots, \zeta)$.

When $|z_2| > |z_1| > 0$ and $\arg z_2 \ge \pi$,

$$\begin{aligned} &\big\langle w_3', \Omega_-(\mathcal{Y})(w_2, z_2) Y_{W_1}^{g_1}(u, z_1) w_1 \big\rangle \\ &= \big\langle w_3', \mathrm{e}^{z_2 L(-1)} \mathcal{Y}\big(Y_{W_1}^{g_1}(u, z_1) w_1, -z_2\big) w_2 \big\rangle \\ &= \big\langle \mathrm{e}^{z_2 L'(1)} w_3', \mathcal{Y}\big(Y_{W_1}^{g_1}(u, (z_1 - z_2) - (-z_2)) w_1, -z_2\big) w_2 \big\rangle \end{aligned} \tag{2.11}$$

converges absolutely and if in addition, $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$, its sum is equal to

$$f^e\big(z_1 - z_2, -z_2; u, w_1, w_2, \mathrm{e}^{z_2 L'(1)} w_3'\big). \tag{2.12}$$

Note that by definition, (2.12) is a single-valued analytic function on the set $\widetilde{M_0^2}$ given by cutting M^2 along the positive real line in the z_1- and $(z_1 - z_2)$-planes and along the negative real line in the z_2-plane, with these positive real lines in the z_1- and $(z_1 - z_2)$-planes attached to the upper half z_1- and $(z_1 - z_2)$-planes and the negative real line in the z_2-plane attached to the lower half z_2-plane. Then the subset

$$\Big\{(z_1, z_2) \in M_0^2 \mid |z_2| > |z_1| > 0,\ \arg z_2 \ge \pi,\ |\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}\Big\} \tag{2.13}$$

of $\widetilde{M_0^2}$ is also a subset of M_0^2. But on the subset of M_0^2 given by $|z_1| > |z_2| > 0$, $\arg z_2 \ge \pi$, by definition, (2.12) is equal to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$. Since $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ on M_0^2 is obtained by analytic extension, we see that (2.12) is equal to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ also on the subset (2.13). Thus the left-hand side of (2.11) is absolutely convergent to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_2| > |z_1| > 0$, $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$ and $\arg z_2 \ge \pi$. Since $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ and the sum of (2.11) are both analytic extensions of their restrictions on the subset given by $|z_2| > |z_1| > 0$, $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$ and $\arg z_2 \ge \pi$, we see that the left-hand side of (2.11) is absolutely convergent to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_2| > |z_1| > 0$ and $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$. But when $\arg z_2 \ge \pi$, $\arg(-z_2) = \arg z_2 - \pi$. Hence in this case, the inequality $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$ becomes $-\frac{3\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$. Also both the left-hand side of (2.11) and $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ are single valued analytic functions in z_1 and z_2 with cuts at $z_1 \in \mathbb{R}_+$ and $z_2 \in \mathbb{R}_+$. Thus when $|z_2| > |z_1| > 0$ and $-\frac{3\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$, the left-hand side of (2.11) is absolutely convergent to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$.

Next we discuss the iterate of $\Omega_-(\mathcal{Y})$ and the twisted vertex operator map $\phi_{g_1}\big(Y_{W_2}^{g_2}\big)$. When $|z_2| > |z_1 - z_2| > 0$ and $\arg z_2 \ge \pi$,

$$\begin{aligned} &\big\langle w_3', \Omega_-(\mathcal{Y})\big(\phi_{g_1}\big(Y_{W_2}^{g_2}\big)(u, z_1 - z_2) w_2, z_2\big) w_1 \big\rangle \\ &= \big\langle w_3', \Omega_-(\mathcal{Y})\big(\big(Y_{W_2}^{g_2}\big)\big(g_1^{-1} u, z_1 - z_2\big) w_2, z_2\big) w_1 \big\rangle \\ &= \big\langle w_3', \mathrm{e}^{z_2 L(-1)} \mathcal{Y}(w_1, -z_2)\big(Y_{W_2}^{g_2}\big)\big(g_1^{-1} u, z_1 - z_2\big) w_2 \big\rangle \\ &= \big\langle \mathrm{e}^{z_2 L'(1)} w_3', \mathcal{Y}(w_1, -z_2)\big(Y_{W_2}^{g_2}\big)\big(g_1^{-1} u, z_1 - z_2\big) w_2 \big\rangle \end{aligned} \tag{2.14}$$

converges absolutely and if in addition, $-\frac{3\pi}{2} < \arg z_1 - \arg(-z_2) < -\frac{\pi}{2}$, its sum is equal to $f^e(z_1 - z_2, -z_2; g_1^{-1} u, w_1, w_2, \mathrm{e}^{z_2 L'(1)} w_3')$. Note that the proofs of [17, Lemmas 4.5 and 4.6] do not use the explicit form of the multivalued analytic functions in the duality property of the twisted intertwining operators introduced in [17]. Then the same proof of [17, Lemma 4.5] shows that the sum of the right-hand side of (2.14)

is equal to $f^{b_{12}^{-1}}\big(z_1 - z_2, -z_2; g_1^{-1}u, w_1, w_2, \mathrm{e}^{z_2 L'(1)}w_3'\big)$ when $|z_2| > |z_1 - z_2| > 0$, $\arg z_2 \geq \pi$ and $\frac{\pi}{2} < \arg z_1 - \arg(-z_2) < \frac{3\pi}{2}$. The same proof of (4.4) in [17, Lemma 4.6] shows that

$$f^{b_{12}^{-1}}\big(z_1 - z_2, -z_2; g_1^{-1}u, w_1, w_2, \mathrm{e}^{z_2 L'(1)}w_3'\big) = f^e\big(z_1 - z_2, -z_2; u, w_1, w_2, \mathrm{e}^{z_2 L'(1)}w_3'\big).$$

Since when $\arg z_2 \geq \pi$, $\arg(-z_2) = \arg z_2 - \pi$ and $\frac{\pi}{2} < \arg z_1 - \arg(-z_2) < \frac{3\pi}{2}$ becomes $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$, we see that the sum of the right-hand side of (2.14) is equal to (2.12) when $|z_2| > |z_1 - z_2| > 0$, $\arg z_2 \geq \pi$ and $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$.

We now use the same argument as above to finish the proof in this case. The subset

$$\Big\{(z_1, z_2) \in M_0^2 \mid |z_2| > |z_1 - z_2| > 0,\ \arg z_2 \geq \pi,\ |\arg(z_1) - \arg z_2| < \frac{\pi}{2}\Big\} \tag{2.15}$$

of $\widetilde{M}_0^2$ is also a subset of M_0^2. By definition, on the subset of M_0^2 given by $|z_1| > |z_2| > 0$, $\arg z_2 \geq \pi$, (2.12) is equal to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$. Since $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ on M_0^2 is obtained by analytic extension, we see that (2.12) is equal to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ also on the subset (2.15). Thus the left-hand side of (2.14) is absolutely convergent to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_2| > |z_1 - z_2| > 0$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$ and $\arg z_2 \geq \pi$. Since $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ and the sum of (2.14) are both analytic extensions of their restrictions on the subset given by $|z_2| > |z_1 - z_2| > 0$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$ and $\arg z_2 \geq \pi$, we see that the left-hand side of (2.14) is absolutely convergent to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_2| > |z_1 - z_2| > 0$ and $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$.

We still need to prove the two other cases for $\Omega_+(\mathcal{Y})$. When $|z_2| > |z_1| > 0$ and $\arg z_2 < \pi$,

$$\begin{aligned}
&\big\langle w_3', \Omega_+(\mathcal{Y})(w_2, z_2)\phi_{g_2^{-1}}\big(Y_{W_1}^{g_1}\big)(u, z_1)w_1\big\rangle \\
&= \big\langle w_3', \mathrm{e}^{z_2 L(-1)}\mathcal{Y}\big(\phi_{g_2^{-1}}\big(Y_{W_1}^{g_1}\big)(u, z_1)w_1, -z_2\big)w_2\big\rangle \\
&= \big\langle \mathrm{e}^{z_2 L'(1)}w_3', \mathcal{Y}\big(\big(Y_{W_1}^{g_1}\big)(g_2 u, z_1)w_1, -z_2\big)w_2\big\rangle
\end{aligned} \tag{2.16}$$

converges absolutely and if in addition, $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$, its sum is equal to

$$f^e\big(z_1 - z_2, -z_2; g_2 u, w_1, w_2, \mathrm{e}^{z_2 L'(1)}w_3'\big). \tag{2.17}$$

The subset

$$\Big\{(z_1, z_2) \in M_0^2 \mid |z_2| > |z_1| > 0,\ \arg z_2 < \pi,\ |\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}\Big\} \tag{2.18}$$

of $\widetilde{M}_0^2$ is also a subset of M_0^2. By definition, on the subset of M_0^2 given by $|z_1| > |z_2| > 0$, $\arg z_2 < \pi$, (2.17) is equal to $g_+^e(z_1, z_2; g_2 u, w_2, w_1, w_3')$. Since $g_+^e(z_1, z_2; g_2 u, w_2, w_1, w_3')$ on M_0^2 is obtained by analytic extension, (2.17) is equal to $g_+^e(z_1, z_2; g_2 u, w_2, w_1, w_3')$ also on the subset (2.18). Thus the left-hand side of (2.16) is absolutely convergent to $g_+^e(z_1, z_2; g_2 u, w_2, w_1, w_3')$ when $|z_2| > |z_1| > 0$, $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$ and $\arg z_2 < \pi$. Since $g_+^e(z_1, z_2; g_2 u, w_2, w_1, w_3')$ and the sum of (2.16) are both analytic extensions of their restrictions on the subset given by $|z_2| > |z_1| > 0$, $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$ and $\arg z_2 < \pi$, the left-hand side of (2.16) is absolutely convergent to $g_+^e(z_1, z_2; g_2 u, w_2, w_1, w_3')$ when $|z_2| > |z_1| > 0$, $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$.

The same proof as that of (4.5) in [17, Lemma 4.6] gives $g_+^e(z_1, z_2; g_2 u, w_1, w_2, w_3') = g_+^{b_{13}^{-1}}(z_1, z_2; u, w_1, w_2, w_3')$, since $\arg z_1 < \pi$, $\arg(-z_2) = \arg z_2 + \pi$ and the inequality $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$ becomes $\frac{\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < \frac{3\pi}{2}$. So the sum of the left-hand side of (2.16) is absolutely convergent to $g_+^{b_{13}^{-1}}(z_1, z_2; u, w_1, w_2, w_3')$ when $|z_2| > |z_1| > 0$ and $\frac{\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < \frac{3\pi}{2}$. Then the same proof

as that of [17, Lemma 4.5] shows that this is equivalent to the sum of left-hand side of (2.16) being equal to $g_+^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_2| > |z_1| > 0$ and $-\frac{3\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$.

Finally, we discuss the iterate of $\Omega_+(\mathcal{Y})$ and the twisted vertex operator map $Y_{W_2}^{g_2}$. When $|z_2| > |z_1 - z_2| > 0$ and $\arg z_2 < \pi$,

$$\begin{aligned}
&\big\langle w_3', \Omega_+(\mathcal{Y})\big(Y_{W_2}^{g_2}(u, z_1 - z_2)w_2, z_2\big)w_1\big\rangle \\
&= \big\langle w_3', \mathrm{e}^{z_2 L(-1)}\mathcal{Y}(w_1, -z_2)Y_{W_2}^{g_2}(u, z_1 - z_2)w_2\big\rangle \\
&= \big\langle \mathrm{e}^{z_2 L'(1)}w_3', \mathcal{Y}(w_1, -z_2)Y_{W_2}^{g_2}(u, z_1 - z_2)w_2\big\rangle \qquad (2.19)
\end{aligned}$$

converges absolutely and if in addition, $-\frac{3\pi}{2} < \arg z_1 - \arg(-z_2) < -\frac{\pi}{2}$, its sum is equal to (2.12). The subset

(2.20)

$$\left\{(z_1, z_2) \in M_0^2 \mid |z_2| > |z_1 - z_2| > 0,\ \arg z_2 < \pi,\ -\frac{3\pi}{2} < \arg z_1 - \arg(-z_2) < -\frac{\pi}{2}\right\}$$

of $\widetilde{M_0^2}$ is also a subset of M_0^2. As we have discussed above, on the subset of M_0^2 given by $|z_1| > |z_2| > 0$, $\arg z_2 < \pi$, (2.12) is equal to $g_+^e(z_1, z_2; u, w_2, w_1, w_3')$. Since $g_+^e(z_1, z_2; u, w_2, w_1, w_3')$ on M_0^2 is obtained by analytic extension, (2.12) is equal to $g_+^e(z_1, z_2; u, w_2, w_1, w_3')$ also on the subset (2.20). Thus the left-hand side of (2.19) is absolutely convergent to $g_+^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_2| > |z_1 - z_2| > 0$, $-\frac{3\pi}{2} < \arg z_1 - \arg(-z_2) < -\frac{\pi}{2}$ and $\arg z_2 < \pi$. When $\arg z_2 < \pi$, $\arg(-z_2) = \pi + \arg z_2$. So in this case, $-\frac{3\pi}{2} < \arg z_1 - \arg(-z_2) < -\frac{\pi}{2}$ is the same as $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$. Since $g_+^e(z_1, z_2; u, w_2, w_1, w_3')$ and the sum of (2.19) are both analytic extensions of their restrictions on the subset given by $|z_2| > |z_1 - z_2| > 0$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$ and $\arg z_2 < \pi$, the left-hand side of (2.19) is absolutely convergent to $g_+^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_2| > |z_1 - z_2| > 0$ and $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$. □

Let $\mathcal{V}_{W_1W_2}^{W_3}$ be the space of twisted intertwining operators of type $\binom{W_3}{W_1W_2}$. Then we have the following.

Corollary 2.2. *The maps*

$$\Omega_+\colon\ \mathcal{V}_{W_1W_2}^{W_3} \to \mathcal{V}_{W_2\phi_{g_2^{-1}}(W_1)}^{W_3}, \qquad \Omega_-\colon\ \mathcal{V}_{W_1W_2}^{W_3} \to \mathcal{V}_{\phi_{g_1}(W_2)W_1}^{W_3},$$

$$\Omega_+^{g_2^{-1}}\colon\ \mathcal{V}_{W_1W_2}^{W_3} \to \mathcal{V}_{W_2\varphi_{g_2^{-1}}(W_1)}^{W_3}, \qquad \Omega_-^{g_1}\colon\ \mathcal{V}_{W_1W_2}^{W_3} \to \mathcal{V}_{\varphi_{g_1}(W_2)W_1}^{W_3}$$

are linear isomorphisms. In particular,

$$\mathcal{V}_{W_1W_2}^{W_3}, \quad \mathcal{V}_{\phi_{g_1}(W_2)W_1}^{W_3}, \quad \mathcal{V}_{W_2\phi_{g_2^{-1}}(W_1)}^{W_3}, \quad \mathcal{V}_{\varphi_{g_1}(W_2)W_1}^{W_3} \quad \textit{and} \quad \mathcal{V}_{W_2\varphi_{g_2^{-1}}(W_1)}^{W_3}$$

are linearly isomorphic.

Proof. It is clear that Ω_+ and Ω_- are inverse of each other. Since $\varphi_{g_1}(W_2)$ and $\varphi_{g_2^{-1}}(W_1)$ are equivalent to $\phi_{g_1}(W_2)$ and $\phi_{g_2^{-1}}(W_1)$, respectively, $\Omega_+^{g_2}$ and $\Omega_-^{g_1}$ are also linear isomorphisms. □

The linear isomorphisms Ω_+, Ω_-, $\Omega_+^{g_2^{-1}}$ and $\Omega_-^{g_1}$ are called *skew-symmetry isomorphisms*.

Let g_1, g_2 be automorphisms of V, W_1, W_2 and W_3 g_1-, g_2- and g_1g_2-twisted V-modules without g_1-, g_2- and g_1g_2-actions and $\mathcal{Y}$ a twisted intertwining operator of type $\binom{W_3}{W_1W_2}$. We define linear maps $A_\pm(\mathcal{Y})\colon W_1 \otimes W_3' \to W_2'\{x\}[\log x]$, $w_1 \otimes w_3' \mapsto A_\pm(\mathcal{Y})(w_1, x)w_3'$ by

$$\langle A_\pm(\mathcal{Y})(w_1, x)w_3', w_2\rangle = \big\langle w_3', \mathcal{Y}\big(\mathrm{e}^{xL(1)}\mathrm{e}^{\pm\pi\mathrm{i}L(0)}\big(x^{-L(0)}\big)^2 w_1, x^{-1}\big)w_2\big\rangle \qquad (2.21)$$

for $w_1 \in W_1$ and $w_2 \in W_2$ and $w_3' \in W_3'$. Similarly to Ω_p for $p \in \mathbb{Z}$, we can also define A_p for $p \in \mathbb{Z}$ by replacing $\pm$ in (2.21) by $(2p+1)$. But we will also not discuss A_p in this paper.

Let $L_{W_1}^s(0)$ be the semisimple part of $L_{W_1}(0)$. From the definition (2.21), for $p \in \mathbb{Z}$, $w_1 \in W_1$, $w_2 \in W_2$, $w_3' \in W_3'$ and $z \in \mathbb{C}^\times$, we have

$$\begin{aligned}
&\big\langle A_\pm(\mathcal{Y})^p(w_1, z)w_3', w_2\big\rangle \\
&= \big\langle A_\pm(\mathcal{Y})^p(w_1, x)w_3', w_2\big\rangle\big|_{x^n = \mathrm{e}^{nl_p(z)}, \log x = l_p(z)} \\
&= \big\langle w_3', \mathcal{Y}\big(\mathrm{e}^{xL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\big(x^{-L_{W_1}(0)}\big)^2 w_1, x^{-1}\big)w_2\big\rangle\big|_{x^n = \mathrm{e}^{nl_p(z)}, \log x = l_p(z)} \\
&= \big\langle w_3', \mathcal{Y}\big(\mathrm{e}^{xL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\big(x^{-L_{W_1}^s(0)}\big)^2 \cdot \\
&\quad \times \big(\mathrm{e}^{-(L_{W_1}(0) - L_{W_1}^s(0))\log x}\big)^2 w_1, x^{-1}\big)w_2\big\rangle\big|_{x^n = \mathrm{e}^{nl_p(z)}, \log x = l_p(z)} \\
&= \big\langle w_3', \mathcal{Y}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\big(\mathrm{e}^{-l_p(z)L_{W_1}^s(0)}\big)^2 \cdot \\
&\quad \times \big(\mathrm{e}^{-(L_{W_1}(0) - L_{W_1}^s(0))l_p(z)}\big)^2 w_1, y\big)w_2\big\rangle\big|_{y^n = \mathrm{e}^{-nl_p(z)}, \log y = -l_p(z)} \\
&= \big\langle w_3', \mathcal{Y}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2l_p(z)L_{W_1}(0)}w_1, y\big)w_2\big\rangle\big|_{y^n = \mathrm{e}^{-nl_p(z)}, \log y = -l_p(z)}.
\end{aligned} \tag{2.22}$$

When $\arg z = 0$, $\arg z^{-1} = \arg z = 0$ and $-l_p(z) = l_{-p}(z^{-1})$. When $\arg z \neq 0$, $\arg z^{-1} = -\arg z + 2\pi$ and $-l_p(z) = l_{-p-1}(z^{-1})$. Hence when $\arg z = 0$, the right-hand side of (2.22) is equal to

$$\begin{aligned}
&\big\langle w_3', \mathcal{Y}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p}(z^{-1})L_{W_1}(0)}w_1, y\big)w_2\big\rangle\big|_{y^n = \mathrm{e}^{nl_{-p}(z^{-1})}, \log y = l_{-p}(z^{-1})} \\
&= \big\langle w_3', \mathcal{Y}^{-p}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p}(z^{-1})L_{W_1}(0)}w_1, z^{-1}\big)w_2\big\rangle
\end{aligned} \tag{2.23}$$

and when $\arg z \neq 0$, it is equal to

$$\begin{aligned}
&\big\langle w_3', \mathcal{Y}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p-1}(z^{-1})L_{W_1}(0)}w_1, y\big)w_2\big\rangle\big|_{y^n = \mathrm{e}^{nl_{-p-1}(z^{-1})}, \log y = l_{-p-1}(z^{-1})} \\
&= \big\langle w_3', \mathcal{Y}^{-p-1}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p-1}(z^{-1})L_{W_1}(0)}w_1, z^{-1}\big)w_2\big\rangle.
\end{aligned} \tag{2.24}$$

From (2.22)–(2.24), for $w_1 \in W_1$, $w_2 \in W_2$, $w_3' \in W_3'$ and $z \in \mathbb{C}^\times$, we have

$$\big\langle A_\pm(\mathcal{Y})^p(w_1, z)w_3', w_2\big\rangle = \big\langle w_3', \mathcal{Y}^{-p}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p}(z^{-1})L_{W_1}(0)}w_1, z^{-1}\big)w_2\big\rangle \tag{2.25}$$

when $\arg z = 0$ and

$$\begin{aligned}
&\big\langle A_\pm(\mathcal{Y})^p(w_1, z)w_3', w_2\big\rangle \\
&= \big\langle w_3', \mathcal{Y}^{-p-1}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p-1}(z^{-1})L_{W_1}(0)}w_1, z^{-1}\big)w_2\big\rangle
\end{aligned} \tag{2.26}$$

when $\arg z \neq 0$.

Now assume that W_1, W_2, and W_3 are g_1-, g_2- and g_1g_2-twisted V-modules with g_1-, g_2- and g_1g_2-actions, respectively. In the case that W_2' has an action of g_1 (which induces a g_1^{-1}-action on W_2), we have the $g_1g_2^{-1}g_1^{-1}$-twisted V-module $\varphi_{g_1}(W_2')$. In this case, we define

$$A_+^{g_1}(\mathcal{Y})\colon\ W_1 \otimes W_3' \to W_2'\{x\}[\log x], \qquad w_1 \otimes w_3' \mapsto A_+^{g_1}(\mathcal{Y})(w_1, x)w_3'$$

by

$$\big\langle A_+^{g_1}(\mathcal{Y})(w_1, x)w_3', w_2\big\rangle = \big\langle w_3', \mathcal{Y}\big(\mathrm{e}^{xL(1)}\mathrm{e}^{\pi\mathrm{i}L(0)}\big(x^{-L(0)}\big)^2 w_1, x^{-1}\big)g_1^{-1}w_2\big\rangle$$

for $w_1 \in W_1$, $w_2 \in W_2$ and $w_3' \in W_3'$. In the case that W_3' has an action of g_1^{-1} (which induces a g_1-action on W_3), we have the $g_1^{-1}g_2^{-1}$-twisted V-module

$\varphi_{g_1^{-1}}(W_3')$. In this case, we define

$$A_-^{g_1^{-1}}(\mathcal{Y})\colon\ W_1\otimes W_3'\to W_2'\{x\}[\log x],\qquad w_1\otimes w_3'\mapsto A_-^{g_1^{-1}}(\mathcal{Y})(w_1,x)w_3'$$

by

$$\big\langle A_-^{g_1^{-1}}(\mathcal{Y})(w_1,x)w_3',w_2\big\rangle=\big\langle g_1w_3',\mathcal{Y}\big(\mathrm{e}^{xL(1)}\mathrm{e}^{-\pi\mathrm{i}L(0)}\big(x^{-L(0)}\big)^2w_1,x^{-1}\big)w_2\big\rangle$$

for $w_1\in W_1$, $w_2\in W_2$ and $w_3'\in W_3'$.

For $w_1\in W_1$, $w_2\in W_2$, $w_3'\in W_3'$ and $z\in\mathbb{C}^\times$, we have from (2.25) and (2.26)

$$\begin{aligned}&\big\langle A_+^{g_1}(\mathcal{Y})^p(w_1,z)w_3',w_2\big\rangle\\&=\big\langle w_3',\mathcal{Y}^{-p}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p}(z^{-1})L_{W_1}(0)}w_1,z^{-1}\big)g_1^{-1}w_2\big\rangle\end{aligned}\tag{2.27}$$

and

$$\begin{aligned}&\big\langle A_-^{g_1^{-1}}(\mathcal{Y})^p(w_1,z)w_3',w_2\big\rangle\\&=\big\langle g_1w_3',\mathcal{Y}^{-p}(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{-\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p}(z^{-1})L_{W_1}(0)}w_1,z^{-1})w_2\big\rangle\end{aligned}\tag{2.28}$$

when $\arg z=0$ and

$$\begin{aligned}&\big\langle A_+^{g_1}(\mathcal{Y})^p(w_1,z)w_3',w_2\big\rangle\\&=\big\langle w_3',\mathcal{Y}^{-p-1}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p-1}(z^{-1})L_{W_1}(0)}w_1,z^{-1}\big)g_1^{-1}w_2\big\rangle\end{aligned}\tag{2.29}$$

and

$$\begin{aligned}&\big\langle A_-^{g_1^{-1}}(\mathcal{Y})^p(w_1,z)w_3',w_2\big\rangle\\&=\big\langle g_1w_3',\mathcal{Y}^{-p-1}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{-\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p-1}(z^{-1})L_{W_1}(0)}w_1,z^{-1}\big)w_2\big\rangle\end{aligned}\tag{2.30}$$

when $\arg z\neq 0$.

Let $\big(W,Y_W^g\big)$ be a g-twisted V-module. When $W_1=V$, $W_2=W_3=W$ and $\mathcal{Y}=Y_W^g$, by definition, $A_+\big(Y_W^g\big)=A_-\big(Y_W^g\big)=A_+^{1_V}\big(Y_W^g\big)=A_-^{1_V}\big(Y_W^g\big)=\big(Y_W^g\big)'$ (see Section 3).

Theorem 2.3. *The linear maps $A_+(\mathcal{Y})$, $A_-(\mathcal{Y})$, $A_+^{g_1}(\mathcal{Y})$ and $A_-^{g_1^{-1}}(\mathcal{Y})$ are twisted intertwining operators of types*

$$\binom{\phi_{g_1}(W_2')}{W_1W_3'},\quad\binom{W_2'}{W_1\phi_{g_1^{-1}}(W_3')},\quad\binom{\varphi_{g_1}(W_2')}{W_1W_3'}\quad\textit{and}\quad\binom{W_2'}{W_1\varphi_{g_1^{-1}}(W_3')},$$

respectively.

Proof. As in the proof of Theorem 2.1, we need only prove the results for $A_+(\mathcal{Y})$ and $A_-(\mathcal{Y})$. The proofs for $A_+^{g_1}(\mathcal{Y})$ and $A_-^{g_1^{-1}}(\mathcal{Y})$ are reduced to the proofs for $A_+(\mathcal{Y})$ and $A_-(\mathcal{Y})$ using the definitions of φ_{g_1} and $\varphi_{g_1^{-1}}$ and (2.27)–(2.30).

The proof for $A_+(\mathcal{Y})$ and $A_-(\mathcal{Y})$ is essentially the same as the proof of [17, Theorem 6.1]. But the proof here is much more complicated because the correlation functions involved are not of the explicit form as in [17]. As in [17], we prove only the duality property since other properties are essentially the same as in the untwisted case.

We first give the multivalued analytic functions with preferred branches in the duality property. We shall denote these multivalued analytic functions for $A_+(\mathcal{Y})$ and $A_-(\mathcal{Y})$ by $h_+(z_1,z_2;u,w_1,w_3',w_2)$ and $h_-(z_1,z_2;u,w_1,w_3',w_2)$, respectively. Let $f(z_1,z_2;u,w_1,w_3',w_2)$ be the multivalued analytic function with the preferred

branch $f^e(z_1, z_2; u, w_1, w_2, w_3')$ in the duality property for the twisted intertwining operator $\mathcal{Y}$. Define

$$h_\pm(z_1, z_2; u, w_1, w_3', w_2)$$

(2.31)
$$= f\big(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1 L_V(1)}\big(-z_1^2\big)^{-L_V(0)}u, \mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\big(z_2^2\big)^{-L_{W_1}(0)}w_1, w_2, w_3'\big)$$

and choose the preferred branch $h_\pm^e(z_1, z_2; u, w_1, w_3', w_2)$ of $h_\pm(z_1, z_2; u, w_1, w_3', w_2)$ as follows: On the subregion $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_1, \arg z_2 \neq 0$ of M_0^2, let

$$h_+^e(z_1, z_2; u, w_1, w_3', w_2) = f^{b_{13}^{-1}b_{23}^{-1}}\big(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z^{-1}L_V(1)}\big(-z_1^2\big)^{-L_V(0)}u,$$
$$\mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\big). \tag{2.32}$$

and

$$h_-^e(z_1, z_2; u, w_1, w_3', w_2) = f^{b_{13}^{-1}b_{23}^{-1}b_{12}^{-1}}\big(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z^{-1}L_V(1)}\big(-z_1^2\big)^{-L_V(0)}u,$$
$$\mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{-\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\big).$$

For general $(z_1, z_2) \in M_0^2$, we define $h_\pm^e(z_1, z_2; u, w_1, w_3', w_2)$ to be the unique analytic extensions on M_0^2.

Let $u \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $w_3' \in W_3'$. We consider $z_1, z_2 \in \mathbb{C}$ satisfying $\big|z_2^{-1}\big| > \big|z_1^{-1}\big| > 0$ (or equivalently $|z_1| > |z_2| > 0$) and $\arg z_1, \arg z_2 \neq 0$. Since $\big|z_2^{-1}\big| > \big|z_1^{-1}\big| > 0$ and $\arg z_1, \arg z_2 \neq 0$, from (2.26), $\big(Y_{W_2}^{g_2}\big)' = A_+\big(Y_{W_2}^{g_2}\big)$ and the duality property for $\mathcal{Y}$, we know that

$$\big\langle w_3', \mathcal{Y}\big(\mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, z_2^{-1}\big)$$
$$\times Y_{W_2}^{g_2}\big(\mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}g_1^{-1}u, z_1^{-1}\big)w_2\big\rangle \tag{2.33}$$

is absolutely convergent and if in addition, $-\frac{3\pi}{2} < \arg\big(z_1^{-1} - z_2^{-1}\big) - \arg z_2^{-1} < -\frac{\pi}{2}$, its sum is equal to

$$f^e\big(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}g_1^{-1}u, \mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\big)$$
$$= f^e\big(z_1^{-1}, z_2^{-1}; g_1^{-1}\mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}u, \mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\big).$$

We know that

$$\big\langle \phi_{g_1}\big(\big(Y_{W_2}^{g_2}\big)'\big)(u, z_1)A_+(\mathcal{Y})(w_1, z_2)w_3', w_2\big\rangle$$
$$= \big\langle \big(Y_{W_2}^{g_2}\big)'\big(g_1^{-1}u, z_1\big)A_+(\mathcal{Y})(w_1, z_2)w_3', w_2\big\rangle$$
$$= \big\langle w_3', \mathcal{Y}^{-1}\big(\mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, z_2^{-1}\big)$$
$$\times \big(Y_{W_2}^{g_2}\big)^{-1}\big(\mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}g_1^{-1}u, z_1^{-1}\big)w_2\big\rangle \tag{2.34}$$

can be obtained using the multivalued analytic function (2.31) on the region $\big|z_2^{-1}\big| > \big|z_1^{-1}\big| > 0$ starting from the value given by (2.33) by letting z_1^{-1} go around 0 clockwise once $\big($corresponding to $b_{13}^{-1}\big)$, then letting z_2^{-1} go around 0 clockwise once $\big($corresponding to $b_{23}^{-1}b_{12}^{-1}\big)$. Then (2.34) also converges absolutely on the region $\big|z_2^{-1}\big| > \big|z_1^{-1}\big| > 0$ and if in addition, $-\frac{3\pi}{2} < \arg\big(z_1^{-1} - z_2^{-1}\big) - \arg z_2^{-1} < -\frac{\pi}{2}$, its sum is equal to

$$f^{b_{13}^{-1}b_{23}^{-1}b_{12}^{-1}}\big(z_1^{-1}, z_2^{-1}; g_1^{-1}\mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}u, \mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\big)$$
$$= f^{b_{12}^{-1}b_{13}^{-1}b_{23}^{-1}}\big(z_1^{-1}, z_2^{-1}; g_1^{-1}\mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}u,$$
$$\mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\big). \tag{2.35}$$

Using the equivariance property for W_1 and the convergence of (1.5) to $f^e(z_1, z_2; u, w_1, w_2, w_3')$ on the region $|z_2| > |z_1 - z_2| > 0$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$, we have

$$f^e\big(z_1, z_2; g_1^{-1}u, w_1, w_2, w_3'\big) = f^{b_{12}}(z_1, z_2; u, w_1, w_2, w_3').$$

Applying $b \in \mathrm{PB}_3$ to both sides of this equality, we obtain

$$f^b\big(z_1, z_2; g_1^{-1}u, w_1, w_2, w_3'\big) = f^{b_{12}b}(z_1, z_2; u, w_1, w_2, w_3') \tag{2.36}$$

for $b \in \mathrm{PB}_3$. By (2.36) with $b = b_{12}^{-1}b_{13}^{-1}b_{23}^{-1}$, we see that (2.35) is equal to

$$f^{b_{13}^{-1}b_{23}^{-1}}\big(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}u, \mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\big),$$

which by definition, is equal to $h_+^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_1, \arg z_2 \neq 0$.

What we need to prove is that when $|z_1| > |z_2| > 0$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$, the sum of the left-hand side of (2.34) is equal to $h_+^e(z_1, z_2; u, w_2, w_1, w_3')$. We choose $z_1, z_2 \in -\mathbb{R}_+$ such that $|z_1| = -z_1 > -z_2 = |z_2| > 0$. Then $\big|z_2^{-1}\big| = -z_2^{-1} > -z_1^{-1} = \big|z_1^{-1}\big| > 0$. In this case, $\arg(z_1 - z_2) = \pi$, $\arg\big(z_1^{-1} - z_2^{-1}\big) = 0$, and $\arg z_1 = \arg z_2 = \arg z_1^{-1} = \arg z_2^{-1} = \pi$. Hence $|\arg(z_1 - z_2) - \arg z_1| = 0 < \frac{\pi}{2}$ and $-\frac{3\pi}{2} < 0 - \pi = \arg\big(z_1^{-1} - z_2^{-1}\big) - \arg z_2^{-1} < -\frac{\pi}{2}$. From the discussion above, for such z_1 and z_2, the sum of the left-hand side of (2.34) is equal to $h_+^e(z_1, z_2; u, w_2, w_1, w_3')$. On the other hand, such z_1 and z_2 satisfy $|z_1| > |z_2| > 0$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$. Since both $h_+^e(z_1, z_2; u, w_2, w_1, w_3')$ and the sum of the left-hand side of (2.34) are analytic extensions of their restrictions on the subset given by (z_1, z_2) for $z_1, z_2 \in -\mathbb{R}_+$ such that $|z_1| = -z_1 > -z_2 = |z_2| > 0$, the sum of the left-hand side of (2.34) is equal to $h_+^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_1| > |z_2| > 0$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$.

Generalizing the convergence and analytic extension of (2.34) above, we can prove that

$$\big\langle \phi_{g_1}\big(\big(Y_{W_2}^{g_2}\big)'\big)(u_1, z_1) \cdots \phi_{g_1}\big(\big(Y_{W_2}^{g_2}\big)'\big)(u_{k-1}, z_{k-1}) A_+(\mathcal{Y})(w_1, z_k)w_3', w_2\big\rangle \tag{2.37}$$

is absolutely convergent on the region $|z_1| > \cdots > |z_k| > 0$ and its sum can be maximally extended to a multivalued analytic function on the region M^k for $k \in \mathbb{Z}_+ + 3$, $u_1, \ldots, u_{k-1} \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $w_3' \in W_3'$. In fact, the same calculations as in (2.34) shows that (2.37) is equal to

$$\begin{aligned}
&\big\langle w_3', \mathcal{Y}^{-1}\big(\mathrm{e}^{z_k L_{W_1}(1)}\mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_k)L_{W_1}(0)}w_1, z_k^{-1}\big)\\
&\quad \times \big(Y_{W_2}^{g_2}\big)^{-1}\big(\mathrm{e}^{z_{k-1}L_V(1)}\big(-z_{k-1}^{-2}\big)^{L_V(0)}g_1^{-1}u_{k-1}, z_{k-1}^{-1}\big)\cdots\\
&\quad \times \big(Y_{W_2}^{g_2}\big)^{-1}\big(\mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}g_1^{-1}u_1, z_1^{-1}\big)w_2\big\rangle
\end{aligned} \tag{2.38}$$

when $\arg z_1, \ldots, \arg z_k \neq 0$. From the duality properties in Definitions 1.1 and 1.8, we know that

$$\begin{aligned}
&\big\langle w_3', \big(Y_{W_2}^{g_2}\big)^{-1}\big(\mathrm{e}^{z_{k-1}L_V(1)}\big(-z_{k-1}^{-2}\big)^{L_V(0)}g_1^{-1}u_{k-1}, z_{k-1}^{-1}\big)\cdots\\
&\quad \times \big(Y_{W_2}^{g_2}\big)^{-1}\big(\mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}g_1^{-1}u_1, z_1^{-1}\big)\\
&\quad \times \mathcal{Y}^{-1}\big(\mathrm{e}^{z_k L_{W_1}(1)}\mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_k)L_{W_1}(0)}w_1, z_k^{-1}\big)w_2\big\rangle
\end{aligned} \tag{2.39}$$

is absolutely convergent on the region $\big|z_{k-1}^{-1}\big| > \cdots > \big|z_1^{-1}\big| > \big|z_k^{-1}\big| > 0$. Then for $M_{ij} \in \mathbb{Z}_+$ $(1 \leq j < i \leq k-1)$ such that

$$x^{M_{ij}}Y_V\big(\mathrm{e}^{z_i L_V(1)}\big(-z_i^{-2}\big)^{L_V(0)}g_1^{-1}u_i, x\big)\mathrm{e}^{z_j L_V(1)}\big(-z_j^{-2}\big)^{L_V(0)}g_1^{-1}u_j \in V[[x]],$$

we have

$$\prod_{1\le j<i\le k-1}\left(z_i^{-1}-z_j^{-1}\right)^{M_{ij}}\langle w_3',\left(Y_{W_2}^{g_2}\right)^{-1}\left(\mathrm{e}^{z_{k-1}L_V(1)}\left(-z_{k-1}^{-2}\right)^{L_V(0)}g_1^{-1}u_{k-1},z_{k-1}^{-1}\right)\cdots$$
$$\times\left(Y_{W_2}^{g_2}\right)^{-1}\left(\mathrm{e}^{z_1L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}g_1^{-1}u_1,z_1^{-1}\right)$$
$$\times\mathcal{Y}^{-1}\left(\mathrm{e}^{z_kL_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_k)L_{W_1}(0)}w_1,z_k^{-1}\right)w_2\rangle \tag{2.40}$$

is absolutely convergent on the region $\left|z_i^{-1}\right|>\left|z_k^{-1}\right|>0$ for $i=1,\dots,k-1$, for $i\ne j$ because $z_i^{-1}=z_j^{-1}$ for $i\ne j$ are poles of (2.39) and are canceled by $\prod_{1\le j<i\le k-1}(z_i^{-1}-z_j^{-1})^{M_{ij}}$ in (2.40). Moreover, the sum of (2.40) can be analytically extended to a maximal multivalued analytic function on $\left\{(z_1,\dots,z_k)\in\mathbb{C}^k\mid z_i\ne z_k,\ i=1,\dots,k-1\right\}$. Using the duality property in Definition 1.8 repeatedly, we see that

$$\prod_{1\le i<j\le k-1}\left(z_i^{-1}-z_j^{-1}\right)^{M_{ij}}\langle w_3',\mathcal{Y}^{-1}\left(\mathrm{e}^{z_kL_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_k)L_{W_1}(0)}w_1,z_k^{-1}\right)$$
$$\times\left(Y_{W_2}^{g_2}\right)^{-1}\left(\mathrm{e}^{z_{k-1}L_V(1)}\left(-z_{k-1}^{-2}\right)^{L_V(0)}g_1^{-1}u_{k-1},z_{k-1}^{-1}\right)\cdots$$
$$\times\left(Y_{W_2}^{g_2}\right)^{-1}\left(\mathrm{e}^{z_1L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}g_1^{-1}u_1,z_1^{-1}\right)w_2\rangle$$

is absolutely convergent on the region $\left|z_i^{-1}\right|<\left|z_k^{-1}\right|$ for $i=1,\dots,k-1$, $z_i^{-1}\ne z_j^{-1}$ for $i\ne j$, and its sum can be analytically extended to a maximal multivalued analytic function on $\left\{(z_1,\dots,z_k)\in\mathbb{C}^k\mid z_i\ne z_k,\ i=1,\dots,k-1\right\}$. Thus (2.38) is absolutely convergent on the region $|z_1|>\cdots>|z_k|>0$ and consequently (2.37) is absolutely convergent on the region $|z_1|>\cdots>|z_k|>0$, $\arg z_1,\dots,\arg z_k\ne 0$. But the absolute convergence of (2.37) on the region $|z_1|>\cdots>|z_k|>0$, $\arg z_1,\dots,\arg z_k\ne 0$ implies the absolute convergence of (2.37) on the region $|z_1|>\cdots>|z_k|>0$. Moreover, its sum has a maximal analytic extension on the region M^k and from the discussion above, we have

$$h(z_1,\dots,z_k)=\sum_{j_1,\dots,j_{k-1}=J'}^{J}\sum_{i=1}^{I}\sum_{m=0}^{M}z_1^{j_1}\cdots z_{k-1}^{j_{k-1}}z_k^{r_i}(\log z_k)^m f_{j_1,\dots,j_{k-1};i,m}\left(z_1^{-1},\dots,,z_k^{-1}\right),$$

where $h(z_1,\dots,z_k)$ is the maximal analytic extension of the sum of (2.37) on the region M^k, $J',J\in\mathbb{Z}$, $r_1,\dots,r_I\in\mathbb{C}$, $M\in\mathbb{N}$ such that

$$\mathrm{e}^{z_lL_V(1)}\left(-z_l^{-2}\right)^{L_V(0)}g_1^{-1}u_l=\sum_{j_l=J'}^{J}z_l^{j_l}u_l^{j_l},\qquad l=1,\dots,k-1,$$

$$\mathrm{e}^{z_kL_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_k)L_{W_1}(0)}w_1=\sum_{i=1}^{I}\sum_{m=0}^{M}z_k^{r_i}(\log z_k)^m w_1^{i,m}$$

for $u_l^{j_l}\in V$ and $w_1^{i,m}\in W_1$, and $f_{j_1,\dots,j_{k-1};i,m}(z_1,\dots,z_k)$ for $J',J\in\mathbb{Z}$, $r_1,\dots,r_I\in\mathbb{C}$, $M\in\mathbb{N}$ are the maximal analytic extensions of the sums of $\langle w_3',Y_{W_2}^{g_2}\left(u_1^{j_1},z_1\right)\cdots Y_{W_2}^{g_2}\left(u_{k-1}^{j_{k-1}},z_{k-1}\right)\mathcal{Y}\left(w_1^{i,m},z_k\right)w_2\rangle$. We will show at the end of this proof that a component-isolated singularity of the form $(z_1,\dots,z_{k-1})-\beta=\delta$ for $\beta\in\mathbb{C}^{k-1}$ and $\delta\in\{0,\infty\}^k$ of any value at $\zeta=z_k$ of $\prod_{1\le i<\le j\le k-1}(z_i-z_j)^{M_{ij}}h(z_1,\dots,z_{k-1},\zeta)$ is regular.

Next we consider the product of $A_+(\mathcal{Y})$ and the twisted vertex operator $\left(Y_{W_3}^{g_2}\right)'$. Let u, w_1, w_2 and w_3' be the same as above. When $\left|z_1^{-1}\right|>\left|z_2^{-1}\right|>0$ (or equivalently $|z_2|>|z_1|>0$),

$$\langle w_3',Y_{W_2}^{g_2}\left(\mathrm{e}^{z_1L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}u,z_1^{-1}\right)$$
$$\times\mathcal{Y}\left(\mathrm{e}^{z_2L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1,z_2^{-1}\right)w_2\rangle \tag{2.41}$$

converges absolutely and if in addition, $\left|\arg\left(z_1^{-1}-z_2^{-1}\right)-\arg z_1^{-1}\right|<\frac{\pi}{2}$, its sum is equal to

$$f^e\left(z_1^{-1},z_2^{-1};\mathrm{e}^{z_1L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}u,\mathrm{e}^{z_2L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1,w_2,w_3'\right).$$

We know that when $\arg z_1,\arg z_2\neq 0$,

$$\begin{aligned}&\left\langle A_+(\mathcal{Y})(w_1,z_2)\left(Y_{W_2}^{g_2}\right)'(u,z_1)w_3',w_2\right\rangle\\&=\left\langle w_3',(Y_{W_2}^{g_2})^{-1}\left(\mathrm{e}^{z_1L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}u,z_1^{-1}\right)\right.\\&\left.\times\,\mathcal{Y}^{-1}\left(\mathrm{e}^{z_2L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1,z_2^{-1}\right)w_2\right\rangle\end{aligned}\tag{2.42}$$

can be obtained using the multivalued analytic function (2.31) on the region $\left|z_1^{-1}\right|>\left|z_2^{-1}\right|>0$ starting from the value given by (2.41) by letting z_2^{-1} go around 0 clockwise once (corresponding to b_{23}^{-1}), then letting z_1^{-1} go around 0 clockwise once (corresponding to $b_{12}^{-1}b_{13}^{-1}$). Then (2.42) also converges absolutely and if in addition, $\left|\arg\left(z_1^{-1}-z_2^{-1}\right)-\arg z_1^{-1}\right|<\frac{\pi}{2}$, its sum is equal to

$$\begin{aligned}&f^{b_{23}^{-1}b_{12}^{-1}b_{13}^{-1}}\left(z_1^{-1},z_2^{-1};\mathrm{e}^{z_1L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}u,\mathrm{e}^{z_2L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1,w_2,w_3'\right)\\&\quad=f^{b_{13}^{-1}b_{23}^{-1}b_{12}^{-1}}\left(z_1^{-1},z_2^{-1};\mathrm{e}^{z_1L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}u,\right.\\&\left.\quad\mathrm{e}^{z_2L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1,w_2,w_3'\right).\end{aligned}\tag{2.43}$$

We now show that the sum of (2.42) is equal to (2.43). We choose $z_1,z_2\in-\mathbb{R}_+$ such that $|z_2|=-z_2>-z_1=|z_1|>0$. Then $\left|z_1^{-1}\right|=-z_1^{-1}>-z_2^{-1}=\left|z_2^{-1}\right|>0$. In this case, $\arg(z_1-z_2)=0$, $\arg\left(z_1^{-1}-z_2^{-1}\right)=\pi$, and $\arg z_1=\arg z_2=\arg z_1^{-1}=\arg z_2^{-1}=\pi$. Hence $-\frac{3\pi}{2}<0-\pi=\arg(z_1-z_2)-\arg z_2<-\frac{\pi}{2}$ and $\left|\arg\left(z_1^{-1}-z_2^{-1}\right)-\arg z_1^{-1}\right|=0<\frac{\pi}{2}$. From the discussion above, for such z_1 and z_2, the sum of (2.42) is equal to (2.43). Using analytic extension, we see that when $|z_2|>|z_1|>0$ and $-\frac{3\pi}{2}<\arg(z_1-z_2)-\arg z_2<-\frac{\pi}{2}$, the sum of (2.42) is equal to (2.43).

On the region $|z_2|>|z_1|>0$, (2.43) is in fact equal to $h_+^e(z_1,z_2;u,w_2,w_1,w_3')$. This can be proved as follows: On the intersection of the region $|z_2|>|z_1|>0$ and M_0^2, the branch $h_+^e(z_1,z_2;u,w_2,w_1,w_3')$ is obtained by analytically extending (2.32) defined on the intersection of the region $|z_1|>|z_2|>0$, $|\arg(z_1-z_2)-\arg z_1|<\frac{\pi}{2}$, $\arg z_1,\arg z_2\neq 0$ and M_0^2. We need to find what is this analytic extension. Let $\xi,\zeta\in-\mathbb{R}_+$ satisfying $\xi<\zeta<0$. Then we have $|\xi|>|\zeta|>0$, $\arg(\xi-\zeta)=\arg\xi=\arg\zeta=\pi$ so that $|\arg(\xi-\zeta)-\arg\xi|=0<\frac{\pi}{2}$. By definition,

$$\begin{aligned}h_+^e(\xi,\zeta;u,w_2,w_1,w_3')=f^{b_{13}^{-1}b_{23}^{-1}}&\left(\xi^{-1},\zeta^{-1};\mathrm{e}^{\xi^{-1}L_V(1)}\left(-\xi^2\right)^{-L_V(0)}u,\right.\\&\left.\mathrm{e}^{\zeta L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log\zeta)L_{W_1}(0)}w_1,w_2,w_3'\right).\end{aligned}\tag{2.44}$$

Let $\gamma=(\gamma_1,\gamma_2)$ be the path from (ξ,ζ) to (ζ,ξ) given by the upper half circle γ_1 centered at $\frac{\xi+\zeta}{2}$ with radius $\frac{-\xi+\zeta}{2}$ from ξ to ζ and the lower half circle γ_2 centered at $\frac{\xi+\zeta}{2}$ with radius $\frac{-\xi+\zeta}{2}$ from ζ to ξ. Since by definition, (ξ,ζ), (ζ,ξ) are in M_0^2 and the path γ does not cross the boundary of M_0^2, γ is a continuous path in M_0^2. So by definition, the value at $(z_1,z_2)=(\zeta,\xi)$ of the multivalued analytic function $h_+(z_1,z_2;u,w_2,w_1,w_3')$ obtained by starting with the value $h_+^e(\xi,\zeta;u,w_2,w_1,w_3')$ and analytically extending along γ is $h_+^e(\zeta,\xi;u,w_2,w_1,w_3')$. On the other hand, we need to find out what the right-hand side of (2.44) is analytically extended to along the same path γ. We have the path $\gamma'=\left(\gamma_1^{-1},\gamma_2^{-1}\right)$ from $\left(\xi^{-1},\zeta^{-1}\right)$ to $\left(\zeta^{-1},\xi^{-1}\right)$, where γ_1^{-1} and γ_2^{-1} are the paths on the complex plane from ξ^{-1} to ζ^{-1} and from ζ^{-1} to ξ^{-1}, respectively, given by $\gamma_1^{-1}(t)=\gamma_1(t)^{-1}$ and $\gamma_2^{-1}(t)=\gamma_1(t)^{-1}$. But γ' is not a continuous path in M_0^2 because when t goes from 0 to 1, $\left(\gamma_1^{-1}(t),\gamma_2^{-1}(t)\right)$ goes from $\left(\xi^{-1},\zeta^{-1}\right)$ to $\left(\zeta^{-1},\xi^{-1}\right)$ along the path γ' but $\gamma_1^{-1}(t)-\gamma_2^{-1}(t)$ when t

goes from 0 to 1 crosses the positive real line clockwise. Crossing the positive real line clockwise corresponds to changing the branch by an action of b_{12}^{-1}. So we see that the value at $\left(z_1^{-1}, z_2^{-1}\right) = \left(\zeta^{-1}, \xi^{-1}\right)$ of the multivalued analytic function

(2.45)
$$f\left(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1 L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)} u, \mathrm{e}^{z_2 L_{W_1}(1)} \mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log z_2) L_{W_1}(0)} w_1, w_2, w_3'\right)$$

of z_1^{-1} and z_2^{-1} obtained by starting with the value

$$f^{b_{13}^{-1} b_{23}^{-1}}\left(\xi^{-1}, \zeta^{-1}; \mathrm{e}^{\xi^{-1} L_V(1)}\left(-\xi^2\right)^{-L_V(0)} u, \mathrm{e}^{\zeta L_{W_1}(1)} \mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log \zeta) L_{W_1}(0)} w_1, w_2, w_3'\right)$$

and going along the path γ' is

$$f^{b_{13}^{-1} b_{23}^{-1} b_{12}^{-1}}\left(\zeta^{-1}, \xi^{-1}; \mathrm{e}^{\zeta^{-1} L_V(1)}\left(-\zeta^2\right)^{-L_V(0)} u, \mathrm{e}^{\xi L_{W_1}(1)} \mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log \xi) L_{W_1}(0)} w_1, w_2, w_3'\right).$$

Since we have (2.44) and the path γ is mapped to γ' under the map $(z_1, z_2) \mapsto \left(z_1^{-1}, z_2^{-1}\right)$, the value $h_+^e(\zeta, \xi; u, w_2, w_1, w_3')$ of the multivalued analytic function $h_+(z_1, z_2; u, w_2, w_1, w_3')$ of z_1 and z_2 at the end point $(z_1, z_2) = (\zeta, \xi)$ of γ and the value of the multivalued analytic function (2.45) of z_1^{-1} and z_2^{-1} at the end point $\left(z_1^{-1}, z_2^{-1}\right) = \left(\zeta^{-1}, \xi^{-1}\right)$ of γ' must be the same, that is,

$$\begin{aligned} h_+^e(\zeta, \xi; u, w_2, w_1, w_3') = f^{b_{13}^{-1} b_{23}^{-1} b_{12}^{-1}}\left(\zeta^{-1}, \xi^{-1}; \mathrm{e}^{\zeta^{-1} L_V(1)}\left(-\zeta^2\right)^{-L_V(0)} u,\right. \\ \left. \mathrm{e}^{\xi L_{W_1}(1)} \mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log \xi) L_{W_1}(0)} w_1, w_2, w_3'\right). \end{aligned}$$

Since a single-valued branch on a subregion of M_0^2 of a multivalued analytic function on M^2 is determined uniquely by the value at one point in this subregion of M_0^2, we see that on the intersection of the region $|z_2| > |z_1| > 0$ and M_0^2,

$$\begin{aligned} h_+^e(z_1, z_2; u, w_2, w_1, w_3') = f^{b_{13}^{-1} b_{23}^{-1} b_{12}^{-1}}\left(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1^{-1} L_V(1)}\left(-z_1^2\right)^{-L_V(0)} u,\right. \\ \left. \mathrm{e}^{z_2 L_{W_1}(1)} \mathrm{e}^{\pm \pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log z_2) L_{W_1}(0)} w_1, w_2, w_3'\right). \end{aligned} \tag{2.46}$$

Since on the region $|z_2| > |z_1| > 0$ and $\left|\arg\left(z_1^{-1} - z_2^{-1}\right) - \arg z_1^{-1}\right| < \frac{\pi}{2}$, the sum of (2.42) is equal to (2.43) (that is, the right-hand side of (2.46)), the sum of (2.42) is indeed equal to $h_+^e(z_1, z_2; u, w_2, w_1, w_3')$ on the same region.

We can prove the absolute convergence of

$$\left\langle \left(Y_{W_2}^{g_2}\right)'(u, z_1) A_-(\mathcal{Y})(w_1, z_2) w_3', w_2\right\rangle, \qquad \left\langle A_-(\mathcal{Y})(w_1, z_2) \phi_{g_1^{-1}}\left(\left(Y_{W_3}^{g_2}\right)'\right)(u, z_1) w_3', w_2\right\rangle$$

in the corresponding regions to $h_-^e(z_1, z_2; u, w_2, w_1, w_3')$ similarly by generalizing the proofs in [17] using the same method above for $A_-(\mathcal{Y})$. Here we omit the details.

Now we study the iterate of $A_\pm(\mathcal{Y})$ and the twisted vertex operator $Y_{W_1}^{g_1}$. When $\arg z_2 \neq 0$, from (2.26), we have

$$\begin{aligned} &\left\langle A_\pm(\mathcal{Y})\left(Y_{W_1}^{g_1}(u, z_1 - z_2) w_1, z_2\right) w_3', w_2\right\rangle \\ &= \left\langle w_3', \mathcal{Y}^{-1}\left(\mathrm{e}^{z_2 L_{W_1}(1)} \mathrm{e}^{\pm \pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log z_2) L_{W_1}(0)} Y_{W_1}^{g_1}(u, z_1 - z_2) w_1, z_2^{-1}\right) w_2\right\rangle. \end{aligned} \tag{2.47}$$

As in [17], we have in the region $|z_2| > |z_1 - z_2| > 0$

$$\begin{aligned} &e^{z_2 L_{W_1}(1)} \mathrm{e}^{\pm \pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log z_2) L_{W_1}(0)} Y_{W_1}^{g_1}(u, z_1 - z_2) w_1 \\ &= Y_{W_1}^{g_1}\left(\mathrm{e}^{z_1 L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)} u, \frac{x x_0}{(x_2 + x_0) x_2}\right)\Big|_{\substack{x_0^n = \mathrm{e}^{n \log(z_1 - z_2)}, \log x_0 = \log(z_1 - z_2), x_2^n = \mathrm{e}^{n \log z_2} \\ \log x_2 = \log z_2, x^n = \mathrm{e}^{\pm n \pi \mathrm{i}}, \log x = \pm \pi \mathrm{i}}} \\ &\qquad \times \mathrm{e}^{z_2 L_{W_1}(1)} \mathrm{e}^{\pm \pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log z_2) L_{W_1}(0)} w_1. \end{aligned}$$

As in the proof of the first part of Theorem 2.1, we see that

$$\mathcal{Y}^{-1}\left(Y_{W_1}^{g_1}\left(\mathrm{e}^{z_1L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}u, \frac{xx_0}{(x_2+x_0)x_2}\right)\Big|_{\substack{x_0^n=\mathrm{e}^{n\log(z_1-z_2)},\log x_0=\log(z_1-z_2),x_2^n=\mathrm{e}^{n\log z_2}\\ \log x_2=\log z_2,x^n=\mathrm{e}^{\pm n\pi\mathrm{i}},\log x=\pm\pi\mathrm{i}}}\right.$$
$$\left.\times\mathrm{e}^{z_2L_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, z_2^{-1}\right)$$

is an iterated series obtained by expanding one variable inside the series obtained from the iterate of $\mathcal{Y}^{-1}$ and $Y_{W_1}^{g_1}$. Using the same method as in the proof of the first part of Theorem 2.1 (including the use of Lemma 5.1), we can prove that the corresponding multiple series is absolutely convergent. Here we omit the detailed proof. In particular, we can calculate the sum of the series using any of the iterated sums associated with the multisum. Then the same calculations as those in the proof of [17, Theorem 6.1] shows that when $|z_1|, |z_2| > |z_1 - z_2| > 0$ and $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$, the right-hand side of (2.47) is equal to

$$\begin{aligned}&\left\langle w_3', \mathcal{Y}^{-1}\left(\left(Y_{W_1}^{g_1}\right)^{m+\frac{1\pm1}{2}}\left(\mathrm{e}^{z_1L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}u, z_1^{-1}-z_2^{-1}\right)\right.\right.\\ &\qquad\left.\left.\times\mathrm{e}^{z_2L_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, z_2^{-1}\right)w_2\right\rangle,\end{aligned} \tag{2.48}$$

where $m \in \mathbb{Z}$ is given by

$$\log(z_1 - z_2) - \log z_1 - \log z_2 + \pi\mathrm{i} = l_{m+1}\left(z_1^{-1} - z_2^{-1}\right).$$

We know that

$$\begin{aligned}&\left\langle w_3', \mathcal{Y}\left(Y_{W_1}^{g_1}\left(\mathrm{e}^{z_1L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}u, z_1^{-1}-z_2^{-1}\right)\mathrm{e}^{z_2L_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\right.\right.\\ &\qquad\left.\left.\times\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, z_2^{-1}\right)w_2\right\rangle\end{aligned} \tag{2.49}$$

is absolutely convergent on the region $\left|z_2^{-1}\right| > \left|z_1^{-1} - z_2^{-1}\right| > 0$ and, if in addition $\left|\arg z_1^{-1} - \arg z_2^{-1}\right| < \frac{\pi}{2}$, the sum is equal to

$$f^e\left(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}u, \mathrm{e}^{z_2L_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\right).$$

Since (2.48) can be obtained from (2.49) using the multivalued analytic function (2.31) on the region $\left|z_2^{-1}\right| > \left|z_1^{-1} - z_2^{-1}\right| > 0$ starting from the value given by (2.49) by letting z_2^{-1} go around 0 clockwise once while $z_1^{-1} - z_2^{-1}$ is fixed (corresponding to $b_{13}^{-1}b_{23}^{-1}$ since to keep $z_1^{-1} - z_2^{-1}$ fixed, z_1^{-1} must also go around 0 clockwise once) and then letting z_1^{-1} go around z_2^{-1} counterclockwise $m + \frac{1\pm1}{2}$ times $\left(\text{corresponding to } b_{12}^{m+\frac{1\pm1}{2}}\right)$, we see that (2.48) is absolutely convergent on the region $\left|z_2^{-1}\right| > \left|z_1^{-1} - z_2^{-1}\right| > 0$ and if $\left|\arg z_1^{-1} - \arg z_2^{-1}\right| < \frac{\pi}{2}$, the sum is equal to

$$\begin{aligned}&f^{b_{13}^{-1}b_{23}^{-1}b_{12}^{m+\frac{1\pm1}{2}}}\left(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}u,\right.\\ &\qquad\left.\mathrm{e}^{z_2L_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\right).\end{aligned}$$

In the case of A_+, we consider the subset S_+ of $(z_1, z_2) \in \mathbb{C}^2$ such that $z_1, z_2 \in -\mathbb{R}_+$, $z_1 < z_2$, and $-z_1^{-1} > z_1^{-1} - z_2^{-1} > 0$. This set is certainly nonempty since for $z_1 = -\frac{1}{2}$, $z_2 = -\frac{1}{3}$, we have $z_1 < z_2$ and $-z_1^{-1} = 2 > 1 = z_1^{-1} - z_2^{-1} > 0$. Then for $(z_1, z_2) \in S_+$, we have

$$|z_1| > |z_2| > 0, \qquad \arg z_1 = \arg z_2 = \arg z_1^{-1} = \arg z_2^{-1} = \arg(z_1 - z_2) = \pi,$$
$$\arg\left(z_1^{-1} - z_2^{-1}\right) = 0, \qquad |\arg(z_1 - z_2) - \arg z_1| = 0 < \frac{\pi}{2},$$
$$\left|z_2^{-1}\right| = -z_2^{-1} > -z_1^{-1} > z_1^{-1} - z_2^{-1} = \left|z_1^{-1} - z_2^{-1}\right| > 0, \qquad \left|\arg z_1^{-1} - \arg z_2^{-1}\right| = 0 < \frac{\pi}{2},$$

and

$$\begin{aligned}&\log(z_1-z_2)-\log z_1-\log z_2+\pi\mathrm{i}\\ &=\log|z_1-z_2|+\arg(z_1-z_2)\mathrm{i}-\log|z_1|-\arg z_1\mathrm{i}-\log|z_2|-\arg z_2\mathrm{i}+\pi\mathrm{i}\\ &=\log\left|\frac{z_1-z_2}{z_1z_2}\right|=\log\left|z_1^{-1}-z_2^{-1}\right|+\arg\bigl(z_1^{-1}-z_2^{-1}\bigr)\\ &=\log\bigl(z_1^{-1}-z_2^{-1}\bigr).\end{aligned}$$

In particular, for $(z_1,z_2)\in S_+$, we have $m+1=0$. Then on S_+, (2.48) is absolutely convergent to

$$\begin{aligned}&f^{b_{13}^{-1}b_{23}^{-1}}\bigl(z_1^{-1},z_2^{-1};\mathrm{e}^{z_1L_V(1)}\bigl(-z_1^{-2}\bigr)^{L_V(0)}u,\mathrm{e}^{z_2L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1,w_2,w_3'\bigr)\\ &\qquad=h_+^e(z_1,z_2;u,w_1,w_3',w_2).\end{aligned}$$

By analytic extension, we see that the sum of (2.48) is equal to $h_+^e(z_1,z_2;u,w_1,w_3',w_2)$ on the subregion of M_0^2 given by $\left|z_2^{-1}\right|>\left|z_1^{-1}-z_2^{-1}\right|>0$, $\left|\arg z_1^{-1}-\arg z_2^{-1}\right|<\frac{\pi}{2}$. By analytic extension again, we see that the sum of (2.47) is equal to $h_+^e(z_1,z_2;u,w_1,w_3',w_2)$ on the subregion of M_0^2 given by $|z_2|>|z_1-z_2|>0$, $|\arg z_1-\arg z_2|<\frac{\pi}{2}$.

In the case of A_-, we consider the subset S_- of $(z_1,z_2)\in\mathbb{C}^2$ such that $z_1,z_2\in-\mathbb{R}_+$, $z_1<z_2$, and $-z_1^{-1}>z_1^{-1}-z_2^{-1}>0$. This set is nonempty since for $z_1=-\frac{1}{2}$, $z_2=-\frac{1}{3}$, we have $z_1<z_2$ and $-z_1^{-1}=2>1=-2+3=z_1^{-1}-z_2^{-1}>0$. Then for for $(z_1,z_2)\in S_-$, we have $|z_1|>|z_2|>0$, $|\arg(z_1-z_2)-\arg z_1|=0<\frac{\pi}{2}$, $\arg(z_1-z_2)=\pi\ge\pi$, $\left|z_2^{-1}\right|=-z_2^{-1}>-z_1^{-1}>z_1^{-1}-z_2^{-1}=\left|z_1^{-1}-z_2^{-1}\right|>0$, $\left|\arg z_1^{-1}-\arg z_2^{-1}\right|=0<\frac{\pi}{2}$, and

$$\log(z_1-z_2)-\log z_1-\log z_2+\pi\mathrm{i}=\log\left|\frac{z_1-z_2}{z_1z_2}\right|\mathrm{i}=\log\bigl(z_1^{-1}-z_2^{-1}\bigr).$$

In particular, for $(z_1,z_2)\in S_-$, we also have $m+1=0$. Then on S_-, (2.48) is absolutely convergent to

$$\begin{aligned}&f^{b_{13}^{-1}b_{23}^{-1}b_{12}^{-1}}\bigl(z_1^{-1},z_2^{-1};\mathrm{e}^{z_1L_V(1)}\bigl(-z_1^{-2}\bigr)^{L_V(0)}u,\mathrm{e}^{z_2L_{W_1}(1)}\mathrm{e}^{-\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1,w_2,w_3'\bigr)\\ &\qquad=h_-^e(z_1,z_2;u,w_1,w_3',w_2).\end{aligned}$$

By analytic extension, we see that the sum of (2.48) is equal to $h_-^e(z_1,z_2;u,w_1,w_3',w_2)$ on the subregion of M_0^2 given by $\left|z_2^{-1}\right|>\left|z_1^{-1}-z_2^{-1}\right|>0$, $\left|\arg z_1^{-1}-\arg z_2^{-1}\right|<\frac{\pi}{2}$. By analytic extension again, we see that the sum of (2.47) is equal to $h_-^e(z_1,z_2;u,w_1,w_3',w_2)$ on the subregion of M_0^2 given by $|z_2|>|z_1-z_2|>0$, $|\arg z_1-\arg z_2|<\frac{\pi}{2}$.

Finally we prove that for any fixed $z_k\in\mathbb{C}^\times$, a component-isolated singularity of the form $(z_1,\dots,z_{k-1})-\beta=\delta$ for $\beta\in\mathbb{C}^{k-1}$ and $\delta\in\{0,\infty\}^k$ of any value at $\zeta=z_k$ of $\prod_{1\le i<\le j\le k-1}(z_i-z_j)^{M_{ij}}h(z_1,\dots,z_{k-1},\zeta)$ is regular. Generalizing the proof above, we can show that

$$\bigl\langle\phi_{g_1}\bigl(\bigl(Y_{W_2}^{g_2}\bigr)'\bigr)(u_1,z_1)\cdots\phi_{g_1}\bigl(\bigl(Y_{W_2}^{g_2}\bigr)'\bigr)(u_{k-2},z_{k-2})A_\pm(\mathcal{Y})(Y_{W_1}^{g_1}(u,z_{k-1}-z_k)w_1,z_k)w_3',w_2\bigr\rangle$$

is absolutely convergent on the region $|z_1|>\cdots>|z_{k-2}|>|z_k|>|z_{k-1}-z_k|>0$ and the maximal analytic extension of its sum on M^k is $h(z_1,\dots,z_k)$. But we have proved above that (2.37) is absolutely convergent on the region $|z_1|>\cdots>|z_k|>0$ and the maximum analytic extension of its sum is also $h(z_1,\dots,z_k)$. Thus $\prod_{1\le i<j\le k-1}(z_i-z_j)^{M_{ij}}h(z_1,\dots,\zeta)$ can be expanded as

$$\begin{aligned}&\prod_{1\le i<j\le k-1}(z_i-z_j)^{M_{ij}}\bigl\langle\phi_{g_1}\bigl(\bigl(Y_{W_2}^{g_2}\bigr)'\bigr)(u_1,z_1)\cdots\phi_{g_1}\bigl(\bigl(Y_{W_2}^{g_2}\bigr)'\bigr)(u_{k-1},z_{k-1})\\ &\qquad\times A_\pm(\mathcal{Y})\bigl(Y_{W_1}^{g_1}(u,z_{k-1}-z_k)w_1,z_k\bigr)w_3',w_2\bigr\rangle,\end{aligned}\tag{2.50}$$

a series in powers of $z_1,\dots,z_{k-2},z_{k-1}-z_k$ and finitely many integral powers of $\log z_1,\dots,\log z_{k-2},\log(z_{k-1}-z_k)$. From the expansion (2.50), we see that the

component-isolated singularity $(z_1, \dots, z_{k-2}, z_{k-1} - z_k) = (\infty, \dots, \infty, 0)$ of the value $\prod_{1 \le i < j \le k-1}(z_i - z_j)^{M_{ij}} h(z_1, \dots, z_k)$ at $\zeta = z_k$ is regular. Using the commutativity for $A_{\pm}(\mathcal{Y})$ (which is already proved), we see that the component-isolated singularity $(z_1, \dots, z_{k-2}, z_{k-1} - z_k) = (0, \dots, 0)$ of the value $\prod_{1 \le i < j \le k-1}(z_i - z_j)^{M_{ij}} h(z_1, \dots, z_k)$ at $\zeta = z_k$ is also regular. The regularity of the other component-isolated singularity of the form above can be proved either similarly or easily by using the relation between $h(z_1, \dots, z_k)$ and $f_{j_1,\dots,j_{k-1};i,m}(z_1, \dots, z_k)$. □

3. Tensor product bifunctors and some natural isomorphisms

In this section, we introduce the notion of twisted $P(z)$-intertwining map and give a definition and a construction of $P(z)$-tensor product of a g_1-twisted module and a g_2-twisted V-module for g_1, g_2 in a group G of automorphisms of V in a category of twisted V-modules under suitable assumptions. Using the skew-symmetry isomorphism Ω_+ given in the preceding section, we construct G-crossed commutativity isomorphisms. We also construct parallel transport isomorphisms. Using G-crossed commutativity isomorphisms and parallel transport isomorphisms, we construct G-crossed braiding isomorphisms. The material in this section is essentially the same as the corresponding material in [13, 24, 27] except that V-modules and intertwining maps are replaced by twisted V-modules and twisted intertwining maps.

Let G be a group of automorphisms of V. Let $\mathcal{C}_1$ be a category of g-twisted V-modules without g-actions for $g \in G$. For $g \in G$, let W be a g-twisted V-module with a g-action and also with an extension of the g-action on W to an action of G. Here by an action of G or a G-action on W, we mean that W is a left G-module. We shall call W simply a g-twisted V-module with a G-action. But note that the g-action on W has additional properties but the actions of other elements of G are not required to have any properties. Let $\mathcal{C}_2$ be a category of g-twisted V-modules with actions of G. In this paper, we do not require that morphisms in $\mathcal{C}_2$ commute with actions of G. But to prove substantial results on $\mathcal{C}_2$, we do need to consider categories whose morphisms commute with the actions of G. See, for example, [22] for such results. The category $\mathcal{C}_1$ can be the category of grading-restricted g-twisted V-modules without g-actions for $g \in G$ and the category $\mathcal{C}_2$ can be the category of grading-restricted g-twisted V-modules with actions of G. We shall use $\mathcal{C}$ to denote either $\mathcal{C}_1$ or $\mathcal{C}_2$. Since many of the constructions in the present paper work for any category satisfying suitable conditions, we shall work with a general category $\mathcal{C}$. But the constructions for $\mathcal{C}_1$ and $\mathcal{C}_2$, especially the construction of the G-crossed commutativity isomorphisms and thus the construction of the G-crossed braiding isomorphisms, are slightly different.

Definition 3.1. *Let $g_1, g_2 \in G$, W_1, W_2, W_3 g_1-, g_2-, g_1g_2-twisted V-modules, respectively, in the category $\mathcal{C}$ and $z \in \mathbb{C}^\times$. A twisted $P(z)$-intertwining map of type $\binom{W_3}{W_1 W_2}$ is a linear map $I\colon W_1 \otimes W_2 \to \overline{W}_3$ given by $I(w_1 \otimes w_2) = \mathcal{Y}(w_1, z)w_2$ for $w_1 \in W_1$ and $w_2 \in W_2$, where $\mathcal{Y}$ is a twisted intertwining operator of type $\binom{W_3}{W_1 W_2}$.*

Using the notion of twisted $P(z)$-intertwining map, we now define the notion of tensor product of two twisted modules in $\mathcal{C}$.

Definition 3.2. Let W_1 and W_2 be g_1- and g_2-twisted V-modules, respectively, in $\mathcal{C}$. A *$P(z)$-product of W_1 and W_2 in $\mathcal{C}$* is a pair (W_3, I) consisting of a g_1g_2-twisted V-module W_3 in $\mathcal{C}$ and a twisted $P(z)$-intertwining map I of type $\binom{W_3}{W_1 W_2}$. A *$P(z)$-tensor product of W_1 and W_2 in $\mathcal{C}$* is a $P(z)$-product $(W_1 \boxtimes_{P(z)} W_2, \boxtimes_{P(z)})$ satisfying the following universal property: For any $P(z)$-product (W_3, I) of W_1 and W_2, there exists a unique module map $f\colon W_1 \boxtimes_{P(z)} W_2 \to W_3$ such that we

have the commutative diagram

$$\begin{array}{ccc} W_1 \otimes W_2 & \xrightarrow{I} & \overline{W}_3 \\ {\scriptstyle \boxtimes_{P(z)}}\downarrow & \nearrow_{\bar{f}} & \\ \overline{W_1 \boxtimes_{P(z)} W_2}, & & \end{array}$$

where $\bar{f}$ is the natural extension of f to $\overline{W_1 \boxtimes_{P(z)} W_2}$.

We now give a construction of $(W_1 \boxtimes_{P(z)} W_2, \boxtimes_{P(z)})$ under a suitable assumption using the same method as in [24, 27].

Given a $P(z)$-product (W_3, I) of W_1 and W_2 in $\mathcal{C}$, for $w_3' \in W_3'$, we have an element $\lambda_{I,w_3'} \in (W_1 \otimes W_2)^*$ defined by $\lambda_{I,w_3'}(w_1 \otimes w_2) = \langle w_3', I(w_1 \otimes w_2)\rangle$ for $w_1 \in W_1$ and $w_2 \in W_2$. Let $W_1 \boxtimes_{P(z)} W_2$ be the subspace of $(W_1 \otimes W_2)^*$ spanned by $\lambda_{I,w_3'}$ for all $P(z)$-products (W_3, I) and $w_3' \in W_3'$. Note that $W_1 \boxtimes_{P(z)} W_2$ in fact depends on the category $\mathcal{C}$. We define a vertex operator map

$$Y^{(g_1g_2)^{-1}}_{W_1 \boxtimes_{P(z)} W_2}\colon\ V \otimes (W_1 \boxtimes_{P(z)} W_2) \to (W_1 \boxtimes_{P(z)} W_2)\{x\}[\log x]$$

by

$$Y^{(g_1g_2)^{-1}}_{W_1 \boxtimes_{P(z)} W_2}(v,x)\lambda_{I,w_3'} = \lambda_{I, Y^{(g_1g_2)^{-1}}_{W_3'}(v,x)w_3'}$$

for $v \in V$ and $\lambda_{I,w_3'} \in W_1 \boxtimes_{P(z)} W_2$.

This vertex operator map is indeed well defined. In fact, assuming that $\lambda_{I,w_3'} = \lambda_{\tilde{I},\tilde{w}_3'}$ for another $P(z)$-product $(\widetilde{W}_3, \tilde{I})$ of W_1 and W_2 in $\mathcal{C}$ and $\tilde{w}_3' \in \widetilde{W}_3'$. Then for $v \in V$, $w_1 \in W_1$ and $w_2 \in W_2$,

$$\begin{aligned} \big\langle w_3', I\big(Y^{g_1}_{W_1}(v,x)w_1 \otimes w_2\big)\big\rangle &= \lambda_{I,w_3'}\big(Y^{g_1}_{W_1}(v,x)w_1 \otimes w_2\big) = \lambda_{\tilde{I},\tilde{w}_3'}\big(Y^{g_1}_{W_1}(v,x)w_1 \otimes w_2\big) \\ &= \big\langle \tilde{w}_3', \tilde{I}\big(Y^{g_1}_{W_1}(v,x)w_1 \otimes w_2\big)\big\rangle. \end{aligned}$$

Let $\mathcal{Y}$ and $\widetilde{Y}$ be the twisted intertwining operators corresponding to I and $\tilde{I}$. We also need the opposite twisted vertex operator map $\big(Y^g_W\big)^o$ of a g-twisted V-module W defined by $\big(Y^g_W\big)^o(v,x) = Y^g_W\big(\mathrm{e}^{xL(1)}\big(-x^{-2}\big)^{-L(0)}v, x^{-1}\big)$ for $v \in V$. Then on the region $|z_1| > |z| > |z_1 - z| > 0$, $|\arg(z_1 - z) - \arg z_1| < \frac{\pi}{2}$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$, we have

$$\begin{aligned} &\big\langle \big(Y^{(g_1g_2)^{-1}}_{W_3'}\big)^o(v,z_1)w_3', I(w_1 \otimes w_2)\big\rangle \\ &= \big\langle w_3', Y^{(g_1g_2)}_{W_3}(v,z)\mathcal{Y}(w_1,z)w_2\big\rangle = \langle w_3', \mathcal{Y}(Y_{W_1}(v,z_1 - z)w_1, z)w_2)\rangle \\ &= \big\langle w_3', I\big(Y^{g_1}_{W_1}(v,z_1 - z)w_1 \otimes w_2\big)\big\rangle = \big\langle \tilde{w}_3', \tilde{I}\big(Y^{g_1}_{W_1}(v,z_1 - z)w_1 \otimes w_2\big)\big\rangle \\ &= \langle \tilde{w}_3', \widetilde{\mathcal{Y}}(Y_{W_1}(v,z_1 - z)w_1, z)w_2)\rangle = \big\langle \tilde{w}_3', Y^{(g_1g_2)}_{\widetilde{W}_3}(v,z)\mathcal{Y}(w_1,z)w_2\big\rangle \\ &= \big\langle \big(Y^{(g_1g_2)^{-1}}_{\widetilde{W}_3'}\big)^o(v,z_1)\tilde{w}_3', I(w_1 \otimes w_2)\big\rangle. \end{aligned}$$

Since the both sides are analytic on the region $|z_1| > |z| > 0$, this equality holds on this larger region. So we can replace the complex variable z_1 by a formal variable x to obtain

$$\big\langle \big(Y^{(g_1g_2)^{-1}}_{W_3'}\big)^o(v,x)w_3', I(w_1 \otimes w_2)\big\rangle = \big\langle \big(Y^{(g_1g_2)^{-1}}_{\widetilde{W}_3'}\big)^o(v,x)\tilde{w}_3', I(w_1 \otimes w_2)\big\rangle.$$

Then from the definition of opposite twisted vertex operator map, we obtain

$$\big\langle Y^{(g_1g_2)^{-1}}_{W_3'}(v,x)w_3', I(w_1 \otimes w_2)\big\rangle = \big\langle Y^{(g_1g_2)^{-1}}_{\widetilde{W}_3'}(v,x)\tilde{w}_3', I(w_1 \otimes w_2)\big\rangle.$$

Thus we have

$$\lambda_{I, Y^{(g_1g_2)^{-1}}_{W_3'}(v,x)w_3'} = \lambda_{\tilde{I}, Y^{(g_1g_2)^{-1}}_{\widetilde{W}_3'}(v,x)\tilde{w}_3'},$$

proving that $Y_{W_1\boxtimes_{P(z)}W_2}^{(g_1g_2)^{-1}}(v,x)$ is indeed well defined.

Proposition 3.3. *The pair* $\left(W_1\boxtimes_{P(z)}W_2, Y_{W_1\boxtimes_{P(z)}W_2}^{(g_1g_2)^{-1}}\right)$ *is a generalized* $(g_1g_2)^{-1}$*-twisted* V*-module without a* $(g_1g_2)^{-1}$*-action.*

Proof. Note that every element of $W_1\boxtimes_{P(z)}W_2$ is a linear combination of elements of the form $\lambda_{I,w_3'}$ for a g_1g_2-twisted V-module W_3 in $\mathcal{C}$, a $P(z)$-intertwining map I of type of $\binom{W_3}{W_1\,W_2}$ and an element $w_3'\in W_3'$. For fixed W_3 and I, the space spanned by all $\lambda_{I,w_3'}$ for $w_3'\in W_3'$ is the image of W_3' under the linear map from W_3' to $W_1\boxtimes_{P(z)}W_2$ given by $w_3'\mapsto\lambda_{I,w_3'}$. This linear map preserves the gradings. So the space of spanned by all $\lambda_{I,w_3'}$ for $w_3'\in W_3'$ is a generalized $(g_1g_2)^{-1}$-twisted V-module without a g-action. Thus $W_1\boxtimes_{P(z)}W_2$ as a sum of generalized $(g_1g_2)^{-1}$-twisted V-modules without a $(g_1g_2)^{-1}$-action is also a generalized $(g_1g_2)^{-1}$-twisted V-module without a $(g_1g_2)^{-1}$-action. □

Assumption 3.4. We assume that the following conditions for $\mathcal{C}$ hold:

(1) For objects W_1 and W_2 in $\mathcal{C}$, the contragredient of $W_1\boxtimes_{P(z)}W_2$ is also in $\mathcal{C}$. (In the case $\mathcal{C}=\mathcal{C}_2$, this means in particular that for objects W_1 and W_2 in $\mathcal{C}_2$, there exists an action of G on $W_1\boxtimes_{P(z)}W_2$ and a decomposition of $W_1\boxtimes_{P(z)}W_2$ as the direct sum of generalized eigenspaces of the (g_1g_2)-action such that the contragredient of $W_1\boxtimes_{P(z)}W_2$ with this action and this decomposition is an object in $\mathcal{C}_2$.)

(2) The double contragredient of an object in $\mathcal{C}$ is equivalent to the object itself.

See Theorem 3.8 for some conditions implying condition (1) in Assumption 3.4. Condition (2) in Assumption 3.4 can be replaced by the condition that all the twisted V-modules in $\mathcal{C}$ is grading-restricted. In fact, the proof in [5] in the untwisted case of the result that the double contragredient of a grading-restricted V-module is equivalent to the V-module itself still works for grading-restricted twisted V-modules. In the untwisted case, it has been shown by the second author in [21] that Assumption 3.4 is enough to construct $P(z)$-tensor product bifunctors. In the case that G is abelain, the second author has shown in [22] that when V is C_2-cofinite and the category is the category of C_2-cofinite twisted modules, condition (1) in Assumption 3.4 is satisfied and condition (2) is a consequence.

From Assumption 3.4 (1), we see that $(W_1\boxtimes_{P(z)}W_2)'$ is in $\mathcal{C}$. We take $W_1\boxtimes_{P(z)}W_2$ to be $(W_1\boxtimes_{P(z)}W_2)'$. We still need to give a twisted $P(z)$-intertwining map $\boxtimes_{P(z)}$ of type $\binom{W_1\boxtimes_{P(z)}W_2}{W_1\,W_2}$ or equivalently, an intertwining operator of the same type.

Let W be a $(g_1g_2)^{-1}$-twisted V-module in $\mathcal{C}$ and $f\colon W\to W_1\boxtimes_{P(z)}W_2$ a V-module map. For $w_1\in W_1$, $w_2\in W_2$ and $w\in W$, we define an element $\mathcal{Y}_f(w_1,z)w_2\in W^*=\overline{W'}$ by

$$\langle w,\mathcal{Y}_f(w_1,z)w_2\rangle=(f(w))(w_1\otimes w_2). \tag{3.1}$$

Then we define

$$\begin{aligned}\mathcal{Y}_f(w_1,x)w_2=\;&x^{L_{W'}(0)}\mathrm{e}^{-(\log z)L_{W_3}(0)}\mathcal{Y}_f\big(x^{-L_{W_1}(0)}\mathrm{e}^{(\log z)L_{W_1}(0)}w_1,z\big)\\ &\times x^{-L_{W_2}(0)}\mathrm{e}^{(\log z)L_{W_2}(0)}w_2\end{aligned} \tag{3.2}$$

for $w_1\in W_1$, $w_2\in W_2$. We now have a linear map $\mathcal{Y}_f\colon W_1\otimes W_2\to W'\{x\}[\log x]$.

Proposition 3.5. *The linear map* $\mathcal{Y}_f\colon W_1\otimes W_2\to W'\{x\}[\log x]$ *given by* (3.1) *and* (3.2) *above is a twisted intertwining operator of type* $\binom{W'}{W_1\,W_2}$*. In particular, in the case that* $W=W_1\boxtimes_{P(z)}W_2$ *and* $f=1_{W_1\boxtimes_{P(z)}W_2}\colon W\to W_1\boxtimes_{P(z)}W_2$ *is the identity map, we obtain a twisted intertwining operator* $\mathcal{Y}_{1_{W_1\boxtimes_{P(z)}W_2}}$ *of type* $\binom{W_1\boxtimes_{P(z)}W_2}{W_1\,W_2}$.

Proof. We first verify the $L(-1)$-derivative property,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\mathcal{Y}_f(w_1,x)w_2 &= \frac{\mathrm{d}}{\mathrm{d}x}x^{L_{W'}(0)}\mathrm{e}^{-(\log z)L_{W'}(0)}\mathcal{Y}_f\\ &\quad\times\big(x^{-L_{W_1'}(0)}\mathrm{e}^{(\log z)L_{W_1'}(0)}w_1,z\big)x^{-L_{W_2}(0)}\mathrm{e}^{(\log z)L_{W_2}(0)}w_2\\ &= x^{L_{W'}(0)-1}\mathrm{e}^{-(\log z)L_{W'}(0)}L_{W'}(0)\mathcal{Y}_f\big(x^{-L_{W_1}(0)}\mathrm{e}^{(\log z)L_{W_1}(0)}w_1,z\big)x^{-L_{W_2}(0)}\\ &\quad\times\mathrm{e}^{(\log z)L_{W_2}(0)}w_2 - x^{L_{W'}(0)}\mathrm{e}^{-(\log z)L_{W'}(0)}\\ &\quad\times\mathcal{Y}_f\big(L_{W_1}(0)x^{-L_{W_1}(0)-1}\mathrm{e}^{(\log z)L_{W_1}(0)}w_1,z\big)x^{-L_{W_2}(0)}\mathrm{e}^{(\log z)L_{W_2}(0)}w_2\\ &\quad\times -x^{L_{W'}(0)}\mathrm{e}^{-(\log z)L_{W'}(0)}\mathcal{Y}_f\big(x^{-L_{W_1}(0)}\mathrm{e}^{(\log z)L_{W_1}(0)}w_1,z\big)\\ &\quad\times L_{W_2}(0)x^{-L_{W_2}(0)-1}\mathrm{e}^{(\log z)L_{W_2}(0)}w_2\\ &= x^{-1}x^{L_{W'}(0)}\mathrm{e}^{-(\log z)L_{W'}(0)}\big(L_{W'}(0)\mathcal{Y}_f\big(x^{-L_{W_1}(0)}\mathrm{e}^{(\log z)L_{W_1}(0)}w_1,z\big)\\ &\quad-\mathcal{Y}_f\big(L_{W_1}(0)x^{-L_{W_1}(0)}\mathrm{e}^{(\log z)L_{W_1}(0)}w_1,z\big)\\ &\quad-\mathcal{Y}_f\big(x^{-L_{W_1}(0)}\mathrm{e}^{(\log z)L_{W_1}(0)}w_1,z\big)L_{W_2}(0)\big)x^{-L_{W_2}(0)}\mathrm{e}^{(\log z)L_{W_2}(0)}w_2. \end{aligned}\tag{3.3}$$

Since f is a V-module map, $f(W)$ is a submodule of $W_1\boxtimes_{P(z)}W_2$. By the definition of $W_1\boxtimes_{P(z)}W_2$, it is spanned by elements of the form $\lambda_{I,w_3'}$ for a g_1g_2-twisted V-module W_3, a $P(z)$-intertwining map I of type $\binom{W_3}{W_1W_2}$ and $w_3'\in W_3'$. In particular, for $w\in W$, $f(w)=\sum_{i=3}^n\lambda_{I^i,w_i'}$, where for $i=3,\dots,n$, w_i' is an element of the contragredient module W_i' of a g_1g_2-twisted V-module W_i, I^i a twisted $P(z)$-intertwining map of type $\binom{W_i}{W_1W_2}$.

Let $\mathcal{Y}^i$ be the twisted intertwining operator of type $\binom{W_i}{W_1W_2}$ such that $I^i=\mathcal{Y}^i(\cdot,z)\cdot$. Then we have

$$\begin{aligned}\langle w,\mathcal{Y}_f(w_1,z)w_2\rangle &= (f(w))(w_1\otimes w_2)=\sum_{i=3}^n\lambda_{I^i,w_i'}(w_1\otimes w_2)=\sum_{i=3}^n\langle w_i',I^i(w_1\otimes w_2)\rangle\\ &=\sum_{i=3}^n\langle w_i',\mathcal{Y}^i(w_1,z)w_2)\rangle\end{aligned}$$

for $w_1\in W_1$ and $w_2\in W_2$. Also,

$$\begin{aligned}(f(L_W(0)w))(w_1\otimes w_2) &= ((L_{W_1\boxtimes_{P(z)}W_2}(0)f(w))(w_1\otimes w_2)\\ &=\sum_{i=3}^n(L_{W_1\boxtimes_{P(z)}W_2}(0)\lambda_{I^i,w_i'})(w_1\otimes w_2)\\ &=\sum_{i=3}^n\mathrm{Res}_x\,x\big(Y_{W'}^{(g_1g_2)^{-1}}(\omega,x)\lambda_{I^i,w_i'}\big)(w_1\otimes w_2)\\ &=\sum_{i=3}^n\mathrm{Res}_x\,x\big(\lambda_{I^i,Y_{W_i'}^{(g_1g_2)^{-1}}(\omega,x)w_i'}\big)(w_1\otimes w_2)\\ &=\sum_{i=3}^n\lambda_{I^i,L_{W_i'}(0)w_i'}(w_1\otimes w_2)\end{aligned}$$

for $w_1\in W_1$ and $w_2\in W_2$. So we have

$$f(L_W(0)w)=\sum_{i=3}^n\lambda_{I^i,L_{W_i'}(0)w_i'}.$$

Thus for $w_1 \in W_1$, $w_2 \in W_2$,

$$\begin{aligned}
&\langle w, (L_{W'}(0)\mathcal{Y}_f(w_1,z) - \mathcal{Y}_f(L_{W_1}(0)w_1,z) - \mathcal{Y}_f(w_1,z)L_{W_2}(0))\, w_2\rangle \\
&= \langle L_W(0)w, \mathcal{Y}_f(w_1,z)w_2\rangle - \langle w, \mathcal{Y}_f(L_{W_1}(0)w_1,z)w_2\rangle - \langle w, \mathcal{Y}_f(w_1,z)L_{W_2}(0)w_2\rangle \\
&= (f(L_W(0)w))(w_1\otimes w_2) - (f(w))(L_{W_1}(0)w_1\otimes w_2) - (f(w))(w_1\otimes L_{W_2}(0)w_2) \\
&\qquad = \sum_{i=3}^n \lambda_{I^i, L_{W_i'}(0)w_i'}(w_1\otimes w_2) - \sum_{i=3}^n \lambda_{I,w_i'}(L_{W_1}(0)w_1\otimes w_2) \\
&\qquad\qquad - \sum_{i=3}^n \lambda_{I,w_i'}(w_1\otimes L_{W_2}(0)w_2) \\
&\qquad = \sum_{i=3}^n \big\langle L_{W_i'}(0)w_i', \mathcal{Y}^i(w_1,z)w_2\big\rangle - \sum_{i=3}^n \big\langle w_i', \mathcal{Y}^i(L_{W_1}(0)w_1,z)w_2\big\rangle \\
&\qquad\qquad - \sum_{i=3}^n \big\langle w_i', \mathcal{Y}^i(w_1,z)L_{W_2}(0)w_2\big\rangle \\
&\qquad = \sum_{i=3}^n \big\langle w_i', \big(L_{W_i}(0)\mathcal{Y}^i(w_1,z) - \mathcal{Y}^i(L_{W_1}(0)w_1,z) - \mathcal{Y}^i(w_1,z)L_{W_2}(0)\big)w_2\big\rangle \\
&\qquad = \sum_{i=3}^n z\big\langle w_i', \mathcal{Y}^i(L_{W_1}(-1)w_1,z)w_2\big\rangle = z\sum_{i=3}^n \lambda_{I^i,w_i'}((L_{W_1}(-1)w_1\otimes w_2) \\
&\qquad\qquad = z(f(w))(L_{W_1}(-1)w_1\otimes w_2) = z\langle w, \mathcal{Y}_f(L_{W_1}(-1)w_1,z)w_2\rangle,
\end{aligned}$$

where we have used the $L(0)$-commutator formula for the twisted intertwining operators $\mathcal{Y}^i$ (see Remark 1.11). Since $w \in W$ and $w_2 \in W_2$ are arbitrary, we obtain

$$L_W(0)\mathcal{Y}_f(w_1,z) - \mathcal{Y}_f(L_{W_1}(0)w_1,z) - \mathcal{Y}_f(w_1,z)L_{W_2}(0) = z\mathcal{Y}_f(L_{W_1}(-1)w_1,z) \tag{3.4}$$

for $w_1 \in W_1$.

Using (3.4), we see that the right-hand side of (3.3) is equal to

$$\begin{aligned}
&x^{-1}x^{L_W(0)}\mathrm{e}^{-(\log z)L_W(0)}z\mathcal{Y}_f\big(L_{W_1}(-1)x^{-L_{W_1}(0)}\mathrm{e}^{(\log z)L_{W_1}(0)}w_1,z\big)x^{-L_{W_2}(0)}\mathrm{e}^{(\log z)L_{W_2}(0)}w_2 \\
&\quad = x^{L_W(0)}\mathrm{e}^{-(\log z)L_W(0)}\mathcal{Y}_f\big(x^{-L_{W_1}(0)}\mathrm{e}^{(\log z)L_{W_1}(0)}L_{W_1}(-1)w_1,z\big)x^{-L_{W_2}(0)}\mathrm{e}^{(\log z)L_{W_2}(0)}w_2 \\
&\qquad = \mathcal{Y}_f(L_{W_1}(-1)w_1,x)w_2,
\end{aligned}$$

proving the $L(-1)$-derivative property.

For $v \in V$, $w_1 \in W_1$ and $w_2 \in W_2$, we have

$$\begin{aligned}
\big(f(Y_W^{(g_1g_2)^{-1}}(v,x)w)\big)(w_1\otimes w_2) &= \big(Y_{W_1\boxtimes_{P(z)}W_2}^{(g_1g_2)^{-1}}(v,x)f(w)\big)(w_1\otimes w_2) \\
&= \sum_{i=3}^n \big(Y_{W_1\boxtimes_{P(z)}W_2}^{(g_1g_2)^{-1}}(v,x)\lambda_{I^i,w_i'}\big)(w_1\otimes w_2) \\
&= \sum_{i=3}^n \lambda_{I^i, Y_{W_i}^{(g_1g_2)^{-1}}(v,x)w_i'}(w_1\otimes w_2).
\end{aligned}$$

Then we obtain

$$f\big(Y_W^{(g_1g_2)^{-1}}(v,x)w\big) = \sum_{i=3}^n \lambda_{I^i, Y_{W_i}^{(g_1g_2)^{-1}}(v,x)w_i'}.$$

For $u \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $w \in W$, we have

$$\begin{aligned}
&\big\langle w, Y_{W'}^{g_1g_2}(u,z_1)\mathcal{Y}_f(w_1,z_2)w_2\big\rangle\\
&= \big\langle w, Y_{W'}^{g_1g_2}(u,z_1)\mathrm{e}^{(\log z_2)L_{W'}(0)}\mathrm{e}^{-(\log z)L_{W'}(0)}\\
&\times \mathcal{Y}_f\big(\mathrm{e}^{-(\log z_2)L_{W_1}(0)}\mathrm{e}^{(\log z)L_{W_1}(0)}w_1,z\big)\mathrm{e}^{-(\log z_2)L_{W_2}(0)}\mathrm{e}^{(\log z)L_{W_2}(0)}w_2\big\rangle\\
&= \big\langle \mathrm{e}^{(\log z_2)L_W(0)}\mathrm{e}^{-(\log z)L_W(0)}Y_W^{(g_1g_2)^{-1}}\big(\mathrm{e}^{z_1L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}u,z_1^{-1}\big)w,\\
&\mathcal{Y}_f\big(\mathrm{e}^{-(\log z_2)L_{W_1}(0)}\mathrm{e}^{(\log z)L_{W_1}(0)}w_1,z\big)\mathrm{e}^{-(\log z_2)L_{W_2}(0)}\mathrm{e}^{(\log z)L_{W_2}(0)}w_2\big\rangle\\
&= \big(f\big(\mathrm{e}^{(\log z_2)L_W(0)}\mathrm{e}^{-(\log z)L_W(0)}Y_W^{(g_1g_2)^{-1}}\big(\mathrm{e}^{z_1L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}u,z_1^{-1}\big)w\big)\big)\\
&\times \big(\mathrm{e}^{-(\log z_2)L_{W_1}(0)}\mathrm{e}^{(\log z)L_{W_1}(0)}w_1\otimes \mathrm{e}^{-(\log z_2)L_{W_2}(0)}\mathrm{e}^{(\log z)L_{W_2}(0)}w_2\big)\\
&= \sum_{i=3}^n \lambda_{I^i,\mathrm{e}^{(\log z_2)L_{W_i'}(0)}\mathrm{e}^{-(\log z)L_{W_i'}(0)}Y_{W_i'}^{(g_1g_2)^{-1}}\left(\mathrm{e}^{z_1L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}u,z_1^{-1}\right)w_i'}\\
&\times \big(\mathrm{e}^{-(\log z_2)L_{W_1}(0)}\mathrm{e}^{(\log z)L_{W_1}(0)}w_1\otimes \mathrm{e}^{-(\log z_2)L_{W_2}(0)}\mathrm{e}^{(\log z)L_{W_2}(0)}w_2\big)\\
&= \sum_{i=3}^n \big\langle \mathrm{e}^{(\log z_2)L_{W_i'}(0)}\mathrm{e}^{-(\log z)L_{W_i'}(0)}Y_{W_i'}^{(g_1g_2)^{-1}}\big(\mathrm{e}^{z_1L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}u,z_1^{-1}\big)w_i',\\
&\mathcal{Y}^i\big(\mathrm{e}^{-(\log z_2)L_{W_1}(0)}\mathrm{e}^{(\log z)L_{W_1}(0)}w_1,z\big)\mathrm{e}^{-(\log z_2)L_{W_2}(0)}\mathrm{e}^{(\log z)L_{W_2}(0)}w_2\big\rangle\\
&= \sum_{i=3}^n \langle w_i', Y_{W_i}^{g_1g_2}(u,z_1)\mathrm{e}^{(\log z_2)L_{W_i}(0)}\mathrm{e}^{-(\log z)L_{W_i}(0)}\\
&\times \mathcal{Y}^i\big(\mathrm{e}^{-(\log z_2)L_{W_1}(0)}\mathrm{e}^{(\log z)L_{W_1}(0)}w_1,z\big)\mathrm{e}^{-(\log z_2)L_{W_2}(0)}\mathrm{e}^{(\log z)L_{W_2}(0)}w_2\big\rangle\\
&= \sum_{i=3}^n \big\langle w_i', Y_{W_i}^{g_1g_2}(u,z_1)\mathcal{Y}^i(w_1,z_2)w_2\big\rangle. \qquad (3.5)
\end{aligned}$$

Similarly, we have

$$\big\langle w, \mathcal{Y}_f(w_1,z_2)Y_{W_2}^{g_2}(u,z_1)w_2\big\rangle = \sum_{i=3}^n \big\langle w_i', \mathcal{Y}^i(w_1,z_2)Y_{W_2}^{g_2}(u,z_1)w_2\big\rangle \qquad (3.6)$$

and

$$\big\langle w, \mathcal{Y}_f\big(Y_{W_2}^{g_1}(u,z_1-z_2)w_1,z_2\big)w_2\big\rangle = \sum_{i=3}^n \big\langle w_i', \mathcal{Y}^i\big(Y_{W_1}^{g_1}(u,z_1-z_2)w_1,z_2\big)w_2\big\rangle. \qquad (3.7)$$

Since $\mathcal{Y}^i$ for $i = 1,\dots,n$ are twisted intertwining operators, the duality property for $\mathcal{Y}$ follows from (3.5), (3.6), (3.7) and the duality properties for $\mathcal{Y}^i$.

The convergence for products of more than two operators follows from the formula

$$\begin{aligned}
&\big\langle w, Y_{W'}^{g_1g_2}(u_1,z_1)\cdots Y_{W'}^{g_1g_2}(u_{k-1},z_{k-1})\mathcal{Y}_f(w_1,z_k)w_2\big\rangle\\
&= \sum_{i=3}^n \big\langle w_i', Y_{W_i}^{g_1g_2}(u_1,z_1)\cdots Y_{W_i}^{g_1g_2}(u_{k-1},z_{k-1})\mathcal{Y}^i(w_1,z_k)w_2\big\rangle,
\end{aligned}$$

whose proof is the same as that of (3.5). □

Let $\boxtimes_{P(z)} = \mathcal{Y}_{1_{W_1\boxtimes_{P(z)}W_2}}(\cdot,z)\cdot$. Then $\boxtimes_{P(z)}$ is a $P(z)$-intertwining map of type $\binom{W_1\boxtimes_{P(z)}W_2}{W_1\ W_2}$. Let $w_1\boxtimes_{P(z)}w_2 = \boxtimes_{P(z)}(w_1\otimes w_2) = \mathcal{Y}(w_1,z)w_2 \in \overline{W_1\boxtimes_{P(z)}W_2}$ for $w_1 \in W_1$ and $w_2 \in W_2$. We call $w_1\boxtimes_{P(z)}w_2$ the *tensor product* of w_1 and w_2. By (3.1), we have

$$\lambda(w_1\otimes w_2) = \langle \lambda, w_1\boxtimes_{P(z)}w_2\rangle \qquad (3.8)$$

for $\lambda \in W_1\boxtimes_{P(z)}W_2$, $w_1 \in W_1$ and $w_2 \in W_2$.

Theorem 3.6. *The pair $(W_1 \boxtimes_{P(z)} W_2, \boxtimes_{P(z)})$ is a $P(z)$-tensor product of W_1 and W_2 in $\mathcal{C}$.*

Proof. Let (W_3, I) be a $P(z)$-product of W_1 and W_2 in $\mathcal{C}$. Then we have a module map $g\colon W_3' \to W_1 \boxtimes_{P(z)} W_2$ given by $g(w_3') = \lambda_{I,w_3'}$ for $w_3' \in W_3'$. By definition, we have $(g(w_3'))(w_1 \otimes w_2) = \lambda_{I,w_3'}(w_1 \otimes w_2) = \langle w_3', I(w_1 \otimes w_2)\rangle$ for $w_1 \in W_1$ and $w_2 \in W_2$. The adjoint of this module map is a module map from $W_1 \boxtimes_{P(z)} W_2$ to W_3''. By condition (2) in Assumption 3.4, we have a canonical module map from W_3'' to W_3. The composition of these two module maps gives a module map $f\colon W_1 \boxtimes_{P(z)} W_2 \to W_3$. By definitions and (3.8),

$$\begin{aligned}\big\langle w_3', (\bar{f} \circ \boxtimes_{P(z)})(w_1 \otimes w_2)\big\rangle &= \big\langle w_3', \bar{f}(w_1 \boxtimes_{P(z)} w_2)\big\rangle = \big\langle g(w_3'), w_1 \boxtimes_{P(z)} w_2\big\rangle \\ &= (g(w_3'))(w_1 \otimes w_2) = \big\langle w_3', I(w_1 \otimes w_2)\big\rangle.\end{aligned}$$

So we obtain $\bar{f} \circ \boxtimes_{P(z)} = I$. □

We have assigned each object (W_1, W_2) in the category $\mathcal{C} \times \mathcal{C}$ an object $W_1 \boxtimes_{P(z)} W_2$ in $\mathcal{C}$. To obtain a functor from $\mathcal{C} \times \mathcal{C}$ to $\mathcal{C}$, we still need to assign a morphism (f_1, f_2) in $\mathcal{C} \times \mathcal{C}$ a morphism $f_1 \boxtimes_{P(z)} f_2$ in $\mathcal{C}$. Recall that in Section 1, we define module maps between a g_1-twisted V-module and a g_2-twisted V-module to be 0 if $g_1 \neq g_2$. So we need only consider module maps or morphisms in $\mathcal{C}$ from g-twisted V-modules to g-twisted V-modules for $g \in G$.

Let W_1, $\widetilde{W}_1$ be g_1-twisted V-modules in $\mathcal{C}$ and W_2, $\widetilde{W}_2$ g_2-twisted V-modules in $\mathcal{C}$. Let

$$f_1\colon\ W_1 \to \widetilde{W}_1 \qquad \text{and} \qquad f_2\colon W_2 \to \widetilde{W}_2$$

be module maps. Let $\widetilde{\mathcal{Y}}$ be the twisted intertwining operator of type $\binom{\widetilde{W}_1 \boxtimes_{P(z)} \widetilde{W}_2}{\widetilde{W}_1\ \widetilde{W}_2}$ such that

$$\tilde{w}_1 \boxtimes_{P(z)} \tilde{w}_2 = \widetilde{\mathcal{Y}}(\tilde{w}_1, z)\tilde{w}_2.$$

Since f_1 and f_2 are module maps, $\mathcal{Y} = \widetilde{\mathcal{Y}} \circ (f_1 \otimes f_2)$ is a twisted intertwining operator of type $\binom{\widetilde{W}_1 \boxtimes_{P(z)} \widetilde{W}_2}{W_1\ W_2}$. Then $I = \mathcal{Y}(\cdot, z)\cdot$ is a twisted $P(z)$-intertwining operator of the same type. Hence we have a $P(z)$-product $\big(\widetilde{W}_1 \boxtimes_{P(z)} \widetilde{W}_2, I\big)$ of W_1 and W_2. By the universal property of the tensor product $\big(W_1 \boxtimes_{P(z)} W_2, \boxtimes_{P(z)}\big)$, there exists a unique module map $f\colon W_1 \boxtimes_{P(z)} W_2 \to \widetilde{W}_1 \boxtimes_{P(z)} \widetilde{W}_2$ such that $I = \bar{f} \circ \boxtimes_{P(z)}$. We define this module map f to be the $P(z)$-tensor product of f_1 and f_2 and denote it by $f_1 \boxtimes_{P(z)} f_2$.

Theorem 3.7. *The assignments given by $(W_1, W_2) \mapsto W_1 \boxtimes_{P(z)} W_2$ and $(f_1, f_2) \mapsto f_1 \boxtimes_{P(z)} f_2$ above is a functor from $\mathcal{C} \times \mathcal{C}$ to $\mathcal{C}$.*

Proof. It is easy to verify $1_{w_1} \boxtimes_{P(z)} 1_{W_2} = 1_{W_1 \boxtimes_{P(z)} W_2}$ and $(f_1 \boxtimes_{P(z)} f_2) \circ (g_1 \boxtimes_{P(z)} g_2) = (f_1 g_1) \boxtimes_{P(z)} (f_2 g_2)$ by using the construction of the tensor products of module maps. We omit the details of the proofs. □

We call this functor the *$P(z)$-tensor product bifunctor*.

We now construct G-crossed commutativity isomorphisms. We assume that $\mathcal{C}_1$ is closed under ϕ_h for $h \in G$ and $\mathcal{C}_2$ is closed under φ_h for $h \in G$, that is, for a g-twisted module W in $\mathcal{C}_1$ (or $\mathcal{C}_2$), $\phi_h(W)$ (or $\varphi_h(W)$) for $h \in G$ is also in $\mathcal{C}_1$ (or $\mathcal{C}_2$). Note that the category of grading-restricted g-twisted V-modules without g-actions (or the category of grading-restricted g-twisted V-modules with actions of G) is indeed closed under ϕ_h (or φ_h) for $h \in G$.

Let W_1 and W_2 be objects of $\mathcal{C}$. Let $\mathcal{Y}$ be the twisted intertwining operator of type $\binom{\phi_{g_1}(W_2) \boxtimes_{P(-z)} W_1}{\phi_{g_1}(W_2)\ W_1}$ when $\mathcal{C} = \mathcal{C}_1$ or the twisted intertwining operator of type $\binom{\varphi_{g_1}(W_2) \boxtimes_{P(-z)} W_1}{\varphi_{g_1}(W_2)\ W_1}$ when $\mathcal{C} = \mathcal{C}_2$ such that

$$w_2 \boxtimes_{P(-z)} w_1 = \mathcal{Y}(w_2, -z)w_1$$

for $w_1 \in W_1$, $w_2 \in \phi_{g_1}(W_2)$ or $w_2 \in \varphi_{g_1}(W_2)$, respectively. Then by Theorem 2.1, $\Omega_+(\mathcal{Y})$ or $\Omega_+^{g_1}(\mathcal{Y})$ is a twisted intertwining operator of type

$$\binom{\phi_{g_1}(W_2) \boxtimes_{P(-z)} W_1}{W_1 \phi_{g_1}^{-1}(\phi_{g_1}(W_2))} = \binom{\phi_{g_1}(W_2) \boxtimes_{P(-z)} W_1}{W_1 W_2},$$

when $\mathcal{C} = \mathcal{C}_1$ or

$$\binom{\varphi_{g_1}(W_2) \boxtimes_{P(-z)} W_1}{W_1 \varphi_{g_1}^{-1}(\varphi_{g_1}(W_2))} = \binom{\varphi_{g_1}(W_2) \boxtimes_{P(-z)} W_1}{W_1 W_2}$$

when $\mathcal{C} = \mathcal{C}_2$. In particular, the pair $(\phi_{g_1}(W_2) \boxtimes_{P(-z)} W_1, \Omega_+(\mathcal{Y})(\cdot, z)\cdot)$ or $(\varphi_{g_1}(W_2) \boxtimes_{P(-z)} W_1, \Omega_+^{g_1}(\mathcal{Y})(\cdot, z)\cdot)$ is a $P(z)$-product of W_1 and W_2 in $\mathcal{C}_1$ or $\mathcal{C}_2$, respectively. By the universal property of the tensor product $W_1 \boxtimes_{P(z)} W_2$, there exists a unique $g_1 g_2$-twisted V-module map

$$\mathcal{R}_{P(z)} \colon\ W_1 \boxtimes_{P(z)} W_2 \to \phi_{g_1}(W_2) \boxtimes_{P(-z)} W_1$$

or

$$\mathcal{R}_{P(z)} \colon\ W_1 \boxtimes_{P(z)} W_2 \to \varphi_{g_1}(W_2) \boxtimes_{P(-z)} W_1$$

such that $\Omega_+(\mathcal{Y})(\cdot, z)\cdot = \overline{\mathcal{R}}_{P(z)} \circ \boxtimes_{P(z)}$ or $\Omega_+^{g_1}(\mathcal{Y})(\cdot, z)\cdot = \overline{\mathcal{R}}_{P(z)} \circ \boxtimes_{P(z)}$, where $\overline{\mathcal{R}}_{P(z)}$ is the natural extension of $\mathcal{R}_{P(z)}$. The $g_1 g_2$-twisted V-module map $\mathcal{R}_{P(z)}$ has an inverse

$$\mathcal{R}_{P(z)}^{-1} \colon\ \phi_{g_1}(W_2) \boxtimes_{P(-z)} W_1 \to W_1 \boxtimes_{P(z)} W_2$$

or

$$\mathcal{R}_{P(z)}^{-1} \colon\ \varphi_{g_1}(W_2) \boxtimes_{P(-z)} W_1 \to W_1 \boxtimes_{P(z)} W_2$$

constructed in the same way as above except that we use Ω_- or $\Omega_-^{g_1}$ instead of Ω_+ or $\Omega_+^{g_1}$, respectively. Then we obtain a natural isomorphism $\mathcal{R}_{P(z)}$ called the *G-crossed commutativity isomorphism*.

As in [13, 27], we also have parallel transport isomorphisms. Let $z_1, z_2 \in \mathbb{C}^\times$ and γ a path in $\mathbb{C}^\times$ from z_1 to z_2. We denote the homotopy class of γ by $[\gamma]$. For the same W_1 and W_2, let $\mathcal{Y}$ be the twisted intertwining operator of type $\binom{W_1 \boxtimes_{P(z_2)} W_2}{W_1 W_2}$ such that $w_1 \boxtimes_{P(z_2)} w_2 = \mathcal{Y}(w_1, z_2) w_2$ for $w_1 \in W_1$, $w_2 \in W_2$. Let $l_p(z_1) = \log|z_1| + \mathrm{i}\arg z_1 + \mathrm{i}2\pi p$ be the value of logarithm of z_1 obtained by analytically extending the value $\log z_2$ along the path γ. Then

$$I = \mathcal{Y}(w_1, x) w_2 \big|_{x^n = \mathrm{e}^{n l_p(z_1)}, \log x = l_p(z_1)}$$

is a $P(z_1)$-intertwining map of type $\binom{W_1 \boxtimes_{P(z_2)} W_2}{W_1 W_2}$ and we have a $P(z_1)$-product $(W_1 \boxtimes_{P(z_2)} W_2, I)$ of W_1 and W_2. By the universal property of the $P(z_1)$-tensor product $W_1 \boxtimes_{P(z_1)} W_2$, there exists a unique $g_1 g_2$-twisted V-module map

$$\mathcal{T}_{[\gamma]} \colon\ W_1 \boxtimes_{P(z_1)} W_2 \to W_1 \boxtimes_{P(z_2)} W_2$$

such that $\overline{\mathcal{T}_{[\gamma]}} \circ \boxtimes_{P(z_1)} = I$. The $g_1 g_2$-twisted V-module map $\mathcal{T}_{[\gamma]}$ is invertible since the same construction also gives a $g_1 g_2$-twisted V-module map

$$\mathcal{T}_{[\gamma^{-1}]} \colon\ W_1 \boxtimes_{P(z_2)} W_2 \to W_1 \boxtimes_{P(z_1)} W_2,$$

which is clearly the inverse of $\mathcal{T}_{[\gamma]}$. Thus the natural transformation $\mathcal{T}_{[\gamma]}$ is a natural isomorphism called the *parallel transport isomorphism from z_1 to z_2 along $[\gamma]$*.

Let γ be a path from -1 to 1 in the closed upper half plane with 0 deleted. For the same W_1 and W_2, we define the *G-crossed braiding isomorphism* $\mathcal{R} \colon W_1 \boxtimes_{P(1)} W_2 \to \phi_{g_1}(W_2) \boxtimes_{P(1)} W_1$ or $\mathcal{R} \colon W_1 \boxtimes_{P(1)} W_2 \to \varphi_{g_1}(W_2) \boxtimes_{P(1)} W_1$ by $\mathcal{R} = \mathcal{T}_{[\gamma]} \circ \mathcal{R}_{P(1)}$.

We now give a result on condition (1) in Assumption 3.4.

Theorem 3.8. *Let $\mathcal{C}_1$ be the category of grading-restricted g-twisted V-modules without g-actions for $g \in G$ and $\mathcal{C}_2$ the category of grading-restricted g-twisted V-modules with actions of G. Let $\mathcal{C}$ be either $\mathcal{C}_1$ or $\mathcal{C}_2$, Assume that the following conditions are satisfied:*

(1) *For $g \in G$, there are only finitely many inequivalent irreducible grading-restricted g-twisted V-modules in $\mathcal{C}$.*
(2) *Every object in $\mathcal{C}$ is a direct sum of irreducible objects in $\mathcal{C}$.*
(3) *For $g_1, g_2 \in G$ and g_1-, g_2-, g_1g_2-twisted V-modules W_1, W_2, W_3 in $\mathcal{C}$, the fusion rule $N_{W_1W_2}^{W_3} = \dim \mathcal{V}_{W_1W_2}^{W_3}$ is finite.*

Then for objects W_1 and W_2 in $\mathcal{C}$, $W_1 \boxtimes_{P(z)} W_2$ is in $\mathcal{C}$.

Proof. Let W_1 and W_2 be g_1- and g_2-twisted V-modules, respectively, in $\mathcal{C}$. From the construction of $W_1 \boxtimes_{P(z)} W_2$, it is a sum of objects in $\mathcal{C}$. By condition (2), $W_1 \boxtimes_{P(z)} W_2$ must be a direct sum of irreducible objects in $\mathcal{C}$. But by condition (1), there are only finitely many irreducible grading-restricted $(g_1g_2)^{-1}$-twisted V-modules in $\mathcal{C}$. If $W_1 \boxtimes_{P(z)} W_2$ is an infinite direct sum of irreducible grading-restricted $(g_1g_2)^{-1}$-twisted V-modules in $\mathcal{C}$, at least one irreducible grading-restricted $(g_1g_2)^{-1}$-twisted V-module W_3 has infinitely many copies in this decomposition of $W_1 \boxtimes_{P(z)} W_2$. But then we have infinitely many linearly independent injective V-module maps from W_3 to the $W_1 \boxtimes_{P(z)} W_2$. But by Proposition 3.5, these infinite injective V-module maps give linearly independent twisted intertwining operator of type $\binom{W_3'}{W_1 W_2}$. Thus the fusion rule $N_{W_1W_2}^{W_3}$ is ∞. By condition (3), this is a contradiction. So $W_1 \boxtimes_{P(z)} W_2$ must be a finite direct sum of irreducible grading-restricted $(g_1g_2)^{-1}$-twisted V-modules in $\mathcal{C}$. In particular, it is grading restricted. In the case $\mathcal{C} = \mathcal{C}_2$, the actions of G on irreducible grading-restricted $(g_1g_2)^{-1}$-twisted V-modules in $\mathcal{C}$ and the decompositions of the finitely many irreducible grading-restricted $(g_1g_2)^{-1}$-twisted V-modules as direct sums of eigenspaces of the actions of $(g_1g_2)^{-1}$ give an action of G and a $(g_1g_2)^{-1}$-grading on $W_1 \boxtimes_{P(z)} W_2$ satisfying the $(g_1g_2)^{-1}$-grading condition. □

Corollary 3.9. *Under the assumptions in Theorem* 3.8*, the categories $\mathcal{C}_1$ and $\mathcal{C}_2$ satisfy Assumption* 3.4*.*

Proof. Note that $\mathcal{C}$ is closed under contragredient. By Theorem 3.8, $W_1 \boxtimes_{P(z)} W_2$ is in $\mathcal{C}$. Thus the contragredient of $W_1 \boxtimes_{P(z)} W_2$ is also in $\mathcal{C}$. So condition (1) in Assumption 3.4 holds. Condition (2) clearly holds for grading-restricted twisted V-modules (see the discussions after Assumption 3.4). □

4. Compatibility condition and grading-restriction condition

In this section, we introduce $P(z)$-compatibility condition and $P(z)$-local grading-restriction condition and use these conditions to give another construction of $W_1 \boxtimes_{P(z)} W_2$ for two twisted V-modules W_1 and W_2. As in Section 4, we take $\mathcal{C}$ to be either $\mathcal{C}_1$ or $\mathcal{C}_2$ satisfying Assumption 3.4. In the untwisted case (the case that $\mathcal{C}$ is the category of (untwisted or 1_V-twisted) V-modules), these conditions and this second construction given in [24, 25, 28] play a crucial role in the proof of the associativity of intertwining operators and the construction of associativity isomorphisms in [9, 30]. It is expected that they will play the same crucial role in the proof of the associativity of twisted intertwining operators and the construction of associativity isomorphisms for the $P(z)$ tensor product bifunctors on the category $\mathcal{C}$ of twisted V-modules.

In the untwisted case, The $P(z)$-compatibility condition is formulated using a formula corresponding to the Jacobi identity for intertwining operators. Even though we can obtain a Jacobi identity for the rational coefficients of the expansions in a suitable basis of products and iterates of twisted intertwining operators with twisted vertex operators as in [10], we do not have a Jacobi identity for the products and iterates of twisted intertwining operators with twisted vertex operators.

Thus we have to use the analytic method to formulate the $P(z)$-compatibility condition and prove the main results. In particular, the formulation and proofs involving the $P(z)$-compatibility condition are completely different from those in [24, 25, 28].

For a fixed $z \in \mathbb{C}_\times$, we need to study multivalued analytic functions on the region

$$M^n(0,z) = \left\{(z_1,\dots,z_n) \in \mathbb{C}^n \;\middle|\; \begin{array}{l} z_i \neq 0,\ z_i \neq z,\ i = 1,\dots,n, \\ z_i \neq z_j \text{ for } i,j = 1,\dots,n, \text{ and } i \neq j \text{if } n > 1 \end{array}\right\}$$

for $n \in \mathbb{Z}_+$ and its subregions

$$\Omega_{m,k,l}(z) = \left\{(z_1,\dots,z_{m+k+l}) \in \mathbb{C}^{m+k+l} \;\middle|\; \begin{array}{l} |z| < |z_m| < \cdots < |z_1| \text{ if } m > 0, \\ 0 < |z_{m+k} - z| < \cdots < |z_{m+1} - z| < |z| \text{ if } k > 0, \\ 0 < |z_{m+k+l}| < \cdots < |z_{m+k+1}| < |z| \text{ if } l > 0, \\ |z_{m+1} - z| + |z_{m+k+1}| < |z| \text{ if } k > 0,\ l > 0, \\ |z_{m+1} - z| + |z_1| < |z| \text{ if } m > 0,\ k > 0 \end{array}\right\},$$

$$\Omega^{(1)}_{m,k,l}(z) = \left\{(z_1,\dots,z_{m+k+l}) \in \Omega_{m,k,l}(z) \;\middle|\; \begin{array}{l} |\arg(z_j - z) - \arg(z_j)| < \dfrac{\pi}{2}, \\ \qquad j = 1,\dots,m, \text{if } m > 0, \\ |\arg(z_j) - \arg(z)| < \dfrac{\pi}{2}, \\ \qquad j = m+1,\dots,m+k \text{ if } k > 0, \\ -\dfrac{3\pi}{2} < \arg(z_j - z) - \arg(z) < -\dfrac{\pi}{2}, \\ \qquad j = m+k+1,\dots,m+k+l \text{ if } l > 0 \end{array}\right\},$$

$$\Omega^{(2)}_{m,k,l}(z) = \left\{(z_1,\dots,z_{m+k+l}) \in \Omega_{m,k,l}(z) \;\middle|\; \begin{array}{l} |\arg(z_j - z) - \arg(z_j)| < \dfrac{\pi}{2}, \\ \qquad j = 1,\dots,m \text{ if } m > 0, \\ |\arg(z_j) - \arg(z)| < \dfrac{\pi}{2}, \\ \qquad j = m+1,\dots,m+k \text{ if } k > 0, \\ \dfrac{\pi}{2} < \arg(z_j - z) - \arg(z) < \dfrac{3\pi}{2}, \\ \qquad j = m+k+1,\dots,m+k+l \text{ if } l > 0 \end{array}\right\}$$

for $m,k,l \in \mathbb{N}$. Also, we define $M^n_0(0,z)$ to be the simply-connected region given by cutting $M^n(0,z)$ along the positive real lines in the z_i-planes, $z_i - z_j$-planes and $z_i - z$-planes, that is, the sets

$$\begin{aligned} &\{(z_1,\dots,z_n) \in M^n(0,z) \mid z_i \in \mathbb{R}_+\}, \qquad i = 1,\dots,n, \\ &\{(z_1,\dots,z_n) \in M^n(0,z) \mid z_i - z_j \in \mathbb{R}_+\}, \qquad i,j = 1,\dots,n, i \neq j, \\ &\{(z_1,\dots,z_n) \in M^n(0,z) \mid z_i - z \in \mathbb{R}_+\}, \qquad i = 1,\dots,n, \end{aligned}$$

with these sets attached to the upper half z_i-planes, $z_i - z_j$-planes and $z_i - z$-planes.

To formulate the $P(z)$-compatibility condition, we need the notion of regular component-isolated singularity of a multivalued analytic function of several variables in Section 1. Let $b = (b_1,\dots,b_n) \in (\mathbb{C} \cup \{\infty\})^n$ and $r = (r_1,\dots,r_n) \in \mathbb{R}^n_+$. Let $I \subset \{1,\dots,n\}$. We use the following notation for polydisks and polycircular

domains

$$\Delta^I(b,r) = \left\{ (z_1,\dots,z_n) \in (\mathbb{C}\cup\{\infty\})^n \;\middle|\; \begin{array}{l} |z_i| > r_i \text{ or } z_i = \infty \text{ if } b_i = \infty, \\ \qquad |z_i - b_i| < r_i \text{ if } b_i \in \mathbb{C}, \text{ for } i \in I, \\ |z_j| > r_j \text{ if } b_j = \infty, \\ \qquad 0 < |z_j - b_j| < r_j \text{ if } b_j \in \mathbb{C} \text{ for } j \notin I \end{array} \right\},$$

$$\Delta(b,r) = \Delta^{\{1,\dots,n\}}(b,r), \qquad \Delta^\times(b,r) = \Delta^\varnothing(b,r).$$

We shall use $b_{z_i,z}$ to denote the homotopy class of loops in $M^l(0,z)$ with z_i going around z counterclockwise once with z_j for $j \neq i$ fixed.

Let g_1 and g_2 be automorphisms of V, W_1, W_2, W_3, g_1-, g_2-, g_1g_2-twisted V-modules, respectively, in the category $\mathcal{C}$ and $z \in \mathbb{C}^\times$. Let I be a twisted $P(z)$-intertwining map of type $\binom{W_3}{W_1 W_2}$ and $w_3' \in W_3'$. Then we have an element $\lambda_{I,w_3'} \in W_1 \boxtimes_{P(z)} W_2$.

Proposition 4.1. *The element $\lambda_{I,w_3'}$ has the following property: For $l \in \mathbb{N}$, $u_1,\dots,u_l \in V$, $w_1 \in W_1$, and $w_2 \in W_2$, there exists a multivalued analytic function*

$$f_l(z_1,\dots,z_l;u_1,\dots,u_l,w_1,w_2;\lambda_{I,w_3'}) \tag{4.1}$$

on $M^l(0,z)$ with a preferred branch

$$f_l^e(z_1,\dots,z_l;u_1,\dots,u_l,w_1,w_2;\lambda_{I,w_3'}) \tag{4.2}$$

on $M_0^l(0,z)$, satisfying the following:

(1) (*a*) *For $i,j = 1,\dots,l$, $i \neq j$, $z_i - z_j = 0$ are poles of (4.1). In particular, there exist $M_{ij} \in \mathbb{Z}_+$ such that*

$$\left(\prod_{1\le i<j\le l} (z_i - z_j)^{M_{ij}}\right) f_l(z_1,\dots,z_l;u_1,\dots,u_l,w_1,w_2;\lambda_{I,w_3'}) \tag{4.3}$$

can be analytically extended to a multivalued analytic function on

$$\{(z_1,\dots,z_l) \in \mathbb{C}^l \mid z_i \neq 0,\ z_i \neq z,\ i = 1,\dots,l\}$$

(*b*) *Any component-isolated singularity of the form $(z_1,\dots,z_l) - \beta = \delta$ for $\beta \in \mathbb{C}^{l-1}$ and $\delta \in \{0,\infty\}^{l-1}$ of (4.3) is regular.*

(2) *For $u_1,\dots,u_l \in V$, $w_1 \in W_1$, and $w_2 \in W_2$,*

(*a*) *The series*

$$\begin{aligned} &\lambda_{I,w_3'}(Y^{g_1}(u_1,z_1 - z)w_1 \otimes w_2) \\ &= \lambda_{I,w_3'}(Y^{g_1}(u_1,x)w_1 \otimes w_2)\big|_{x^n = \mathrm{e}^{nl_0(z_1-z)},\log x = l_0(z_1-z)} \end{aligned}$$

is absolutely convergent on the region $\Omega_{0,1,0}(z)$. Moreover, it is convergent to $f_1^e(z_1;u_1,w_1,w_2;\lambda_{I,w_3'})$ on the region $\Omega_{0,1,0}^{(1)}(z) = \Omega_{0,1,0}^{(2)}(z)$.

(*b*) *For $l \in \mathbb{N}$, the multiple series*

$$\begin{aligned} &\lambda_{I,w_3'}(w_1 \otimes Y^{g_2}(u_1,z_1)\cdots Y^{g_2}(u_l,z_l)w_2) \\ &= \lambda_{I,w_3'}(w_1 \otimes Y^{g_2}(u_1,x_1)\cdots Y^{g_2}(u_l,x_l)w_2)|_{x_i^n = \mathrm{e}^{nl_0(z_1)},\log x_i = l_0(z_1),i=1,\dots,l} \end{aligned}$$

in powers and logarithms of $z_1,\dots,z_l$ is absolutely convergent on the region $\Omega_{0,0,l}(z)$. Moreover, it is absolutely convergent to (4.2) on the region $\Omega_{0,0,l}^{(1)}(z)$ and to

$$f_l^{b_{z_1,z}^{-1}\cdots b_{z_l,z}^{-1}}(z_1,\dots,z_l;u_1,\dots,u_l,w_1,w_2;\lambda)$$

on the region $\Omega_{0,0,l}^{(2)}(z)$.

Proof. This result can be easily verified by using the definitions of $\lambda_{I,w_3'}$ and $P(z)$-intertwining maps and the properties of twisted intertwining operators. We omit the details. But see Remarks 4.2, 4.3, and 4.4 for some subtle points. □

Let g_1 and g_2 be automorphisms of V, W_1, W_2 g_1-, g_2-twisted V-modules, respectively, in the category $\mathcal{C}$ and $z \in \mathbb{C}^\times$. Motivated by Proposition 4.1, we formulate the following condition for $\lambda \in (W_1 \otimes W_2)^*$:

$P(z)$-compatibility condition.: An element $\lambda \in (W_1 \otimes W_2)^*$ is said to be $P(z)$-*compatible* if for $l \in \mathbb{N}$, $u_1, \dots, u_l \in V$, $w_1 \in W_1$, and $w_2 \in W_2$, there exists a multivalued analytic function

$$f_l(z_1, \dots, z_l; u_1, \dots, u_l, w_1, w_2; \lambda) \tag{4.4}$$

on $M^l(0, z)$ with a preferred branch

$$f_l^e(z_1, \dots, z_l; u_1, \dots, u_l, w_1, w_2; \lambda) \tag{4.5}$$

on $M_0^l(0, z)$, satisfying the following:

(1) (a) For $i, j = 1, \dots, l$, $i \neq j$, $z_1 - z_j = 0$ are poles of (4.4). In particular, there exists $M_{ij} \in \mathbb{Z}_+$ such that

$$\prod_{1 \leq i < j \leq n} (z_i - z_j)^{M_{ij}} f_l(z_1, \dots, z_n; u_1, \dots, u_n, w_1, w_2; \lambda) \tag{4.6}$$

can be analytically extended to a multivalued analytic function on

$$\big\{(z_1, \dots, z_l) \in \mathbb{C}^l \mid z_i \neq 0,\ z_i \neq z,\ i = 1, \dots, l\big\}.$$

(b) Any component-isolated singularity of the form $(z_1, \dots, z_l) - \beta = \delta$ for $\beta \in \mathbb{C}^{l-1}$ and $\delta \in \{0, \infty\}^{l-1}$ of (4.6) is regular.

(2) For $u_1, \dots, u_l \in V$, $w_1 \in W_1$, and $w_2 \in W_2$,

(a) The series

$$\lambda(Y^{g_1}(u_1, z_1 - z)w_1 \otimes w_2)$$
$$= \lambda(Y^{g_1}(u_1, x)w_1 \otimes w_2)|_{x^n = \mathrm{e}^{nl_0(z_1 - z)}, \log x = l_0(z_1 - z)}$$

is absolutely convergent on the region $\Omega_{0,1,0}(z)$. Moreover, it is convergent to $f_1^e(z_1; u_1, w_1, w_2; \lambda)$ on the region $\Omega^{(1)}_{0,1,0}(z) = \Omega^{(2)}_{0,1,0}(z)$.

(b) For $l \in \mathbb{N}$, the multiple series

$$\lambda(w_1 \otimes Y^{g_2}(u_1, z_1) \cdots Y^{g_2}(u_l, z_l)w_2) \tag{4.7}$$
$$= \lambda(w_1 \otimes Y^{g_2}(u_1, x_1) \cdots Y^{g_2}(u_l, x_l)w_2)|_{x_i^n = \mathrm{e}^{nl_0(z_1)}, \log x_i = l_0(z_1), i = 1, \dots, l}$$

in powers and logarithms of $z_1, \dots, z_l$ is absolutely convergent on the region $\Omega_{0,0,l}(z)$. Moreover, it is absolutely convergent to (4.5) on the region $\Omega^{(1)}_{0,0,l}(z)$ and to

$$f_l^{b^{-1}_{z_1,z} \cdots b^{-1}_{z_l,z}}(z_1, \dots, z_l; u_1, \dots, u_l, w_1, w_2; \lambda)$$

on the region $\Omega^{(2)}_{0,0,l}(z)$.

We denote the subspace of $P(z)$-compatible functionals in $(W_1 \otimes W_2)^*$ as

$$\mathrm{COMP}_{P(z)}((W_1 \otimes W_2)^*)$$

or COMP for short.

Remark 4.2. Note that by 2 (b) of Proposition 4.1 or 2 (b) of the $P(z)$-compatibility condition and the associativity of twisted vertex operators, M_{ij} in Proposition 4.1 or in the $P(z)$-compatibility condition depend only on u_i and u_j.

Remark 4.3. In 2 (b) of Proposition 4.1 and 2 (b) of the $P(z)$-compatibility condition, the reason that we involve two different regions $\Omega^{(1)}_{0,0,l}(z)$ and $\Omega^{(2)}_{0,0,l}(z)$ is because either of these two regions could be empty. Notice that here z is a *fixed* nonzero complex number. Actually, when $\arg z \in [0, \pi/2]$, the region $\Omega^{(1)}_{0,0,l}(z)$ is empty. When $\arg z \in [3\pi/2, 2\pi)$, the region $\Omega^{(2)}_{0,0,l}(z)$ is empty. When $\Omega^{(1)}_{0,0,l}(z)$ and $\Omega^{(2)}_{0,0,l}(z)$ are both nonempty, the absolute convergences of (4.7) on these two regions are equivalent.

Remark 4.4. Because of the definition of the domain of (4.5), i.e., $M_0^l(0, z)$, and the fact that its singularities at $z_i = z_j$ for $i \neq j$ are poles, branches of (4.4) can be indexed by elements in the fundamental group of the space

$$\{(z_1, \dots, z_n) \in \mathbb{C}^n \mid z_i \neq 0,\ z_i \neq z,\ i = 1, \dots, n\} = \prod_{i=1}^{n} \{z_i \in \mathbb{C} \mid z_i \neq 0,\ z_i \neq z\}.$$

A set of generators of this fundamental group can be chosen to be $b_{z_i,0}$, $b_{z_i,z}$, $i = 1, \dots, n$. For each i, the elements $b_{z_i,0}$ and $b_{z_i,z}$ correspond to b_{13} and b_{12} defined in Section 1, and they freely generate $\pi_1(\{z_i \in \mathbb{C} \mid z_i \neq 0, z_i \neq z\})$. Notice that

$$\pi_1\left(\prod_{i=1}^{n} \{z_i \in \mathbb{C} \mid z_i \neq 0,\ z_i \neq z\}\right) = \prod_{i=1}^{n} \pi_1\big(\{z_i \in \mathbb{C} \mid z_i \neq 0,\ z_i \neq z\}\big) = \prod_{i=1}^{n} \langle b_{i,0}, b_{i,z} \rangle.$$

For $\lambda \in \mathrm{COMP}$, we want to define $Y_{P(z)}^{(g_1 g_2)^{-1}}(u, x)\lambda \in (W_1 \otimes W_2)^*\{x\}[\log x]$. We first define $Y_{P(z)}^{(g_1 g_2)^{-1}}(u, x)\lambda \in (W_1 \otimes W_2)^*\{x\}[\log x]$ for λ in a larger subspace of $(W_1 \otimes W_2)^*$ than COMP. Let $\mathrm{COM}_{P(z)}((W_1 \otimes W_2)^*)$ or simply COM be the subspace of $(W_1 \otimes W_2)^*$ consisting of λ satisfying 1 (a), (b) and 2 (b) in the $P(z)$-compatibility condition. By definition, COMP $\subset$ COM. To define $Y_{P(z)}^{(g_1 g_2)^{-1}}(u, x)\lambda \in (W_1 \otimes W_2)^*\{x\}[\log x]$ for $u \in V$ and $\lambda \in \mathrm{COM}$ is equivalent to define

$$Y_{P(z)}^{(g_1 g_2)^{-1}}\big(\mathrm{e}^{xL(1)}\big(-x^2\big)^{-L(0)}u, x^{-1}\big)\lambda = \big(Y_{P(z)}^{(g_1 g_2)^{-1}}\big)^o(u, x)\lambda \in (W_1 \otimes W_2)^*\{x\}[\log x].$$

As $z_1 = \infty$ is a regular singular point of $f_1(z_1; u_1, w_1, w_2; \lambda)$, we know that there exist unique $a_{i,n,j}(u_1, w_1, w_2; \lambda) \in \mathbb{C}$ and $r_i \in \mathbb{C}$, for $i, j = 0, \dots, K$ and $n \in \mathbb{N}$, such that on $\Omega^{(1)}_{1,0,0}(z) = \Omega^{(2)}_{1,0,0}(z)$ (i.e., the region given by $|z_1| > |z|$, $|\arg(z_1 - z) - \arg(z_1)| < \frac{\pi}{2}$),

$$f_1^e(z_1; u, w_1, w_2; \lambda) = \sum_{i,j=0}^{K} \sum_{n \in \mathbb{N}} a_{i,n,j}(u, w_1, w_2; \lambda) z_1^{r_i - n} (\log z_1)^j.$$

For $i, j = 0, \dots, K$, $n \in \mathbb{N}$ and $u \in V$, we define $\big(Y_{P(z)}^{(g_1 g_2)^{-1}}\big)^o_{-r_i+n-1,j}(u)\lambda \in (W_1 \otimes W_2)^*$ by

$$\big(\big(Y_{P(z)}^{(g_1 g_2)^{-1}}\big)^o_{-r_i+n-1,k}(u)\lambda\big)(w_1 \otimes w_2) = a_{i,n,j}(u, w_1, w_2; \lambda)$$

for $w_1 \in W_1$ and $w_2 \in W_2$. Then we define $Y_{P(z)}^{(g_1 g_2)^{-1}}\big(\mathrm{e}^{xL(1)}\big(-x^2\big)^{L(0)}u, x^{-1}\big)\lambda$ to be

$$\sum_{i,j=0}^{N} \sum_{n \in \mathbb{N}} \big(Y_{P(z)}^{(g_1 g_2)^{-1}}\big)^o_{-r_i+n-1,j}(u)\lambda x^{r_i - n} (\log x)^j \in (W_1 \otimes W_2)^*\{x\}[\log x],$$

that is,

$$\begin{aligned} &\big(Y_{P(z)}^{(g_1 g_2)^{-1}}\big(\mathrm{e}^{xL(1)}\big(-x^2\big)^{-L(0)}u, x^{-1}\big)\lambda\big)(w_1 \otimes w_2) \\ &\qquad = \sum_{i,j=0}^{N} \sum_{n \in \mathbb{N}} a_{i,n,j}(u, w_1, w_2; \lambda) x^{r_i - n} (\log x)^j. \end{aligned} \tag{4.8}$$

By definition, on the region $|z_1| > |z|$ $\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u,z_1)\lambda\big)(w_1\otimes w_2)$ is absolutely convergent and its sum on $\Omega^{(1)}_{1,0,0}(z) = \Omega^{(2)}_{1,0,0}(z)$ is equal to $f_1^e(z_1;u,w_1,w_2;\lambda)$. For simplicity, let

$$\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o_{m,k}(u) = 0$$

for $m \in \mathbb{C}$, $m \neq -r_i + n - 1$ for $i = 0,\dots,N$ and $n \in \mathbb{N}$ and $k = 0,\dots,N$. Then we have

$$\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u,x) = \sum_{m\in\mathbb{C}}\sum_{k=0}^{N}\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o_{m,k}(u)x^{-m-1}(\log x)^k.$$

We have the following result.

Proposition 4.5. *The space* COM *is invariant under the action of the components of the twisted vertex operators* $Y_{P(z)}^{(g_1g_2)^{-1}}(u,x)$ *for* $u \in V$.

Proof. We need to show that for $n_1 \in \mathbb{C}$, $k = 1,\dots,K$, $u_1 \in V$ and $\lambda \in \mathrm{COM}$,

$$\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o_{n_1,k_1}(u_1)\lambda \in \mathrm{COM}\,. \tag{4.9}$$

For $u_2,\dots,u_l \in V$, we have

$$\begin{aligned} Y^{g_2}(u_2,x_2)\cdots Y^{g_2}(u_l,x_l) = &\sum_{n_2,\dots,n_l\in\mathbb{C}}\sum_{k_2,\dots,k_l=0}^{K} Y^{g_2}_{n_2,k_2}(u_2)\cdots Y^{g_2}_{n_l,k_l}(u_l)\\ &\times x_2^{-n_2-1}\cdots x_{l+1}^{-n_l-1}(\log x_2)^{k_2}\cdots(\log x_l)^{k_l}. \end{aligned}$$

Since $\lambda \in \mathrm{COM}$, for $u_2,\dots,u_l \in V$, $n_2,\dots,n_l \in \mathbb{C}$, $k_2,\dots,k_l = 0,\dots,K$, $w_1 \in W_1$ and $w_2 \in W_2$,

$$\lambda\big(w_1 \otimes Y^{g_2}(u_1,z_1)Y^{g_2}_{n_2,k_2}(u_2)\cdots Y^{g_2}_{n_l,k_l}(u_l)w_2\big)$$

is absolutely convergent on the region $\Omega_{0,0,1}(z)$ and is absolutely convergent on the region $\Omega^{(1)}_{0,0,1}(z)$ and $\Omega^{(2)}_{0,0,1}(z)$ to the preferred single-valued branches

$$f_1^e\big(z_1;u_1,w_1,Y^{g_2}_{n_2,k_2}(u_2)\cdots Y^{g_2}_{n_l,k_l}(u_l)w_2\big), \qquad f_1^{b^{-1}_{z_1,z}}\big(z_1;u_1,w_1,Y^{g_2}_{n_2,k_2}(u_2)\cdots Y^{g_2}_{n_l,k_l}(u_l)w_2\big),$$

respectively, on $M_0^1(0,z)$ of the multivalued analytic function

$$f_1\big(z_1;u_1,w_1,Y^{g_2}_{n_2,k_2}(u_2)\cdots Y^{g_2}_{n_l,k_l}(u_l)w_2\big) \tag{4.10}$$

defined on $M^1(0,z)$. We discuss only the case that $\Omega^{(1)}_{0,0,1}(z)$ is nonempty (see Remark 4.3). By definition, $\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o_{-r_i+n-1,j}(u)\lambda$ is obtained by expanding (4.10) on the region $\Omega^{(1)}_{1,0,0}(z) = \Omega^{(2)}_{1,0,0}(z)$ as a series in suitable powers of z_1 and $\log z_1$ and then taking the corresponding coefficients.

We now consider the series

$$f_1^e(z_1;u_1,w_1,Y^{g_2}(u_2,z_2)\cdots Y^{g_2}(u_l,z_l)w_2) \tag{4.11}$$

in suitable powers of $z_2,\dots,z_l$ and $\log z_1,\dots,\log z_l$. The series (4.11) on the region $\Omega^{(1)}_{0,0,1}(z)$ can be further expanded as the series

$$\lambda(w_1 \otimes Y^{g_2}(u_1,z_1)Y^{g_2}(u_2,z_2)\cdots Y^{g_2}(u_l,z_l)w_2). \tag{4.12}$$

But (4.12) is absolutely convergent on the region $\Omega_{0,0,l}(z)$ and is absolutely convergent either to

$$f_l^e(z_1,z_2,\dots,z_l;u_1,\dots,u_l,w_1,w_2;\lambda) \tag{4.13}$$

on the region $\Omega^{(1)}_{0,0,l}(z)$ or to

$$f_l^{b^{-1}_{z_1,z}\cdots b^{-1}_{b_l,z}}(z_1,z_2,\dots,z_l;u_1,\dots,u_l,w_1,w_2;\lambda) \tag{4.14}$$

on the region $\Omega^{(2)}_{0,0,l}(z)$. Thus for z_1 satisfying $|z| > |z_1| > 0$, the series (4.11) as the sum of (4.12) viewed as a series in suitable powers of z_1 and $\log z_1$ must be absolutely convergent on the region $|z_1| > |z_2| > \cdots > |z_l| > 0$ and if in addition, $-\frac{3\pi}{2} < \arg(z_i - z) < -\frac{\pi}{2}$ or $\frac{\pi}{2} < \arg(z_i - z) < \frac{3\pi}{2}$ for $i = 1, \dots, l+1$, its sum must also be equal to (4.13) or (4.14), respectively. But (4.13) can also be expanded on the region $|z_1| > |z| > |z_2| > \cdots > |z_l| > 0$ as a series in suitable powers of $z_1, \dots, z_l$ and $\log z_1, \dots, \log z_l$. This fact can be seen as follows: By 1 (a) in the $P(z)$-compatibility condition, we know that $\prod_{1 \le i < j \le n}(z_i - z_j)^{M_{ij}} f_l(z_1, z_2, \dots, z_n; u_1, \dots, u_n, w_1, w_2; \lambda)$ can be analytically extended to an analytic function on the region

$$\big\{(z_1, \dots, z_l) \in \mathbb{C}^l \mid z_i \neq 0,\ z_i \neq z,\ i = 1, \dots, l\big\}.$$

Moreover, this analytic function has a regular singularity at $(\infty, 0, \dots, 0)$. In particular, this function can be analytically expanded on the region $|z| < |z_1|$, $0 < |z_i| < |z|$ for $i = 2, \dots, l$ as a series in suitable powers of $z_1, \dots, z_l$ and $\log z_1, \dots, \log z_l$. Thus (4.13) can be expanded on the region $|z_1| > |z| > |z_2| > \cdots > |z_l| > 0$ as a series in suitable powers of $z_1, \dots, z_l$ and $\log z_1, \dots, \log z_l$. By definition, this expansion is equal to

$$\big(\big(Y_{P(z)}^{(g_1 g_2)^{-1}}\big)^o(u_1, z_1)\lambda\big)(w_1 \otimes Y^{g_2}(u_2, z_2) \cdots Y^{g_2}(u_l, z_l)w_2). \tag{4.15}$$

The coefficients of the powers of z_1 and $\log z_1$ in (4.15) are exactly

$$\big(\big(Y_{P(z)}^{(g_1 g_2)^{-1}}\big)^o_{-r_i+n-1,j}(u_1)\lambda\big)(w_1 \otimes Y^{g_2}(u_2, z_2) \cdots Y^{g_2}(u_l, z_l)w_2)$$

for $i, j = 0, \dots, N$, $n \in \mathbb{N}$. Since these are expansions of single-valued branches of multivalued analytic functions on $\Omega_{0,0,l}$, they are absolutely convergent to these multivalued analytic functions on the same region and are absolutely convergent to their corresponding single-valued branches on $\Omega^{(1)}_{0,0,l}$ or $\Omega^{(2)}_{0,0,l}$. Moreover, since $f_l(z_1, \dots, z_l; u_1, \dots, u_l, w_1, w_2; \lambda)$ satisfies 1 (a) and 1(b) in $P(z)$-compatibility condition, the coefficients of its expansion on the region $|z_1| > |z| > |z_2| > \cdots > |z_l| > 0$ also satisfy these conditions. This finishes the proof of (4.9). □

Proposition 4.6. *For $u_1, \dots, u_{m+l} \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $\lambda \in \mathrm{COM}$, the series*

$$\begin{aligned}&\big(\big(Y_{P(z)}^{(g_1 g_2)^{-1}}\big)^o(u_m, z_m) \cdots \big(Y_{P(z)}^{(g_1 g_2)^{-1}}\big)^o(u_1, z_1)\lambda\big)\\ &\quad\times\big(w_1 \otimes Y^{g_2}(u_{m+1}, z_{m+1}) \cdots Y^{g_2}(u_{m+l}, z_{m+l})w_2\big)\end{aligned} \tag{4.16}$$

is absolutely convergent on the region $|z_1| > \cdots > |z_m| > |z| > |z_{m+1}| > \cdots > |z_{l+m}| > 0$ and its sum is equal to

$$f^e_{m+l}(z_1, \dots, z_{m+l}; u_1, \dots, u_{m+l}; w_1, w_2; \lambda) \tag{4.17}$$

on $\Omega^{(1)}_{m,0,l}(z)$ or to

$$f^{b^{-1}_{z_{m+1},z} \cdots b^{-1}_{z_{m+l},z}}_{m+l}(z_1, \dots, z_{m+l}; u_1, \dots, u_{m+l}; w_1, w_2; \lambda) \tag{4.18}$$

on $\Omega^{(2)}_{m,0,l}(z)$. Moreover, we have the following commutativity for $Y_{P(z)}^{(g_1 g_2)^{-1}}$: For $u_1, \dots, u_m \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $\lambda \in \mathrm{COM}$, the series

$$\big(Y_{P(z)}^{(g_1 g_2)^{-1}}(u_1, z_1) \cdots Y_{P(z)}^{(g_1 g_2)^{-1}}(u_m, z_m)\lambda\big)(w_1 \otimes w_2) \tag{4.19}$$

is absolutely convergent on the region $|z^{-1}| > |z_1| > \cdots > |z_m| > 0$ and for $\sigma \in S_m$, the sums of (4.19) *and*

$$\big(Y_{P(z)}^{(g_1 g_2)^{-1}}(u_{\sigma(1)}, z_{\sigma(1)}) \cdots Y_{P(z)}^{(g_1 g_2)^{-1}}(u_{\sigma(m)}, z_{\sigma(l)})\lambda\big)(w_1 \otimes w_2) \tag{4.20}$$

are analytic extensions of each other. We also have the weak commutativity for $Y_{P(z)}^{(g_1g_2)^{-1}}$*: For* $u, v \in V$*, there exists* $M \in \mathbb{Z}_+$ *depending only on* u *and* v *such that*

$$\begin{aligned}&(x_1-x_2)^M Y_{P(z)}^{(g_1g_2)^{-1}}(u,x_1)Y_{P(z)}^{(g_1g_2)^{-1}}(v,x_2)\\ &=(x_1-x_2)^M Y_{P(z)}^{(g_1g_2)^{-1}}(v,x_2)Y_{P(z)}^{(g_1g_2)^{-1}}(u,x_1).\end{aligned}\tag{4.21}$$

Proof. To prove the convergence of (4.16), we use induction on m. In the case $m = 0$, the convergence of (4.16) is given by condition (2) (b). Assume that the convergence of (4.16) in the case that m is $m-1$ holds. Then

$$\begin{aligned}&\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u_{m-1},z_{m-1})\cdots\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u_1,z_1)\lambda\big)\\ &\quad\times\big(w_1\otimes Y^{g_2}(u_m,z_m)\cdots Y^{g_2}(u_{m+l},z_{m+l})w_2\big)\\ &=\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u_{m-1},x_{m-1})\cdots\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u_1,x_1)\lambda\big)\\ &\quad\times\big(w_1\otimes Y^{g_2}(u_m,x_m)\cdots Y^{g_2}(u_{m+l},x_{m+l})w_2\big)\big|_{x_i^n=\mathrm{e}^{n\log z_i},\log x_i=\log z_i,i=1,\dots,m+l}\end{aligned}\tag{4.22}$$

is absolutely convergent on the region $|z_1| > \cdots > |z_{m-1}| > |z| > |z_m| > \cdots > |z_{m+l}| > 0$ and its sum is equal to (4.17) on $\Omega^{(1)}_{m-1,0,l+1}(z)$ or to

$$f_l^{b^{-1}_{z_m,z}\cdots b^{-1}_{z_{m+l},z}}(z_1,\dots,z_{m+l};u_1,\dots,u_{m+l};w_1,w_2;\lambda)\tag{4.23}$$

on $\Omega^{(2)}_{m-1,0,l+1}(z)$.

Write

$$\begin{aligned}&\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u_{m-1},x_{m-1})\cdots\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u_1,x_1)\\ &=\sum_{n_{m-1},\dots,n_1\in\mathbb{C}}\sum_{k_{m-1},\dots,k_1=0}^{K}\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o_{n_{m-1},k_{m-1}}(u_{m-1})\cdots\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o_{n_1,k_1}(u_1)\\ &\qquad\times x_{m-1}^{-n_{m-1}-1}\cdots x_1^{-n_1-1}(\log x_{m-1})^{k_{m-1}}\cdots(\log x_1)^{k_1}\end{aligned}$$

and

$$\begin{aligned}&Y^{g_2}(u_{m+1},x_{m+1})\cdots Y^{g_2}(u_{m+l},x_{m+l})\\ &=\sum_{n_{m+1},\dots,n_{m+l}\in\mathbb{C}}\sum_{k_{m+1},\dots,k_{m+l}=0}^{K}Y^{g_2}_{n_{m+1},k_{m+1}}(u_{m+1})\cdots Y^{g_2}_{n_{m+l},k_{m+l}}(u_{m+l})\\ &\qquad\times x_2^{-n_{m+1}-1}\cdots x_{m+l}^{-n_{m+l}-1}(\log x_{m+1})^{k_{m+1}}\cdots(\log x_{m+l})^{k_{m+l}}.\end{aligned}$$

By Proposition (4.5),

$$\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o_{n_{m-1},k_{m-1}}(u_{m-1})\cdots\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o_{n_1,k_1}(u_1)\lambda\in\mathrm{COM}.$$

Then

$$\begin{aligned}&\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o_{n_{m-1},k_{m-1}}(u_{m-1})\cdots\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o_{n_1,k_1}(u_1)\lambda\big)\\ &\times\big(w_1\otimes Y^{g_2}(u_m,z_m)Y^{g_2}_{n_{m+1},k_{m+1}}(u_m)\cdots Y^{g_2}_{n_{m+l},k_{m+l}}(u_{m+l})w_2\big)\end{aligned}$$

is absolutely convergent on the region $|z| > |z_m| > 0$ and its sum is equal to

$$\begin{aligned}&f^e\big(z_m;u_m,w_1,Y^{g_2}_{n_{m+1},k_{m+1}}(u_m)\cdots Y^{g_2}_{n_{m+l},k_{m+l}}(u_{m+l})w_2;\\ &\qquad\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o_{n_{m-1},k_{m-1}}(u_{m-1})\cdots\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o_{n_1,k_1}(u_1)\lambda\big)\end{aligned}\tag{4.24}$$

on the region $|z| > |z_m| > 0$, $-\frac{3\pi}{2} < \arg(z_m - z) - \arg z < -\frac{\pi}{2}$ and to

$$f^{b_{z_m,z}^{-1}}\big(z_m; u_m, w_1, Y^{g_2}_{n_{m+1},k_{m+1}}(u_m)\cdots Y^{g_2}_{n_{m+l},k_{m+l}}(u_{m+l})w_2;$$
$$\big(Y^{(g_1g_2)^{-1}}_{P(z)}\big)^o_{n_{m-1},k_{m-1}}(u_{m-1})\cdots\big(Y^{(g_1g_2)^{-1}}_{P(z)}\big)^o_{n_1,k_1}(u_1)\lambda\big)$$

on the region $|z| > |z_m| > 0$, $\frac{\pi}{2} < \arg(z_m - z) - \arg z < \frac{3\pi}{2}$. By definition,

$$\big(\big(Y^{(g_1g_2)^{-1}}_{P(z)}\big)^o(u_m, z_m)\big(Y^{(g_1g_2)^{-1}}_{P(z)}\big)^o_{n_{m-1},k_{m-1}}(u_{m-1})\cdots\big(Y^{(g_1g_2)^{-1}}_{P(z)}\big)^o_{n_1,k_1}(u_1)\lambda\big)$$
$$\times\big(w_1 \otimes Y^{g_2}_{n_{m+1},k_{m+1}}(u_m)\cdots Y^{g_2}_{n_{m+l},k_{m+l}}(u_{m+l})w_2\big) \tag{4.25}$$

is absolutely convergent on the region $|z_m| > |z|$ and its sum is equal to (4.24) on the region $|z_m| > |z|$, $|\arg(z_m - z) - \arg z_m| < \frac{\pi}{2}$.

But we know that

$$\sum_{n_1,\dots,n_{m-1},n_{m+1},\dots,n_{m+l}\in\mathbb{C}}\ \sum_{k_1,\dots,k_{m-1},k_{m+1},\dots,k_{m+l}=0}^{K} f^e\big(z_m; u_m, w_1, Y^{g_2}_{n_{m+1},k_{m+1}}(u_m)\cdots$$
$$\times Y^{g_2}_{n_{m+l},k_{m+l}}(u_{m+l})w_2; \big(Y^{(g_1g_2)^{-1}}_{P(z)}\big)^o_{n_{m-1},k_{m-1}}(u_{m-1})\cdots\big(Y^{(g_1g_2)^{-1}}_{P(z)}\big)^o_{n_1,k_1}(u_1)\lambda\big)$$
$$\times z_1^{-n_1-1}\cdots z_{m-1}^{-n_{m-1}-1}z_{m+1}^{-n_1-1}\cdots z_{m+l}^{-n_{m+l}-1}$$
$$\times(\log z_1)^{k_1}\cdots(\log z_{m-1})^{k_{m-1}}(\log z_{m+1})^{k_{m+1}}\cdots(\log z_{m+l})^{k_{m+l}}$$
$$= \sum_{n_1,\dots,n_{m-1},n_{m+1},\dots,n_{m+l}\in\mathbb{C}}\ \sum_{k_1,\dots,k_{m-1},k_{m+1},\dots,k_{m+l}=0}^{K}\big(\big(Y^{(g_1g_2)^{-1}}_{P(z)}\big)^o_{n_{m-1},k_{m-1}}(u_{m-1})\cdots$$
$$\times\big(Y^{(g_1g_2)^{-1}}_{P(z)}\big)^o_{n_1,k_1}(u_1)\lambda\big)\big(w_1 \otimes Y^{g_2}(u_m, z_m)Y^{g_2}_{n_{m+1},k_{m+1}}(u_m)\cdots Y^{g_2}_{n_{m+l},k_{m+l}}(u_{m+l})w_2\big)$$
$$= z_1^{-n_1-1}\cdots z_{m-1}^{-n_{m-1}-1}z_{m+1}^{-n_1-1}\cdots z_{m+l}^{-n_{m+l}-1}$$
$$\times(\log z_1)^{k_1}\cdots(\log z_{m-1})^{k_{m-1}}(\log z_{m+1})^{k_{m+1}}\cdots(\log z_{m+l})^{k_{m+l}} \tag{4.26}$$

as an iterated series of the multi-series (4.22) is absolutely convergent on the region

$$|z_1| > \cdots > |z_{m-1}| > |z| > |z_m| > \cdots > |z_{m+l}| > 0$$

and its sum is equal to (4.17) on $\Omega^{(1)}_{m-1,0,l+1}(z)$ or to (4.23) on $\Omega^{(2)}_{m-1,0,l+1}(0,z)$. We discuss only the case that $\Omega^{(1)}_{m-1,0,l+1}(z)$ is nonempty. The case that $\Omega^{(2)}_{m-1,0,l+1}(0,z)$ is nonempty is similar and is omitted. The sum of (4.26) being equal to (4.17) on $\Omega^{(1)}_{m-1,0,l+1}(z)$ means that the expansion of (4.17) on $\Omega^{(1)}_{m-1,0,l+1}(z)$ can also be written as the iterated series (4.26). By the discussion above, the coefficients of the left-hand side of (4.26) is equal to (4.25) on the region $|z_m| > |z|$, $|\arg(z_m - z) - \arg z_m| < \frac{\pi}{2}$. So the expansion of (4.17) on the region $\Omega^{(1)}_{m,0,l}(z)$ can be written as the iterated series

$$\sum_{n_1,\dots,n_{m-1},n_{m+1},\dots,n_{m+l}\in\mathbb{C}}\ \sum_{k_1,\dots,k_{m-1},k_{m+1},\dots,k_{m+l}=0}^{K}\big(\big(Y^{(g_1g_2)^{-1}}_{P(z)}\big)^o(u_m, z_m)$$
$$\times\big(Y^{(g_1g_2)^{-1}}_{P(z)}\big)^o_{n_{m-1},k_{m-1}}(u_{m-1})\cdots\big(Y^{(g_1g_2)^{-1}}_{P(z)}\big)^o_{n_1,k_1}(u_1)\lambda\big)$$
$$\times\big(w_1 \otimes Y^{g_2}_{n_{m+1},k_{m+1}}(u_m)\cdots Y^{g_2}_{n_{m+l},k_{m+l}}(u_{m+l})w_2\big)$$
$$\times z_1^{-n_1-1}\cdots z_{m-1}^{-n_{m-1}-1}z_{m+1}^{-n_1-1}\cdots z_{m+l}^{-n_{m+l}-1}$$
$$\times(\log z_1)^{k_1}\cdots(\log z_{m-1})^{k_{m-1}}(\log z_{m+1})^{k_{m+1}}\cdots(\log z_{m+l})^{k_{m+l}}. \tag{4.27}$$

Since the expansion of (4.17) on the region $\Omega^{(1)}_{m,0,l}(z)$ must be absolutely convergent as a multisum, we see that the multi-series (4.16) corresponding to (4.27) must be absolutely convergent on the region $\Omega^{(1)}_{m,0,l}(z)$ to (4.17). Similarly, we can show that (4.16) is absolutely convergent on the region $\Omega^{(2)}_{m,0,l}(z)$ to (4.18).

This convergence result implies in particular the absolute convergence of (4.19) on the region $|z^{-1}| > |z_1| > \cdots > |z_l| > 0$. Using the commutativity for the twisted vertex operators Y^{g_2}, we see that for $\sigma \in S_m$, the sums of (4.19) and (4.20) are analytic extensions of each other.

For $u, v \in V$, since λ satisfies the condition 1 (a) in the $P(z)$-compatibility condition, there exists $M \in \mathbb{Z}_+$ depending on u and v only (see Remark 4.2) such that $z_1 - z_2 = 0$ is not a singularity of $(z_1 - z_2)^M f_2^e(z_1, z_2; u, v, w_1, w_2; \lambda)$ and we have

$$(z_1 - z_2)^M f_2^e(z_1, z_2; u, v, w_1, w_2; \lambda) = (z_1 - z_2)^M f_2^e(z_2, z_1; v, u, w_1, w_2; \lambda). \tag{4.28}$$

Since the expansion of $f_2^e(z_1, z_2; u, v, w_1, w_2; \lambda)$ on the region $\Omega_{2,0,0}^{(1)}(z)$ is

$$\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_2)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u, z_1)\lambda\big)(w_1 \otimes w_2),$$

we see that

$$\begin{aligned}
&(z_1 - z_2)^M\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_2)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u, z_1)\lambda\big)(w_1 \otimes w_2)\\
&= \sum_{i=0}^{M}\sum_{n\in\mathbb{C}}\sum_{k=0}^{K_1}\sum_{n_1\in\mathbb{C}}\sum_{k_1=0}^{K_1}\binom{M}{i}\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o_{n_1,k_1}(v)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o_{n,k}(u)\lambda\big)(w_1 \otimes w_2)\cdot\\
&\qquad \times \mathrm{e}^{(n+M-i)\log z_1}(\log z_1)^k \mathrm{e}^{(n_1+i)\log z_2}(\log z_2)^{k_1}
\end{aligned} \tag{4.29}$$

must be convergent absolutely to the left-hand side of (4.28) on $\Omega_{2,0,0}^{(1)}(z)$ (the region given by $|z_1| > |z_2| > |z|$, $|\arg(z_1 - z) - \arg z_1| < \frac{\pi}{2}$, $|\arg(z_2 - z) - \arg z_2| < \frac{\pi}{2}$). On the other hand, $(z_1 - z_2)^M f_2^e(z_1, z_2; u, v, w_1, w_2; \lambda)$ is analytic at $z_1 - z_2 = 0$. So (4.29) is in fact absolutely convergent to the left-hand side of (4.28) on the region $|z_1|, |z_2| > |z|$, $|\arg(z_1 - z) - \arg z_1| < \frac{\pi}{2}$, $|\arg(z_2 - z) - \arg z_2| < \frac{\pi}{2}$. Thus

$$(z_1 - z_2)^M\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u, z_1)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_2)\lambda\big)(w_1 \otimes w_2) \tag{4.30}$$

is absolutely convergent to the right-hand side of (4.28) also on the region $|z_1|, |z_2| > |z|$, $|\arg(z_1 - z) - \arg z_1| < \frac{\pi}{2}$, $|\arg(z_2 - z) - \arg z_2| < \frac{\pi}{2}$. From (4.28), (4.29) and (4.30), we obtain

$$\begin{aligned}
&(z_1 - z_2)^M\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_2)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u, z_1)\lambda\big)(w_1 \otimes w_2)\\
&\quad = (z_1 - z_2)^M\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u, z_1)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_2)\lambda\big)(w_1 \otimes w_2)
\end{aligned}$$

on the region $|z_1|, |z_2| > |z|$, $|\arg(z_1 - z) - \arg z_1| < \frac{\pi}{2}$, $|\arg(z_2 - z) - \arg z_2| < \frac{\pi}{2}$ for $\lambda \in \mathrm{COM}$, $u_1, v \in V$, $w_1 \in W_1$ and $w_2 \in W_2$, which, by the definition of $\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u, x)$ for $u \in V$ above, is equivalent to (4.21). □

Since $\mathrm{COMP} \subset \mathrm{COM}$, by Proposition 4.5, $Y_{P(z)}^{(g_1g_2)^{-1}}(u, x)\lambda$ for $u \in V$ and $\lambda \in \mathrm{COMP}$ is in $\mathrm{COM}\{x\}[\log x]$. We now prove the following stronger result.

Proposition 4.7. *The space* COMP *is invariant under the action of the components of the twisted vertex operators* $Y_{P(z)}^{(g_1g_2)^{-1}}(u, x)$ *for* $u \in V$.

Proof. Let $\lambda \in \mathrm{COMP}$. Then λ satisfies 2 (a) in the $P(z)$-compatibility condition. We need only show that for $v \in V$, $n \in \mathbb{C}$, $k = 0, \dots, K$, $\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o_{n,k}(v)\lambda$ also satisfies 2 (a) in the $P(z)$-compatibility condition.

By the definition of $\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u, z_1)$ for $u \in V$ and the $P(z)$-compatibility condition for λ,

$$\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u, z_1)\lambda\big)\big(w_1 \otimes Y_{n,k}^{g_2}(v)w_2\big)$$

and $\lambda\big(Y^{g_1}(u, z_1 - z)w_1 \otimes Y_{n,k}^{g_2}(v)w_2\big)$ for $v \in V$, $n \in \mathbb{C}$, $k = 0, \dots, K$, $w_1 \in W_1$ and $w_2 \in W_2$ are absolutely convergent to $f_1^e\big(z_1; u, w_1, Y_{n,k}^{g_2}(v)w_2; \lambda\big)$ on the region $|z_1| >$

$|z|$, $|\arg(z_1 - z) - \arg z_1| < \frac{\pi}{2}$ and the region $|z| > |z_1 - z| > 0$, $|\arg z_1 - \arg z| < \frac{\pi}{2}$, respectively. By Proposition 4.6,

$$\begin{aligned}&\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u, z_1)\lambda\big)(w_1 \otimes Y^{g_2}(v, z_2)w_2)\\ &= \sum_{n\in\mathbb{C}}\sum_{k=0}^{K}\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u, z_1)\lambda\big)\big(w_1 \otimes Y_{n,k}^{g_2}(v)w_2\big)\mathrm{e}^{n\log z_2}(\log z_2)^k\end{aligned}$$

is absolutely convergent on the region $|z_1| > |z| > |z_2| > 0$ and its sum is equal to

$$f_2^e(z_1, z_2; u, v; w_1, w_2; \lambda) \tag{4.31}$$

on $\Omega_{1,0,1}^{(1)}(z)$ and to

$$f_2^{b_{z_2,z}^{-1}}(z_1, z_2; u, v; w_1, w_2; \lambda) \tag{4.32}$$

on $\Omega_{1,0,1}^{(2)}(z)$. Thus

$$\begin{aligned}&\sum_{n\in\mathbb{C}}\sum_{k=0}^{K}\lambda\big(Y^{g_1}(u, z_1 - z)w_1 \otimes Y_{n,k}^{g_2}(v)w_2\big)\mathrm{e}^{n\log z_2}(\log z_2)^k\\ &= \lambda\big(Y^{g_1}(u, z_1 - z)w_1 \otimes Y^{g_2}(v, z_2)w_2\big)\end{aligned} \tag{4.33}$$

is absolutely convergent on the region $|z_1| > |z| > |z_1 - z| + |z_2| > 0$ and its sum is equal to (4.31) on the region $|z_1| > |z| > |z_1 - z| + |z_2| > 0$, $|\arg(z_1 - z) - \arg z_1| < \frac{\pi}{2}$, $|\arg z_1 - \arg z| < \frac{\pi}{2}$, $-\frac{3\pi}{2} < \arg(z_2 - z) - \arg z < -\frac{\pi}{2}$ and is equal to (4.32) on the region $|z_1| > |z| > |z_1 - z| + |z_2| > 0$, $|\arg(z_1 - z) - \arg z_1| < \frac{\pi}{2}$, $|\arg z_1 - \arg z| < \frac{\pi}{2}$, $\frac{\pi}{2} < \arg(z_2 - z) - \arg z < \frac{3\pi}{2}$. On the other hand, since $(z_1 - z, z_2) = (0, 0)$ is a regular singular point of (4.31) and (4.32), we can expand them on the regions $|z| > |z_2|, |z_1 - z| > 0$ to obtain a series of the same form as (4.33). Thus we see that (4.33) must be absolutely convergent on the region $|z| > |z_2|, |z_1 - z| > 0$ and its sum is equal to (4.31) and (4.32) on the regions $|z| > |z_1 - z| + |z_2| > 0$, $|\arg z_1 - \arg z| < \frac{\pi}{2}$, $-\frac{3\pi}{2} < \arg(z_2 - z) - \arg z < -\frac{\pi}{2}$ and on the region $|z| > |z_1 - z| + |z_2| > 0$, $|\arg z_1 - \arg z| < \frac{\pi}{2}$, $\frac{\pi}{2} < \arg(z_2 - z) - \arg z < \frac{3\pi}{2}$, respectively.

The right-hand side of (4.33) is equal to

$$\sum_{n\in\mathbb{C}}\sum_{k=0}^{K}\lambda\big(Y_{n,k}^{g_1}(u)w_1 \otimes Y^{g_2}(v, z_2)w_2\big)\mathrm{e}^{n\log(z_1 - z)}(\log(z_1 - z))^k. \tag{4.34}$$

We know that the series $\lambda\big(Y_{n,k}^{g_1}(u)w_1 \otimes Y^{g_2}(v, z_2)w_2\big)$ is absolutely convergent on the region $|z| > |z_2| > 0$ and its sum is equal to $f_1^e\big(z_2; v, Y_{n,k}^{g_1}(u)w_1, w_2; \lambda\big)$ on the region $|z| > |z_2| > 0$, $-\frac{3\pi}{2} < \arg(z_2 - z) - \arg z_2 < -\frac{\pi}{2}$ and to $f_1^{b_{z_2,z}}\big(z_2; v, Y_{n,k}^{g_1}(u)w_1, w_2; \lambda\big)$ on the region $|z| > |z_2| > 0$, $\frac{\pi}{2} < \arg(z_2 - z) - \arg z_2 < \frac{3\pi}{2}$. We also know that the series

$$\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_2)\lambda\big)\big(Y_{n,k}^{g_1}(u)w_1 \otimes w_2\big)$$

is absolutely convergent to $f_1^e\big(z_2; v, Y_{n,k}^{g_1}(u)w_1, w_2; \lambda\big)$ on the region $|z_2| > |z| > 0$, $|\arg(z_2 - z) - \arg z_2| < \frac{\pi}{2}$. From this discussion, (4.34) and the convergence of (4.33), we see that

$$\begin{aligned}&\sum_{n\in\mathbb{C}}\sum_{k=0}^{K}\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_2)\lambda\big)(Y_{n,k}^{g_1}(u)w_1 \otimes w_2)\mathrm{e}^{n\log(z_1 - z)}(\log(z_1 - z))^k\\ &= \big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_2)\lambda\big)\big(Y^{g_1}(u, z_1 - z)w_1 \otimes w_2\big)\end{aligned} \tag{4.35}$$

is absolutely convergent on the region $|z_2| > |z| > |z_1 - z| > 0$ and its sum is equal to (4.31) and (4.32) on the region $|z_2| > |z| > |z_1 - z| > 0$, $|\arg(z_2 - z) - \arg z_2| <$

$\frac{\pi}{2}$, $|\arg z_1 - \arg z| < \frac{\pi}{2}$ and on the region $|z_2| > |z| > |z_1 - z| > 0$, $|\arg(z_2 - z) - \arg z_2| < \frac{\pi}{2}$, $\frac{\pi}{2} < \arg(z_2 - z) - \arg z < \frac{3\pi}{2}$, respectively.

On the other hand, by Proposition 4.6,

$$\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u, z_1)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_2)\lambda\big)(w_1 \otimes w_2) \tag{4.36}$$

is absolutely convergent on the region $|z_2| > |z_1| > |z|$ and its sum is equal to (4.31) on the region $|z_2| > |z_1| > |z|$, $|\arg(z_1 - z) - \arg z_1| < \frac{\pi}{2}$, $|\arg(z_2 - z) - \arg z_2| < \frac{\pi}{2}$. Taking the coefficients of $\big(Y_{P(z)}^{(g_1g_2)^{-1}}(v, z_2)\big)^o$ in both (4.35) and (4.36), we see that

$$\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)_{n,k}^o(v)\lambda\big)\big(Y^{g_1}(u, z_1 - z)w_1 \otimes w_2\big)$$

and

$$\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u, z_1)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)_{n,k}^o(v)\lambda\big)(w_1 \otimes w_2)$$

are absolutely convergent to the coefficients of $\mathrm{e}^{n \log z_2}(\log z_2)^k$ in the expansion of (4.31) near the singularity $z_2 = \infty$ on the region $|z| > |z_1 - z| > 0$, $|\arg z_1 - \arg z| < \frac{\pi}{2}$ and on the region $|z_1| > |z|$, $|\arg(z_1 - z) - \arg z_1| < \frac{\pi}{2}$, respectively. This is equivalent to 2 (a) in the $P(z)$-compatibility condition for $\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)_{n,k}^o(v)\lambda$. □

For a $P(z)$-intertwining map I of type $\binom{W_3}{W_1 W_2}$ and an element w_3' of W_3', the element $\lambda_{I,w_3'} \in \mathrm{COM}$ also has the following property.

Proposition 4.8. *Consider the subspace $W_{\lambda_{I,w_3'}}$ of* COM *obtained by applying the coefficients of the vertex operators $Y_{P(z)}^{(g_1g_2)^{-1}}(u, x)$ for all $u \in V$ to $\lambda_{I,w_3'}$. Then $W_{\lambda_{I,w_3'}}$ equipped with $Y_{P(z)}^{(g_1g_2)^{-1}}$ is a generalized $(g_1g_2)^{-1}$-twisted V-module. Moreover, $W_{\lambda_{I,w_3'}}$ is a generalized $(g_1g_2)^{-1}$-twisted V-submodule of the generalized $(g_1g_2)^{-1}$-twisted V-module $W_1 \boxtimes_{P(z)} W_2$.*

Proof. We need only verify that the map given by $w_3' \mapsto \lambda_{I,w_3'}$ commutes with the vertex operators on W_3' and COM. This verification is straightforward and is omitted. □

Note that the conformal element ω is in the fixed point subalgebra $V^{\langle g_1, g_2\rangle}$ of V under the group $\langle g_1, g_2\rangle$ generated by g_1 and g_2. Also note that W_1 and W_2 are $V^{\langle g_1, g_2\rangle}$-modules. Then we can apply the construction and results in [28] to $V^{\langle g_1, g_2\rangle}$-modules W_1 and W_2. In particular, [28, equation (5.85)] give a vertex operator $Y'_{P(z)}(u, x)$ for $u \in V^{\langle g_1, g_2\rangle}$ acting on $(W_1 \otimes W_2)^*$. It is clear that on COM, $Y'_{P(z)}(u, x) = Y_{P(z)}^{(g_1g_2)^{-1}}(u, x)$ for $u \in V^{\langle g_1, g_2\rangle}$. In particular, we have the vertex operator $Y'_{P(z)}(\omega, x)$ acting on $(W_1 \otimes W_2)^*$ and is equal to $Y_{P(z)}^{(g_1g_2)^{-1}}(\omega, x)$ on COMP. Taking the coefficient of x^{-2} in $Y'_{P(z)}(\omega, x)$, we obtain an operator $L'_{P(z)}(0)$ on $(W_1 \otimes W_2)^*$, which is equal to the coefficient of x^{-2} in $Y_{P(z)}^{(g_1g_2)^{-1}}(\omega, x)$ on COMP.

Proposition 4.9. *For $v \in V$, the $L(0)$-commutator formula*

$$\big[L'_{P(z)}(0), Y_{P(z)}^{(g_1g_2)^{-1}}(v, x)\big] = x\frac{\mathrm{d}}{\mathrm{d}x}Y_{P(z)}^{(g_1g_2)^{-1}}(v, x) + Y_{P(z)}^{(g_1g_2)^{-1}}(L_V(0)v, x) \tag{4.37}$$

holds on COM.

Proof. We have the $L(0)$-commutator formula

$$\big[L_{W_2}(0), Y^{g_2}(v, x)\big] = x\frac{\mathrm{d}}{\mathrm{d}x}Y^{g_2}(v, x) + Y^{g_2}(L_V(0)v, x).$$

For $\lambda \in \mathrm{COM}$, by the definition of $L'_{P(z)}(0)$ [28, formula (5.110)], we have

$$(L'_{P(z)}(0)\lambda)(w_1 \otimes w_2) = \lambda((zL_{W_1}(-1) + L_{W_1}(0))w_1 \otimes w_2) + \lambda(w_1 \otimes L_{W_2}(0)w_2)$$

for $w_1 \in W_1$ and $w_2 \in W_2$. Also, the series $\lambda\big(w_1 \otimes Y^{g_2}(v, z_1)w_2\big)$ is absolutely convergent on the region $|z| > |z_1| > 0$ and can be analytically extended to a multivalued analytic function $f_1(z_1; v, w_1, w_2; \lambda)$ on $M^1(0, z)$ with a preferred single-valued branch $f_1^e(z_1; v, w_1, w_2; \lambda)$ on the region $M_0^1(0, z)$. Moreover, it is convergent to $f_1^e(z_1; v, w_1, w_2; \lambda)$ on the region $|z| > |z_1| > 0$, $-\frac{3\pi}{2} < \arg(z_1 - z) - \arg z < -\frac{\pi}{2}$ and to $f_1^{z_1,z}(z_1; v, w_1, w_2; \lambda)$ on the region $|z| > |z_1| > 0$, $\frac{\pi}{2} < \arg(z_1 - z) - \arg z < \frac{3\pi}{2}$.

By the definition of $Y_{P(z)}^{(g_1g_2)^{-1}}$, we have $\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_1)\lambda\big)(w_1 \otimes w_2) = f_1^e(z_1; v, w_1, w_2; \lambda)$ on the region $|z_1| > |z|$, $|\arg(z_1 - z) - \arg z_1| < \frac{\pi}{2}$. Then we have

$$\begin{aligned}
&\big(L'_{P(z)}(0)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_1)\lambda\big)(w_1 \otimes w_2)\\
&\quad = \big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_1)\lambda\big)\big((zL_{W_1}(-1) + L_{W_1}(0))w_1 \otimes w_2\big)\\
&\qquad + \big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_1)\lambda\big)(w_1 \otimes L_{W_2}(0)w_2)\\
&\quad = f_1^e(z_1; v, (zL_{W_1}(-1) + L_{W_1}(0))w_1, w_2; \lambda) + f_1^e(z_1; v, w_1, L_{W_2}(0)w_2; \lambda)
\end{aligned}$$

on the region $|z_1| > |z|$, $|\arg(z_1 - z) - \arg z_1| < \frac{\pi}{2}$. Also we have

$$\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_1)L'_{P(z)}(0)\lambda\big)(w_1 \otimes w_2) = f_1^e(z_1; v, w_1, w_2; L'_{P(z)}(0)\lambda).$$

But on the region $|z| > |z_1| > 0$, $-\frac{3\pi}{2} < \arg(z_1 - z) - \arg z < -\frac{\pi}{2}$, we have

$$\begin{aligned}
&f_1^e(z_1; v, w_1, w_2; L'_{P(z)}(0)\lambda)\\
&\quad = (L'_{P(z)}(0)\lambda)\big(w_1 \otimes Y^{g_2}(v, z_1)w_2\big)\\
&\quad = \lambda\big((zL_{W_1}(-1) + L_{W_1}(0))w_1 \otimes Y^{g_2}(v, z_1)w_2\big) + \lambda\big(w_1 \otimes L_{W_2}(0)Y^{g_2}(v, z_1)w_2\big)\\
&\quad = \lambda\big((zL_{W_1}(-1) + L_{W_1}(0))w_1 \otimes Y^{g_2}(v, z_1)w_2\big) + \lambda\big(w_1 \otimes Y^{g_2}(v, z_1)L_{W_2}(0)w_2\big)\\
&\qquad + \lambda\left(w_1 \otimes z_1\frac{\mathrm{d}}{\mathrm{d}z_1}Y^{g_2}(v, z_1)w_2\right) + \lambda\big(w_1 \otimes Y^{g_2}(L_V(0)v, z_1)w_2\big)\\
&\quad = f_1^e(z_1; v, (zL_{W_1}(-1) + L_{W_1}(0))w_1, w_2; \lambda) + f_1^e(z_1; v, w_1, L_{W_2}(0)w_2; \lambda)\\
&\qquad + z_1\frac{\mathrm{d}}{\mathrm{d}z_1}f_1^e(z_1; v, w_1, w_2; \lambda) + f_1^e(z_1; L_V(0)v, w_1, w_2; \lambda).
\end{aligned}$$

Then on the region $|z_1| > |z|$, $|\arg(z_1 - z) - \arg z_1| < \frac{\pi}{2}$, we have

$$\begin{aligned}
&\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_1)L'_{P(z)}(0)\lambda\big)(w_1 \otimes w_2)\\
&\quad = f_1^e(z_1; v, w_1, w_2; L'_{P(z)}(0)\lambda)\\
&\quad = f_1^e(z_1; v, (zL_{W_1}(-1) + L_{W_1}(0))w_1, w_2; \lambda) + f_1^e(z_1; v, w_1, L_{W_2}(0)w_2; \lambda)\\
&\qquad + z_1\frac{\mathrm{d}}{\mathrm{d}z_1}f_1^e(z_1; v, w_1, w_2; \lambda) + f_1^e(z_1; L_V(0)v, w_1, w_2; \lambda)\\
&\quad = \big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_1)\lambda\big)\big((zL_{W_1}(-1) + L_{W_1}(0))w_1 \otimes w_2\big)\\
&\qquad + \big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_1)\lambda\big)(w_1 \otimes L_{W_2}(0)w_2) + z_1\frac{\mathrm{d}}{\mathrm{d}z_1}\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_1)\lambda\big)(w_1 \otimes w_2)\\
&\qquad + \big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(L_V(0)v, z_1)\lambda\big)(w_1 \otimes w_2)\\
&\quad = \big(L'_{P(z)}(0)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_1)\lambda\big)(w_1 \otimes w_2) + \left(z_1\frac{\mathrm{d}}{\mathrm{d}z_1}\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_1)\lambda\right)(w_1 \otimes w_2)\\
&\qquad + \big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(L_V(0)v, z_1)\lambda\big)(w_1 \otimes w_2).
\end{aligned}$$

Thus we obtain

$$\big[\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_1), L'_{P(z)}(0)\big] = z_1\frac{\mathrm{d}}{\mathrm{d}z_1}\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_1) + \big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(L_V(0)v, z_1),$$

which is equivalent to (4.37). □

Because (4.37) holds, the coefficient $\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)_{n,k}(u)$ of $x^{-n-1}(\log x)^k$ in $Y_{P(z)}^{(g_1g_2)^{-1}}(u,x)$ must have weight $\operatorname{wt} u - n - 1$ when $u \in V$ is homogeneous. In particular, if $\lambda \in \mathrm{COM}$ is a finite sum of generalized eigenvectors of $L'_{P(z)}(0)$, the subspace W_λ of COM obtained by applying the coefficients of the vertex operators $Y_{P(z)}^{(g_1g_2)^{-1}}(u,x)$ for all $u \in V$ to λ must be a direct sum of generalized eigenspaces of $L'_{P(z)}(0)$, that is, $W_\lambda = \amalg_{n\in\mathbb{C}}(W_\lambda)_{[n]}$, where for $n \in \mathbb{C}$, $(W_\lambda)_{[n]}$ is the generalized eigenspace of $L'_{P(z)}(0)$ with eigenvalue n.

Proposition 4.8 together with Assumption 3.4 in particular says that in the case that $\mathcal{C}$ is a category of g-twisted V-modules with G-actions, for $\lambda = \lambda_{I,w_3'}$, there is a G-action (in particular, an action of $(g_1g_2)^{-1}$) on $W_{\lambda_{I,w_3'}}$ such that for $n \in \mathbb{C}$,

$$(W_{\lambda_{I,w_3'}})_{[n]} = \coprod_{\alpha\in\mathbb{C}/\mathbb{Z}} (W_{\lambda_{I,w_3'}})_{[n]}^{[\alpha]},$$

where for $\alpha \in \mathbb{C}/\mathbb{Z}$, $(W_{\lambda_{I,w_3'}})_{[n]}^{[\alpha]}$ is the generalized eigenspace of the action $(g_1g_2)^{-1}$ with eigenvalues $\mathrm{e}^{2\pi\alpha\mathrm{i}}$ for $\alpha \in \mathbb{C}/\mathbb{Z}$ and

$$(g_1g_2)^{-1}Y_{W_{\lambda_{I,w_3'}}}(v,x)w = Y_{W_{\lambda_{I,w_3'}}}\big((g_1g_2)^{-1}v,x\big)(g_1g_2)^{-1}w$$

for $v \in V$ and $w \in W_{\lambda_{I,w_3'}}$.

Moreover, such an action of $(g_1g_2)^{-1}$ can be extended to the action of $(g_1g_2)^{-1}$ on $W_1 \boxtimes_{P(z)} W_2$. It is possible that the action of $(g_1g_2)^{-1}$ on $W_1 \boxtimes_{P(z)} W_2$ can be further extended to subspaces of COM larger than on $W_1 \boxtimes_{P(z)} W_2$ such that (4.38) holds. By Zorn's lemma, there exists a maximal subspace of COM containing $W_1 \boxtimes_{P(z)} W_2$ with action of $(g_1g_2)^{-1}$ such that the restriction of the action to $W_1 \boxtimes_{P(z)} W_2$ is equal to the action of $(g_1g_2)^{-1}$ on $W_1 \boxtimes_{P(z)} W_2$ and

$$(g_1g_2)^{-1}Y_{P(z)}^{(g_1g_2)^{-1}}(v,x)w = Y_{P(z)}^{(g_1g_2)^{-1}}\big((g_1g_2)^{-1}v,x\big)(g_1g_2)^{-1}w \tag{4.38}$$

holds for $v \in V$ and w in this maximal subspace.

Using the vertex operators $Y_{P(z)}^{(g_1g_2)^{-1}}(u,x)$ for $u \in V$ and $L'_{P(z)}(0)$ and motivated by Proposition 4.8 (especially, the discussion above on the action of $(g_1g_2)^{-1}$ on $W_{\lambda_{I,w_3'}}$), we also introduce the following condition for $\lambda \in \mathrm{COM}$:

***P(z)*-local-grading-restriction condition.**

(a) The *$P(z)$-grading condition*: λ is a (finite) sum of generalized eigenvectors for the operator $L'_{P(z)}(0)$.

(b) Let W_λ be the smallest subspace of $(W_1 \otimes W_2)^*$ containing λ and stable under the action of all the coefficients of the vertex operators $Y_{P(z)}^{(g_1g_2)^{-1}}(u,x)$ for all $u \in V$. Then $\dim(W_\lambda)_{[n]} < \infty$, $(W_\lambda)_{[n]} = 0$, for $\Re(n)$ sufficiently negative.

We denote the subspace of $P(z)$-local grading restricted functionals in $\mathrm{COM} \subset (W_1 \otimes W_2)^*$ as $\mathrm{LGR}_{P(z)}((W_1 \otimes W_2)^*_{-1})$, or LGR for short. Clearly, the space LGR is closed under the action of $Y_{P(z)}^{(g_1g_2)^{-1}}(u,x)$, $u \in V$.

Theorem 4.10. *For λ satisfying the $P(z)$-compatibility condition and the $P(z)$-local-grading-restriction condition, the graded space W_λ equipped with the twisted vertex operator map $Y_{P(z)}^{(g_1g_2)^{-1}}$ is a grading-restricted $(g_1g_2)^{-1}$-twisted generalized V-module. If for every element $\lambda \in (W_1 \otimes W_2)^*$ satisfying both the $P(z)$-compatibility condition and the $P(z)$-local grading restriction condition, W_λ is a $(g_1g_2)^{-1}$-twisted generalized V-submodule of a $(g_1g_2)^{-1}$-twisted generalized V-module in* COMP *whose contragredient is in $\mathcal{C}$ (this holds in particular if $\mathcal{C}$ is the category of all grading-restricted g-twisted generalized V-modules without g-actions for $g \in G$ or the category of all grading-restricted g-twisted generalized V-modules with G-actions*

for $g \in G$), then an element $\lambda \in (W_1 \otimes W_2)^*$ is in $W_1 \boxtimes_{P(z)} W_2$ if and only if λ satisfies the $P(z)$-compatibility condition and the $P(z)$-local-grading-restriction condition. In other words, $W_1 \boxtimes_{P(z)} W_2 = \mathrm{COMP} \cap \mathrm{LGR}$.

Proof. The identity property follows immediately from the definition of the twisted vertex operator map $Y_{P(z)}^{(g_1g_2)^{-1}}$. The $L(0)$-grading condition follows from the $P(z)$-local-grading-restriction property. The $L(-1)$-derivative property follows from the definition of $Y_{P(z)}^{(g_1g_2)^{-1}}$ and the $L(-1)$-derivative property of the twisted vertex operator Y^{g_2}. We omit the details of the proofs of these properties.

We prove the equivariance property for $Y_{P(z)}^{(g_1g_2)^{-1}}$ now. It is equivalent to

$$\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o\big)^{p+1}(g_1g_2u, z_1)\tilde{\lambda} = \big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o\big)^{p}(u, z_1)\tilde{\lambda} \tag{4.39}$$

for $u \in V$, $\tilde{\lambda} \in W_\lambda$. By the definition of $\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o$, we know that for $w_1 \in W_1$, $w_2 \in W_2$,

$$\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u, z)\tilde{\lambda}\big)(w_1 \otimes w_2)$$

is absolutely convergent on the region $|z_1| > |z|$, $|\arg(z_1 - z) - \arg(z_1)| < \frac{\pi}{2}$ to $f_1^e\big(z_1; u, w_1, w_2; \tilde{\lambda}\big)$. Let b_3 be the homotopy class containing a loop given by a circle centered at 0 with radius larger than $|z|$ in the counterclockwise direction on the complex z_1 plane. Then (4.39) is equivalent to

$$f_1^{b_3}\big(z_1; g_1g_2u, w_1, w_2; \tilde{\lambda}\big) = f_1^e\big(z_1; u, w_1, w_2; \tilde{\lambda}\big) \tag{4.40}$$

for $u \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $\tilde{\lambda} \in W_\lambda$.

By the equivariance property for Y^{g_2} and the convergence of $\tilde{\lambda}(w_1 \otimes Y^{g_2}(v, z_1)w_2)$ to the branch $f_1^e\big(z_1; v, w_1, w_2; \tilde{\lambda}\big)$ for $v \in V$ on the region $|z| > |z_1| > 0$, $-\frac{3\pi}{2} < \arg(z_1 - z) - \arg z_1 < -\frac{\pi}{2}$, we obtain

$$f_1^{b_{z_1,0}}\big(z_1; g_2v, w_1, w_2; \tilde{\lambda}\big) = f_1^e\big(z_1; v, w_1, w_2; \tilde{\lambda}\big).$$

Using the equivariance property for Y^{g_2} and the convergence of $\tilde{\lambda}\big(Y^{g_1}(v, z_1 - z)w_1 \otimes w_2\big)$, we obtain similarly

$$f_1^{b_{z_1,z}}\big(z_1; g_1v, w_1, w_2; \tilde{\lambda}\big) = f_1^e\big(z_1; v, w_1, w_2; \tilde{\lambda}\big).$$

Applying any homotopy class b of loops in the z_1 complex plane with z and 0 deleted to both sides of this equality, we obtain

$$f_1^{b_{z_1,z}b}\big(z_1; g_1v, w_1, w_2; \tilde{\lambda}\big) = f_1^b\big(z_1; v, w_1, w_2; \tilde{\lambda}\big)$$

for $v \in V$. Then we have

$$f_1^{b_{z_1,z}b_{z_1,0}}\big(z_1; g_1g_2u, w_1, w_2; \tilde{\lambda}\big) = f_1^{b_{z_1,0}}(z_1; g_2u, w_1, w_2; \tilde{\lambda}) = f_1^e\big(z_1; u, w_1, w_2; \tilde{\lambda}\big).$$

But it is easy to see that $b_{z_1,z}b_{z_1,0} = b_3$. Thus we have proved (4.40).

For $u, v \in V$, $w \in W_\lambda$, $w' \in W_\lambda'$, by Proposition 4.6, we have

$$\begin{aligned}&(x_1 - x_2)^M\big\langle w', Y_{P(z)}^{(g_1g_2)^{-1}}(u, x_1)Y_{P(z)}^{(g_1g_2)^{-1}}(v, x_2)w\big\rangle\\ &\quad = (x_1 - x_2)^M\big\langle w', Y_{P(z)}^{(g_1g_2)^{-1}}(v, x_2)Y_{P(z)}^{(g_1g_2)^{-1}}(u, x_1)w\big\rangle,\end{aligned} \tag{4.41}$$

where $M \in \mathbb{Z}_+$ depends only on u and v. Since W_λ is lower bounded, by (4.41), the left-hand side of (4.41) has only finitely many terms in complex powers of x_1, x_2 and integer powers of $\log x_1$, $\log x_2$. Then

$$\big\langle w', Y_{P(z)}^{(g_1g_2)^{-1}}(u, x_1)Y_{P(z)}^{(g_1g_2)^{-1}}(v, x_2)w\big\rangle$$

is equal to this finite sum multiplied by $(x_1 - x_2)^{-M}$, which is expanded in non-negative powers of x_2. Thus we have a multivalued function of the form

$$f(z_1, z_2) = \sum_{i,j,k,l=0}^{N} a_{ijkl} z_1^{m_i} z_2^{n_j} (\log z_1)^k (\log z_2)^l (z_1 - z_2)^{-M},$$

where $m_i, n_i \in \mathbb{C}$ for $i = 0, \dots, N$, with the preferred branch

$$f^e(z_1, z_2) = \sum_{i,j,k,l=0}^{N} a_{ijkl} \mathrm{e}^{m_i \log z_1} \mathrm{e}^{n_j \log z_2} (\log z_1)^k (\log z_2)^l (z_1 - z_2)^{-M}$$

such that

$$\big\langle w', Y_{P(z)}^{(g_1g_2)^{-1}}(u, z_1) Y_{P(z)}^{(g_1g_2)^{-1}}(v, z_2) w \big\rangle$$

is absolutely convergent on the region $|z_1| > |z_2| > 0$ to $f^e(z_1, z_2)$. From (4.41), we also obtain the commutativity, that is,

$$\big\langle w', Y_{P(z)}^{(g_1g_2)^{-1}}(v, x_2) Y_{P(z)}^{(g_1g_2)^{-1}}(u, x_1) w \big\rangle$$

is absolutely convergent on the region $|z_2| > |z_1| > 0$ to $f^e(z_1, z_2)$.

We now prove the associativity for $Y_{P(z)}^{(g_1g_2)^{-1}}$. Since the associativity for $Y_{P(z)}^{(g_1g_2)^{-1}}$ is equivalent to the associativity for $\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o$, we prove this associativity. For $u, v \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $\tilde{\lambda} \in W_\lambda$, by Proposition 4.6,

$$\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v, z_2)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u, z_1)\tilde{\lambda}\big)(w_1 \otimes w_2) \tag{4.42}$$

is absolutely convergent on the region $|z_1| > |z_2| > |z|$ and its sum is equal to

$$f_2^e\big(z_1, z_2; u, v, w_1, w_2; \tilde{\lambda}\big) \tag{4.43}$$

on the region $|z_1| > |z_2| > |z|$, $|\arg(z_1 - z) - \arg z_1| < \frac{\pi}{2}$, $|\arg(z_2 - z) - \arg z_2| < \frac{\pi}{2}$. By the definition of $\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o$, $\big(\big(Y_{P(z)}^{(g_1g_2)}\big)^o((Y_V)_n(u)v, z_2)\tilde{\lambda}\big)(w_1 \otimes w_2)$ is absolutely convergent on the region $|z_2| > |z|$ and its sum is equal to

$$f_1^e\big(z_2; (Y_V)_n(u)v, w_1, w_2; \tilde{\lambda}\big) \tag{4.44}$$

on the region $|z_2| > |z|$, $|\arg(z_2 - z) - \arg z_2| < \frac{\pi}{2}$. Also $\tilde{\lambda}\big(w_1 \otimes Y^{g_2}((Y_V)_n(u)v, z_2)w_2\big)$ is absolutely convergent on the region $|z| > |z_2| > 0$ and its sum is equal to (4.44) on the region $|z| > |z_2| > 0$, $-\frac{3\pi}{2} < \arg(z_2 - z) - \arg z < -\frac{\pi}{2}$ and is equal to

$$f_1^{b_{z_2,z}^{-1}}\big(z_2; (Y_V)_n(u)v, w_1, w_2; \tilde{\lambda}\big)$$

on the region $|z| > |z_2| > 0$, $\frac{\pi}{2} < \arg(z_2 - z) - \arg z < \frac{3\pi}{2}$.

By the $P(z)$-compatibility condition for $\tilde{\lambda}$, $\tilde{\lambda}\big(w_1 \otimes Y^{g_2}(u, z_1)Y^{g_2}(v, z_2)w_2\big)$ is absolutely convergent to (4.43) on the region $|z| > |z_1| > |z_2| > 0$, $-\frac{3\pi}{2} < \arg(z_1 - z) - \arg z, \arg(z_2 - z) - \arg z < -\frac{\pi}{2}$ and to

$$f_2^{b_{z_1,z}^{-1} b_{z_2,z}^{-1}}\big(z_1, z_2; u, v, w_1, w_2; \tilde{\lambda}\big) \tag{4.45}$$

on the region $|z| > |z_1| > |z_2| > 0$, $\frac{\pi}{2} < \arg(z_1 - z) - \arg z, \arg(z_2 - z) - \arg z < \frac{3\pi}{2}$, respectively. By the associativity of the twisted vertex operator map Y^{g_2},

$$\begin{aligned}
&\sum_{k=0}^{K} \sum_{m \in \mathbb{C}} \Big(\sum_{n \in \mathbb{Z}} \tilde{\lambda}\big(w_1 \otimes Y_{m,k}^{g_2}((Y_V)_n(u)v)w_2\big)(z_1 - z_2)^{-n-1} \Big) \mathrm{e}^{(-m-1)\log z_2} (\log z_2)^k \\
&\quad = \sum_{k=0}^{K} \sum_{m \in \mathbb{C}} \tilde{\lambda}\big(w_1 \otimes Y_{m,k}^{g_2}(Y_V(u, z_1 - z_2)v, z_2)w_2\big) \mathrm{e}^{(-m-1)\log z_2} (\log z_2)^k \\
&\quad = \tilde{\lambda}\big(w_1 \otimes Y^{g_2}(Y_V(u, z_1 - z_2)v, z_2)w_2\big) = \tilde{\lambda}\big(w_1 \otimes Y^{g_2}(u, z_1)Y^{g_2}(v, z_2)w_2\big)
\end{aligned} \tag{4.46}$$

on the region $|z| > |z_1| > |z_2| > |z_1 - z_2| > 0$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$. But since $\tilde{\lambda}$ satisfies the $P(z)$-compatibility condition, we know that the right-hand side of (4.46) is equal to (4.43) on the region $|z| > |z_1| > |z_2| > 0$, $-\frac{3\pi}{2} < \arg(z_1 - z) - \arg z, \arg(z_2 - z) - \arg z < -\frac{\pi}{2}$ and is equal to (4.45) on the region $|z| > |z_2| > |z_1| > |z_1 - z_2| > 0$, $-\frac{3\pi}{2} < \arg(z_1 - z) - \arg z, \arg(z_2 - z) - \arg z < -\frac{\pi}{2}$. Moreover, by 1(b) of the $P(z)$-compatibility condition, the component-isolated singularity at $z_2 = \infty$, $z_1 - z_2 = 0$ of the multivalued analytic function $f\big(z_1, z_2; u, v, w_1, w_2; \tilde{\lambda}\big)$ is regular. Thus it can be expanded as a multiseries in powers of z_2, $z_1 - z_2$ and nonnegative integral powers of $\log z_2$ and $\log(z_1 - z_2)$ on the region $|z| > |z_2| > |z_1 - z_2| > 0$. By (4.46), the single-valued branches (4.43) and (4.45) of $f\big(z_1, z_2; u, v, w_1, w_2; \tilde{\lambda}\big)$ are expanded as iterated series of the same form on the region $|z| > |z_1| > |z_2| > |z_1 - z_2| > 0$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$. Thus the corresponding multiseries of this iterated series must be the expansion of a single-valued branch of $f\big(z_1, z_2; u, v, w_1, w_2; \tilde{\lambda}\big)$ and is thus absolutely convergent as a multiseries on the region $|z| > |z_2| > |z_1 - z_2| > 0$. In particular, the other iterated series

$$\sum_{n\in\mathbb{Z}} \left(\sum_{k=0}^{K} \sum_{m\in\mathbb{C}} \tilde{\lambda}\Big(w_1 \otimes Y^{g_2}_{m,k}((Y_V)_n(u)v)w_2 \Big) \mathrm{e}^{(-m-1)\log z_2} (\log z_2)^k \right) (z_1 - z_2)^{-n-1}$$

(4.47)

$$= \sum_{n\in\mathbb{Z}} \tilde{\lambda}(w_1 \otimes Y^{g_2}((Y_V)_n(u)v, z_2)w_2)(z_1 - z_2)^{-n-1}$$

of this multiseries is also absolutely convergent on the region $|z| > |z_2| > |z_1 - z_2| > 0$ to the same single-valued analytic functions. Thus we have proved that (4.47) is absolutely convergent on the region $|z| > |z_1| > |z_2| > 0$ and its sum is equal to (4.43) on the region $|z| > |z_1| > |z_2| > 0$, $-\frac{3\pi}{2} < \arg(z_1 - z) - \arg z, \arg(z_2 - z) - \arg z < -\frac{\pi}{2}$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$ and is equal to (4.45) on the region $|z| > |z_1| > |z_2| > 0$, $\frac{\pi}{2} < \arg(z_1 - z) - \arg z, \arg(z_2 - z) - \arg z < \frac{3\pi}{2}$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$. Then by the definition of $\big(Y^{(g_1 g_2)^{-1}}_{P(z)}\big)^o$,

$$\begin{aligned} &\big(\big(Y^{(g_1 g_2)^{-1}}_{P(z)}\big)^o (Y_V(u, z_1 - z_2)v, z_2)\tilde{\lambda}\big)(w_1 \otimes w_2) \\ &\quad = \sum_{n\in\mathbb{Z}} \big(\big(Y^{(g_1 g_2)^{-1}}_{P(z)}\big)^o \big((Y_V)^n(u)v, z_2\big)\tilde{\lambda}\big)(w_1 \otimes w_2)(z_1 - z_2)^{-n-1} \\ &\quad = \sum_{n\in\mathbb{Z}} f_1^e\big(z_2; (Y_V)_n(u)v, w_1, w_2; \tilde{\lambda}\big)(z_1 - z_2)^{-n-1} \end{aligned} \tag{4.48}$$

is in fact the expansion of (4.43) as a Laurent series in $z_1 - z_2$ near $z_1 - z_2 = 0$ and then to expand the coefficients as a series in powers of z_2 and $\log z_2$ near $z_2 = \infty$. Thus we have shown that (4.42) and the left-hand side of (4.48) are convergent on the region $|z_1| > |z_2| > |z|$ and $|z_2| > |z_1 - z_2|, |z|$, respectively, and their sums are equal to (4.43) on the region $|z_1| > |z_2| > |z|$, $|\arg(z_1 - z) - \arg z_1|, |\arg(z_2 - z) - \arg z_2| < \frac{\pi}{2}$, $|z_2| > |z_1 - z_2|, |z|$, $|\arg(z_2 - z) - \arg z_2| < \frac{\pi}{2}$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$.

Since λ satisfies the $P(z)$-local-grading-restriction condition, $W^*_\lambda = \overline{W'_\lambda}$. Then we have a linear map $I\colon W_1 \otimes W_2 \to W^*_\lambda = \overline{W'_\lambda}$ given by $\big\langle I(w_1 \otimes w_2), \tilde{\lambda}\big\rangle = \tilde{\lambda}(w_1 \otimes w_2)$ for $w_1 \in W_1$, $w_2 \in W_2$ and $\tilde{\lambda} \in W_\lambda$. Then (4.42) is absolutely convergent on the region $|z_1| > |z_2| > |z|$ means that the series $\big\langle I(w_1 \otimes w_2), \big(Y^{(g_1 g_2)^{-1}}_{P(z)}\big)^o(v, z_2)\big(Y^{(g_1 g_2)^{-1}}_{P(z)}\big)^o(u, z_1)\tilde{\lambda}\big\rangle$ is absolutely convergent on the same region. Let $L_{P(z)}(0)$ be the adjoint operator

of $L'_{P(z)}(0)$. Then by (4.37), we have

$$\begin{aligned}&\big\langle x^{L_{P(z)}(0)}I(w_1\otimes w_2),\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v,x_2)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u,x_1)\tilde\lambda\big\rangle\\&\quad=\big\langle I(w_1\otimes w_2),x^{L'_{P(z)}(0)}\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v,x_2)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u,x_1)\tilde\lambda\big\rangle\\&\quad=\big\langle I(w_1\otimes w_2),\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o\big(x^{-L_V(0)}v,x_2x^{-1}\big)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o\\&\qquad\times\big(x^{-L_V(0)}u,x_1x^{-1}\big)x^{L'_{P(z)}(0)}\tilde\lambda\big\rangle.\end{aligned}\tag{4.49}$$

Substituting $\mathrm{e}^{n\log\xi}$, $\log\xi$, $\mathrm{e}^{n\log z_1}$, $\log z_1$, $\mathrm{e}^{n\log z_2}$, $\log z_2$ for x^n, $\log x$, x_1^n, $\log x_1$, x_2^n, $\log x_2$, respectively, in (4.49), we see that the left-hand side (4.49) becomes

$$\big\langle \mathrm{e}^{(\log\xi)L_{P(z)}(0)}I(w_1\otimes w_2),\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v,z_2)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u,z_1)\tilde\lambda\big\rangle\tag{4.50}$$

and the right-hand side becomes a single-valued branch of a multivalued analytic function of ξ, z_1 and z_2 defined on the region $\left|z_1\xi^{-1}\right|>\left|z_2\xi^{-1}\right|>|z|$. The analytic function obtained from the right-hand side of (4.49) can be expanded as an absolutely convergent multiseries in powers of ξ, z_1, z_2 and nonnegative integral powers of $\log\xi$, $\log z_1$, $\log z_2$ on the region $0<|\xi|<1$, $|z_1|>|z_2|>|z|$. In particular, the iterated series corresponding to this multiseries is also convergent absolutely. Thus we see that the coefficients of (4.50) as a series in the variable ξ is an absolutely convergent series in the variable z_1 and z_2 on the region $|z_1|>|z_2|>|z|$. In particular, for $n\in\mathbb{C}$,

$$\big\langle \pi_nI(w_1\otimes w_2),\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v,z_2)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u,z_1)\tilde\lambda\big\rangle$$

is absolutely convergent on the region $|z_1|>|z_2|>|z|$. On the other hand, $\pi_nI(w_1\otimes w_2)$ for $n\in\mathbb{C}$, $w_1\in W_1$ and $w_2\in W_2$ span W'_λ. In fact, if these elements do not span W'_λ, then there exists a nonzero element $\tilde\lambda\in W_\lambda$ such that $\big\langle I(w_1\otimes w_2),\tilde\lambda\big\rangle=0$ for $w_1\in W_1$ and $w_2\in W_2$. But this means that $\tilde\lambda(w_1\otimes w_2)=0$ for $w_1\in W_1$ and $w_2\in W_2$. Hence $\tilde\lambda=0$. Contradiction. Thus we see that

$$\big\langle w',\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(v,z_2)\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(u,z_1)\tilde\lambda\big\rangle\tag{4.51}$$

is absolutely convergent on the region $|z_1|>|z_2|>|z|$ for $w'\in W'_\lambda$. Since W_λ and W'_λ are lower bounded, the series (4.51) must have only finitely many terms in negative real parts of powers of z_2 and finitely many terms in positive real parts of z_1. Then the absolute convergence of (4.51) on the region $|z_1|>|z_2|>|z|$ means that it must be absolutely convergent on the region $|z_1|>|z_2|>0$. Similarly, using the left-hand side of (4.48), we see that

$$\big\langle w',\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o(Y_V(u,z_1-z_2)v,z_2)\tilde\lambda\big\rangle\tag{4.52}$$

is absolutely convergent on the region $|z_2|>|z_1-z_2|>0$. Since the sums of (4.42) and the left-hand side of (4.48) are equal to the single-valued branches on the intersection of the corresponding regions, we see that the sums of (4.51) and (4.52) are also equal on the intersection of the corresponding regions. Thus we have proved the associativity for $\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o$, which is equivalent to the associativity for $Y_{P(z)}^{(g_1g_2)^{-1}}$. This finishes the proof that W_λ equipped with $Y_{P(z)}^{(g_1g_2)^{-1}}$ is a grading-restricted generalized $(g_1g_2)^{-1}$-twisted V-module.

By Propositions 4.1 and 4.8, an element of $W_1\boxtimes_{P(z)}W_2$ satisfies the $P(z)$-compatibility condition and the $P(z)$-local-grading-restriction condition. We now assume that for an element λ of $(W_1\otimes W_2)^*$ satisfying the $P(z)$-compatibility condition and the $P(z)$-local-grading-restriction, W_λ is a $(g_1g_2)^{-1}$-twisted generalized V-submodule of a $(g_1g_2)^{-1}$-twisted generalized V-module W in COMP whose contragredient W' is in $\mathcal{C}$. We want to prove $\lambda\in W_1\boxtimes_{P(z)}W_2$.

Since W^* is linearly isomorphic to $\overline{W'}$, we shall identify W^* with $\overline{W'}$. We define a linear map $I\colon W_1 \otimes W_2 \to W^*$ by $\langle \mu, I(w_1 \otimes w_2)\rangle = \mu(w_1 \otimes w_2)$ for $\mu \in W$, $w_1 \in W_1$ and $w_2 \in W_2$. We define a linear map $\mathcal{Y}_I\colon W_1 \otimes W_2 \to W'\{x\}[\log x]$, $w_1 \otimes w_2 \mapsto \mathcal{Y}_I(w_1, x)w_2$ by

$$\mathcal{Y}_I(w_1, x)w_2 = x^{L(0)}\mathrm{e}^{-(\log z)L(0)} I\big(x^{-L(0)}\mathrm{e}^{(\log z)L(0)}w_1 \otimes x^{-L(0)}\mathrm{e}^{(\log z)L(0)}w_2\big)$$

for $w_1 \in W_1$ and $w_2 \in W_2$.

From the definitions of $\mathcal{Y}_I$ and $L'_{P(z)}(0)$ on COMP $\subset$ COM, it is easy to see that $\mathcal{Y}_I$ satisfies the $L(-1)$-derivative property.

For $u_1, \dots, u_{k-1} \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $w \in W$, we have

$$\begin{aligned}
&\big\langle w, Y_{P(z)}^{(g_1g_2)^{-1}}(u_1, z_1)\cdots Y_{P(z)}^{(g_1g_2)^{-1}}(u_{k-1}, z_{k-1})\mathcal{Y}_I(w_1, z_k)w_2\big\rangle\\
&\quad= \big\langle w, Y_{P(z)}^{(g_1g_2)^{-1}}(u_1, z_1)\cdots Y_{P(z)}^{(g_1g_2)^{-1}}(u_{k-1}, z_{k-1})\\
&\qquad\times \mathrm{e}^{(\log z_k - \log z)L(0)} I\big(\mathrm{e}^{-(\log z_k - \log z)L(0)}w_1 \otimes \mathrm{e}^{-(\log z_k - \log z)L(0)}w_2\big)\big\rangle\\
&\quad= \big\langle \mathrm{e}^{(\log z_k - \log z)L(0)}w, Y_{P(z)}^{(g_1g_2)^{-1}}\big(z_k^{-L(0)}z^{L(0)}u_1, \mathrm{e}^{-(\log z_k - \log z - \log z_1)}\big)\\
&\qquad\times Y_{P(z)}^{(g_1g_2)^{-1}}\big(z_k^{-L(0)}z^{L(0)}u_{k-1}, \mathrm{e}^{-(\log z_k - \log z - \log z_{k-1})}\big)\\
&\qquad\times I\big(\mathrm{e}^{-(\log z_k - \log z)L(0)}w_1 \otimes \mathrm{e}^{-(\log z_k - \log z)L(0)}w_2\big)\big\rangle\\
&\quad= \big\langle \big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o\big(z_k^{-L(0)}z^{L(0)}u_{k-1}, \mathrm{e}^{-(\log z_k - \log z - \log z_{k-1})}\big)\cdots\\
&\qquad\times \big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o\big(z_k^{-L(0)}z^{L(0)}u_1, \mathrm{e}^{-(\log z_k - \log z - \log z_1)}\big)\mathrm{e}^{(\log z_k - \log z)L(0)}w,\\
&\qquad I\big(\mathrm{e}^{-(\log z_k - \log z)L(0)}w_1 \otimes \mathrm{e}^{-(\log z_k - \log z)L(0)}w_2\big)\big\rangle\\
&\quad= \big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o\big(z_k^{-L(0)}z^{L(0)}u_{k-1}, \mathrm{e}^{-(\log z_k - \log z - \log z_{k-1})}\big)\cdots\\
&\qquad\times \big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o\big(z_k^{-L(0)}z^{L(0)}u_1, \mathrm{e}^{-(\log z_k - \log z - \log z_1)}\big)\mathrm{e}^{(\log z_k - \log z)L(0)}w\big)\\
&\qquad\times \big(\mathrm{e}^{-(\log z_k - \log z)L(0)}w_1 \otimes \mathrm{e}^{-(\log z_k - \log z)L(0)}w_2\big).
\end{aligned} \tag{4.53}$$

By (4.8), the right-hand side of (4.53) is equal to the series obtained by expanding the function

$$\begin{aligned}
&f_{k-1}^e\big(\xi_1, \dots, \xi_{k-1}; z_k^{-L(0)}z^{L(0)}u_1, \dots, z_k^{-L(0)}z^{L(0)}u_{k-1},\\
&\mathrm{e}^{-(\log z_k - \log z)L(0)}w_1, \mathrm{e}^{-(\log z_k - \log z)L(0)}w_2; \mathrm{e}^{(\log z_k - \log z)L(0)}w\big)
\end{aligned}$$

on the region $|\xi_1| > \cdots > |\xi_{k-1}| > |z|$ as series in powers of $\xi_1, \dots, \xi_{k-1}$ and nonnegative integer powers of $\log \xi_1, \dots, \log \xi_1$ and then substituting $\mathrm{e}^{-n(\log z_k - \log z - \log z_1)}$, $\dots$, $\mathrm{e}^{-n(\log z_k - \log z - \log z_{k-1})}$ $(n \in \mathbb{C})$ for $\mathrm{e}^{n\log \xi_1}, \dots, \mathrm{e}^{n\log \xi_{k-1}}$ and $\log z_k - \log z - \log z_1, \dots, \log z_k - \log z - \log z_{k-1}$ for $\log \xi_1, \dots, \log \xi_{k-1}$, respectively. Then the right-hand side of (4.53) is absolutely convergent on the region $\big|zz_1z_k^{-1}\big| > \cdots > \big|zz_{k-1}z_k^{-1}\big| > |z|$ or equivalently the region $|z_1| > \cdots > |z_{k-1}| > |z_k| > 0$ and can be analytically extended to a multivalued analytic function on $M^{k-1}(0, z)$ with a preferred branch. By (4.53),

$$\big\langle w, Y_{P(z)}^{(g_1g_2)^{-1}}(u_1, z_1)\cdots Y_{P(z)}^{(g_1g_2)^{-1}}(u_{k-1}, z_{k-1})\mathcal{Y}_I(w_1, z_k)w_2\big\rangle$$

is absolutely convergent on the region $|z_1| > \cdots > |z_{k-1}| > |z_k| > 0$ and can be analytically extended to a multivalued analytic function $g(z_1, \dots, z_k)$ on $M^k(0, z)$ with a preferred branch. Moreover, since a component-isolated singularity of the

multivalued analytic function

$$\prod_{1\le i\le j\le k-1}(\xi_i-\xi_j)^{M_{ij}} f\big(\xi_1,\dots,\xi_{k-1}; z_k^{-L(0)}z^{L(0)}u_1,\dots,z_k^{-L(0)}z^{L(0)}u_{k-1},\\ \mathrm{e}^{-(\log z_k-\log z)L(0)}w_1,\mathrm{e}^{-(\log z_k-\log z)L(0)}w_2;\mathrm{e}^{(\log z_k-\log z)L(0)}w\big)$$

of the form $(\xi_1,\dots,\xi_{k-1})-\beta=\delta$ for $\beta\in\mathbb{C}^{k-1}$ and $\delta\in\{0,\infty\}^{k-1}$ is regular, we see that a component-isolated singularity of the multivalued analytic function $g(z_1,\dots,z_k)$ of the form $\big(zz_1z_k^{-1},\dots,zz_{k-1}z_k^{-1}\big)-\beta=\delta$ for $\beta\in\mathbb{C}^{k-1}$ and $\delta\in\{0,\infty\}^{k-1}$ is regular. But this is the same as saying that for fixed z_k, a component-isolated singularity of $\prod_{1\le i\le j\le k-1}(z_i-z_j)^{M_{ij}}g(z_1,\dots,z_k)$ of the form $(z_1,\dots,z_{k-1})-\beta'=\delta'$ for $\beta'\in\mathbb{C}^{k-1}$ and $\delta'\in\{0,\infty\}^{k-1}$.

We now prove the duality property for $\mathcal{Y}_I$. By (4.53), for $w\in W$, $v\in V$, $w_1\in W_1$ and $w_2\in W_2$,

$$\begin{aligned}&\big\langle w, Y_{P(z)}^{(g_1g_2)^{-1}}(v,z_1)\mathcal{Y}_I(w_1,z_2)w_2\big\rangle\\ &\quad=\big(\big(Y_{P(z)}^{(g_1g_2)^{-1}}\big)^o\big(z_2^{-L(0)}z^{L(0)}v,\mathrm{e}^{-(\log z_2-\log z-\log z_1)}\big)\mathrm{e}^{(\log z_2-\log z)L(0)}w\big)\\ &\quad\quad\times\big(\mathrm{e}^{-(\log z_2-\log z)L(0)}w_1\otimes\mathrm{e}^{-(\log z_2-\log z)L(0)}w_2\big),\end{aligned}\tag{4.54}$$

which is absolutely convergent on the region $\big|z_2^{-1}zz_1\big|>|z|$ or equivalently $|z_1|>|z_2|>0$ and its sum is equal to

$$f_1^e\big(\mathrm{e}^{-(\log z_2-\log z-\log z_1)};z_2^{-L(0)}z^{L(0)}v,\mathrm{e}^{-(\log z_2-\log z)L(0)}w_1,\mathrm{e}^{-(\log z_2-\log z)L(0)}w_2;\\ \mathrm{e}^{(\log z_2-\log z)L(0)}w\big)\tag{4.55}$$

on the region $\big|z_2^{-1}zz_1\big|>|z|$, $\big|\arg\big(z_2^{-1}zz_1-z\big)-\arg z_2^{-1}zz_1\big|<\frac{\pi}{2}$. For $z_1,z_2\in\mathbb{R}_+^2$ satisfying $z_1>z_2>0$, we have $\big|z_2^{-1}zz_1\big|=z_1z_2^{-1}|z|>|z|$ and $\big|\arg\big(z_2^{-1}zz_1-z\big)-\arg z_2^{-1}zz_1\big|=\big|\arg\big(z_1z_2^{-1}-1\big)z-\arg\big(z_1z_2^{-1}\big)z\big|\arg z-\arg z|=0<\frac{\pi}{2}$. Hence the sum of the left-hand side of (4.54) for such z_1 and z_2 is equal to (4.55). But for an absolutely convergent series of the form of the left-hand side of (4.54), its sum must be equal to one single valued branch on the region $|z_1|>|z_2|>0$, $|\arg(z_1-z_2)-\arg z_1|<\frac{\pi}{2}$. Thus the sum of the left-hand side of (4.54) is equal to (4.55) on the region $|z_1|>|z_2|>0$, $|\arg(z_1-z_2)-\arg z_1|<\frac{\pi}{2}$.

By definition,

$$\begin{aligned}&\big\langle w,\mathcal{Y}_I(w_1,z_2)Y^{g_2}(v,z_1)w_2\big\rangle\\ &\quad=\big\langle\mathrm{e}^{(\log z_2-\log z)L(0)}w,\mathcal{Y}_I\big(\mathrm{e}^{-(\log z_2-\log z)L(0)}w_1,z\big)\\ &\quad\quad\times Y^{g_2}\big(z_2^{-L(0)}z^{L(0)}v,\mathrm{e}^{-(\log z_2-\log z-\log z_1)}\big)\mathrm{e}^{-(\log z_2-\log z)L(0)}w_2\big\rangle\\ &\quad=\big(\mathrm{e}^{(\log z_2-\log z)L(0)}w\big)\big(\mathrm{e}^{-(\log z_2-\log z)L(0)}w_1\\ &\quad\quad\otimes Y^{g_2}\big(z_2^{-L(0)}z^{L(0)}v,\mathrm{e}^{-(\log z_2-\log z-\log z_1)}\big)\mathrm{e}^{-(\log z_2-\log z)L(0)}w_2\big)\end{aligned}\tag{4.56}$$

is absolutely convergent on the region $|z|>\big|z_2^{-1}zz_1\big|>0$ or equivalently $|z_2|>|z_1|>0$ and its sum is equal to (4.55) on the region $|z_2|>|z_1|>0$, $-\frac{3\pi}{2}<\arg\big(z_2^{-1}zz_1-z\big)-\arg z<-\frac{\pi}{2}$. Using the same argument as above, we can show that the sum of the left-hand side of (4.56) must be equal to (4.55) on the region $|z_2|>|z_1|>0$, $-\frac{3\pi}{2}<\arg(z_1-z_2)-\arg z_2<-\frac{\pi}{2}$.

By 2 (a) in the compatibility condition, we see that

$$\begin{aligned}&\big\langle w, \mathcal{Y}_I\big(Y^{g_1}(v, z_1 - z_2)w_1, z_2\big)w_2\big\rangle\\ &\quad= \big\langle \mathrm{e}^{(\log z_2 - \log z)L(0)}w,\\ &\times \mathcal{Y}_I\big(Y^{g_1}\big(z_2^{-L(0)}z^{L(0)}v, z_2^{-1}zz_1 - z\big)\mathrm{e}^{-(\log z_2 - \log z)L(0)}w_1, z\big)\mathrm{e}^{-(\log z_2 - \log z)L(0)}w_2\big\rangle\\ &= \big(\mathrm{e}^{(\log z_2 - \log z)L(0)}w\big)\big(Y^{g_1}\big(z_2^{-L(0)}z^{L(0)}v, z_2^{-1}zz_1 - z\big)\mathrm{e}^{-(\log z_2 - \log z)L(0)}w_1\\ &\quad\otimes \mathrm{e}^{-(\log z_2 - \log z)L(0)}w_2\big),\end{aligned} \tag{4.57}$$

which is absolutely convergent on the region $|z| > \left|z_2^{-1}zz_1 - z\right| > 0$ or equivalently $|z_2| > |z_1 - z_2| > 0$ and its sum is equal to (4.55) on the region $|z| > \left|z_2^{-1}zz_1 - z\right| > 0$, $\left|\arg z_2^{-1}zz_1 - \arg z\right| < \frac{\pi}{2}$. Using the same argument as above, we can show that the sum of the left-hand side of (4.57) must be equal to (4.55) on the region $|z_2| > |z_1 - z_2| > 0$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$. Thus the duality property for $\mathcal{Y}_I$ is proved and $\mathcal{Y}_I$ is a twisted intertwining operator of type $\binom{W'}{W_1 W_2}$. Then I is a twisted $P(z)$-intertwining map of the same type.

Now we have $\lambda(w_1 \otimes w_2) = \langle \lambda, I(w_1 \otimes w_2)\rangle = \lambda_{I,\lambda}(w_1 \otimes w_2)$ for $w_1 \in W_1$ and $w_2 \in W_2$. In particular, $\lambda = \lambda_{I,\lambda}$. Since W' is in $\mathcal{C}$, by definition, $\lambda_{I,\lambda} \in W_1 \boxtimes_{P(z)} W_2$. $\square$

5. A convergence lemma

Let A be a finite subset of $\mathbb{C}/\mathbb{Z}$, $R_\mu \in \mu$ for $\mu \in A$, D a subset of $\cup_{\mu\in A}(R_\mu - \mathbb{N})$, $\Delta \in \mathbb{C}$ and $a_{n,j,i} \in \mathbb{C}$ for $n \in D$, $j = 0, \dots, M$ and $i = 1, \dots, N$. Consider the triple series

$$\sum_{n\in D}\sum_{j=0}^{M}\sum_{i=0}^{N} a_{n,j,i}\mathrm{e}^{(-\Delta+n+1)\log z_1}(\log z_1)^j \mathrm{e}^{(-n-1)\log z_2}(\log z_2)^i \tag{5.1}$$

for $z_1, z_2 \in \mathbb{C}^\times$.

For any z_1, z_2 satisfying $|z_1| > |z_2|$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$, we have

$$\mathrm{e}^{\alpha \log(z_1 - z_2)} = \sum_{k\in\mathbb{N}} \binom{\alpha}{k}(-1)^k \mathrm{e}^{(\alpha-k)\log z_1} z_2^k, \tag{5.2}$$

$$\log(z_1 - z_2) = \log z_1 + \sum_{k\in\mathbb{Z}_+} \frac{(-1)^k}{k} z_1^{-k} z_2^k, \tag{5.3}$$

and for $z_2 \neq 0$, we have $\mathrm{e}^{\alpha l_{q_2}(-z_2)} = \mathrm{e}^{\alpha\pi\mathrm{i}}\mathrm{e}^{\alpha \log z_2}$, $l_{q_2}(-z_2) = \log z_2 + \pi\mathrm{i}$, where $\alpha \in \mathbb{C}$ and $q_2 = 0, 1$ if $\arg z_2 < \pi$, $\arg z_2 \geq \pi$, respectively. Note that in our notations, $\log z = l_0(z)$ for $z \in \mathbb{C}^\times$.

For $n \in D$, $j = 0, \dots, M$, $k \in \mathbb{Z}_{\geq 0}$, $s = 0, \dots, j$, define $b_{n,j,k,s} \in \mathbb{C}$ as the coefficients of the following formal power series expansion

$$(x + y)^{-\Delta+n+1}\log(x + y)^j = \sum_{k\in\mathbb{N}}\sum_{s=0}^{j} b_{n,j,k,s} x^{-\Delta+n+1-k} y^k (\log x)^s, \tag{5.4}$$

where x and y are formal variables. From (5.2), (5.3) and (5.4), when $|z_1| > |z_2|$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$, we have the following expansion:

$$\begin{aligned}&\mathrm{e}^{(-\Delta+n+1)\log(z_1 - z_2)}(\log(z_1 - z_2))^j\\ &= \sum_{k\in\mathbb{N}}\sum_{s=0}^{j}(-1)^k b_{n,j,k,s}\mathrm{e}^{(-\Delta+n+1-k)\log z_1} z_2^k (\log z_1)^s.\end{aligned} \tag{5.5}$$

Now consider

(5.6)
$$\sum_{n\in D}\sum_{j=0}^{M}\sum_{i=0}^{N} a_{n,j,i}\mathrm{e}^{(-\Delta+n+1)\log(z_1-z_2)}(\log(z_1-z_2))^j\mathrm{e}^{(-n-1)\log(-z_2)}(\log(-z_2))^i.$$

Using (5.5), we can further expand each term in the right-hand side of (5.6) so that the right-hand side of (5.6) becomes the iterated sum

$$\begin{aligned}&\sum_{n\in D}\sum_{j=0}^{M}\sum_{i=0}^{N} a_{n,j,i}\left(\sum_{k\in\mathbb{N}}\sum_{s=0}^{j}(-1)^k b_{n,j,k,s}\mathrm{e}^{(-\Delta+n+1-k)\log z_1}z_2^k(\log z_1)^s\right)\\ &\qquad\times\mathrm{e}^{(-n-1)\log(-z_2)}(\log(-z_2))^i\\ &=\sum_{n\in D}\sum_{j=0}^{M}\sum_{i=0}^{N}\left(\sum_{k\in\mathbb{N}}\sum_{s=0}^{j} a_{n,j,i}(-1)^k b_{n,j,k,s}\mathrm{e}^{(-\Delta+n+1-k)\log z_1}z_2^k(\log z_1)^s\right.\\ &\qquad\left.\times\ \mathrm{e}^{(-n-1)\log(-z_2)}(\log(-z_2))^i\right),\end{aligned}\tag{5.7}$$

where the inner sum is absolutely convergent on the region $|z_1|>|z_2|>0$.

We are interested in the convergence of the multisum

$$\begin{aligned}&\sum_{n\in D}\sum_{j=0}^{M}\sum_{i=0}^{N}\sum_{k\in\mathbb{N}}\sum_{s=0}^{j} a_{n,j,i}(-1)^k b_{n,j,k,s}\mathrm{e}^{(-\Delta+n+1-k)\log z_1}z_2^k\\ &\qquad\times(\log z_1)^s\mathrm{e}^{(-n-1)\log(-z_2)}(\log(-z_2))^i\end{aligned}\tag{5.8}$$

and the corresponding series

$$\begin{aligned}&\sum_{m\in D-\mathbb{N}}\sum_{s=0}^{M}\sum_{i=0}^{N}\left(\sum_{j=s}^{M}\sum_{\substack{n-k=m\\ n\in D,k\in\mathbb{N}}} a_{n,j,i}b_{n,j,k,s}\right)\mathrm{e}^{(-\Delta+m+1)\log z_1}\\ &\qquad\times(\log z_1)^s\mathrm{e}^{(-m-1)\log(-z_2)}(\log(-z_2))^i\end{aligned}\tag{5.9}$$

in powers of z_1 and z_2 and nonnegative integral powers of $\log z_1$ and $\log z_2$.

Lemma 5.1. *Assume that the triple series* (5.1) *and the series obtained from* (5.1) *by taking derivatives of each term in* (5.1) *with respect to z_1 and z_2 are absolutely convergent on the region given by $|z_1|>|z_2|>0$. Then the multisum* (5.8) *is absolutely convergent on the region $|z_1|>2|z_2|>0$. Assume in addition that* (5.1) *is convergent on the region $|z_1|>|z_2|>0$, $|\arg(z_1-z_2)-\arg z_1|<\frac{\pi}{2}$ to a single-valued analytic branch $f^e(z_1,z_2)$ on M_0^2 of a maximally extended multivalued analytic function on M^2 such that $f^e(z_1-z_2,-z_2)$ has no singular point in the region $|z_1|>|z_2|>0$. Then the iterated sum* (5.9) *is absolutely convergent on the region $|z_1|>|z_2|>0$, $|\arg z_1-\arg(z_1-z_2)|<\frac{\pi}{2}$ to $f^e(z_1-z_2,-z_2)$.*

Proof. Let $\tilde{n}=R_\mu-n$. Then the sum $\sum_{n\in D}$ in (5.1) can be written as the same as $\sum_{\mu\in A}\sum_{\tilde{n}\in\mathbb{N}}$. So the series (5.1) can be written as

$$\sum_{\mu\in A}\sum_{\tilde{n}\in\mathbb{N}}\sum_{j=0}^{M}\sum_{i=0}^{N} a_{R_\mu-\tilde{n},j,i}\mathrm{e}^{(-\Delta+R_\mu-\tilde{n}+1)\log z_1}(\log z_1)^j\mathrm{e}^{(-R_\mu+\tilde{n}-1)\log z_2}(\log z_2)^i.$$

By assumption, the series

$$\sum_{\mu\in A}\sum_{\tilde{n}\in\mathbb{N}}\sum_{j=0}^{M}\sum_{i=0}^{N} a_{R_\mu-\tilde{n},j,i}\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}\mathrm{e}^{(-\Delta+R_\mu-\tilde{n}+1)\log z_1}(\log z_1)^j\mathrm{e}^{(-R_\mu+\tilde{n}-1)\log z_2}(\log z_2)^i.\tag{5.10}$$

for $p, q \in \mathbb{N}$ is absolutely convergent on the region $|z_1| > |z_2| > 0$. For any $r > 1$, consider

$$\sum_{\mu\in A}\sum_{\tilde{n}\in\mathbb{N}}\sum_{j=0}^{M}\sum_{i=0}^{N} a_{R_\mu-\tilde{n},j,i}\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}\mathrm{e}^{(-\Delta+R_\mu-\tilde{n}+1)\log z_1}(\log z_1)^j \times\mathrm{e}^{(-R_\mu+\tilde{n}-1)\log z_2}(\log z_2)^i r^{\tilde{n}}. \tag{5.11}$$

Then the absolute convergence of (5.10) on the region $|z_1| > |z_2| > 0$ is equivalent to the absolute convergence of (5.11) on the region $|z_1| > r|z_2| > 0$. But the absolute convergence of (5.11) on the region $|z_1| > r|z_2| > 0$ is in turn equivalent to the absolute convergence of

$$\sum_{\tilde{n}\in\mathbb{N}} a_{R_\mu-\tilde{n},j,i}\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}\left(\frac{z_2}{z_1}\right)^{\tilde{n}} r^{\tilde{n}} \tag{5.12}$$

on the same region $|z_1| > r|z_2| > 0$. But as a power series in $\frac{z_2}{z_1}$, (5.12) with $p, q = 0$ has a removable singularity at $\frac{z_2}{z_1} = 0$. Then we see that (5.12) with $p, q = 0$ is absolutely convergent on the region $|\frac{z_2}{z_1}| < \frac{1}{r}$ and is hence also uniformly convergent on the closed region $|\frac{z_2}{z_1}| \le r_2$ for any positive $r_2 < \frac{1}{r}$. In particular, by the property of power series, (5.12) for $p, q \in \mathbb{N}$ is also absolutely convergent on the region $|\frac{z_2}{z_1}| < \frac{1}{r}$ and is uniformly convergent on the closed region $|\frac{z_2}{z_1}| \le r_2$. Thus for such r_2, (5.11) is a (finite) linear combination of the uniformly convergent series (5.12) on the region $|\frac{z_2}{z_1}| \le r_2$ with analytic functions of z_1 and z_2 on M_0^2 as coefficients.

Substituting $z_1 - z_2$ and $-z_2$ for z_1 and z_2, respectively, in (5.11), we see that

$$\sum_{\mu\in A}\sum_{\tilde{n}\in\mathbb{N}}\sum_{j=0}^{M}\sum_{i=0}^{N} a_{n,j,i}(-1)^q\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}\mathrm{e}^{(-\Delta+n+1)\log(z_1-z_2)}(\log(z_1-z_2))^j \times\mathrm{e}^{(-n-1)\log(-z_2)}(\log(-z_2))^i r^{\tilde{n}} \tag{5.13}$$

is absolutely convergent on the region $|z_1 - z_2| > r|z_2| > 0$ and is a (finite) linear combination of uniformly convergent series on the region $|\frac{z_2}{z_1-z_2}| \le r_2$ with analytic coefficients on M_0^2 for any positive $r_2 < \frac{1}{r}$. Now we expand each term in (5.13) using (5.5) and then passing the derivative $\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}$ to each term since the right-hand side of (5.5) is essentially a power series in $\frac{z_2}{z_1}$ to obtain the iterated sum

$$\begin{aligned}
&\sum_{\mu\in A}\sum_{\tilde{n}\in\mathbb{N}}\sum_{j=0}^{M}\sum_{i=0}^{N} a_{R_\mu-\tilde{n},j,i}\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}\left(\left(\sum_{k\in\mathbb{N}}\sum_{s=0}^{j}(-1)^k b_{R_\mu-\tilde{n},j,k,s}\right.\right.\\
&\left.\left.\times\mathrm{e}^{(-\Delta+R_\mu-\tilde{n}+1-k)\log z_1} z_2^k(\log z_1)^s\right)\mathrm{e}^{(-R_\mu+\tilde{n}-1)\log(-z_2)}(\log(-z_2))^i r^{\tilde{n}}\right)\\
=&\sum_{\mu\in A}\sum_{\tilde{n}\in\mathbb{N}}\sum_{j=0}^{M}\sum_{i=0}^{N}\left(\sum_{k\in\mathbb{N}}\sum_{s=0}^{j} a_{R_\mu-\tilde{n},j,i}(-1)^k b_{R_\mu-\tilde{n},j,k,s}\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}\mathrm{e}^{(-\Delta+R_\mu-\tilde{n}+1-k)\log z_1}\right.\\
&\left.\times\ z_2^k(\log z_1)^s\mathrm{e}^{(-R_\mu+\tilde{n}-1)\log(-z_2)}(\log(-z_2))^i r^{\tilde{n}}\right),
\end{aligned} \tag{5.14}$$

where the inner sum is absolutely convergent on the region $|z_1| > |z_2| > 0$. Then as a subseries of (5.14) divided by a suitable function of the form $\mathrm{e}^{\lambda\log z_1}\mathrm{e}^{\mu\log(-z_2)}(\log(-z_2))^\nu$

for some $\lambda, \mu \in \mathbb{C}$, $\nu \in \mathbb{N}$, the series

$$\sum_{\tilde{n}\in\mathbb{N}}\left(\sum_{k\in\mathbb{N}}\sum_{s=0}^{j}(-1)^{\tilde{n}+k}a_{R_\mu-\tilde{n},j,i}b_{R_\mu-\tilde{n},j,k,s}\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}\left(\frac{z_2}{z_1}\right)^{\tilde{n}+k}(\log z_1)^s\right)r^{\tilde{n}}$$

$$=\sum_{\tilde{n}\in\mathbb{N}}\left(\sum_{\tilde{k}\in\mathbb{N}}\sum_{s=0}^{j}(-1)^{\tilde{k}}a_{R_\mu-\tilde{n},j,i}b_{R_\mu-\tilde{n},j,\tilde{k}-\tilde{n},s}\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}\left(\frac{z_2}{z_1}\right)^{\tilde{k}}(\log z_1)^s\right)r^{\tilde{n}}, \tag{5.15}$$

where $b_{R_\mu-\tilde{n},j,\tilde{k}-\tilde{n},s}$ is defined to be 0 when $\tilde{k}<\tilde{n}$, is also absolutely convergent on the region $|z_1-z_2|>r|z_2|$ and is uniformly convergent on the closed region $\left|\frac{z_2}{z_1-z_2}\right|\le r_2$ for any positive $r_2<\frac{1}{r}$, where the inner sum is absolutely convergent in the region $|z_1|>|z_2|$. In the region $|z_1|>(1+r)|z_2|>0$, we have $|z_1-z_2|\ge |z_1|-|z_2|>r|z_2|>0$. Then (5.15) is absolutely convergent on the region $|z_1|>(1+r)|z_2|$ or $|\frac{z_2}{z_1}|<\frac{1}{1+r}$ with the inner sum absolutely convergent in the region $|z_1|>|z_2|$ or $|\frac{z_2}{z_1}|<1$. Note that for fixed $\frac{z_2}{z_1}$, z_1 can be any complex number and thus $\log z_1$ can also be any nonzero complex number whose imaginary part is in $[0,2\pi)$. Moreover, (5.15) is convergent absolutely on the same region when $\log z_1$ is replaced by any choice of branches of the logarithm of z_1. This means that $\zeta_1=\frac{z_2}{z_1}$ and $\zeta_2=\log z_1$ can be viewed as independent variables and, after the change of variables from z_1, z_2 to ζ_1 and ζ_2,

$$\sum_{\tilde{n}\in\mathbb{N}}\left(\sum_{\tilde{k}\in\mathbb{N}}\sum_{s=0}^{j}(-1)^{\tilde{k}}a_{R_\mu-\tilde{n},j,i}b_{R_\mu-\tilde{n},j,\tilde{k}-\tilde{n},s}\left(-\zeta_1\mathrm{e}^{\zeta_2}\frac{\partial}{\partial\zeta_1}+\mathrm{e}^{-\zeta_2}\frac{\partial}{\partial\zeta_2}\right)^p\right.$$

$$\left.\times\left(\mathrm{e}^{-\zeta_2}\frac{\partial}{\partial\zeta_1}\right)^q\zeta_1^{\tilde{k}}\zeta_2^s\right)\zeta_3^{\tilde{n}} \tag{5.16}$$

is absolutely convergent on the region given by $|\zeta_1|<\frac{1}{1+r}$, $\zeta_2\in\mathbb{C}$ and $|\zeta_3|<r$. From (5.16), we see that

$$\sum_{\tilde{n}\in\mathbb{N}}\left(\sum_{\tilde{k}\in\mathbb{N}}\sum_{s=0}^{j}(-1)^{\tilde{k}}a_{R_\mu-\tilde{n},j,i}b_{R_\mu-\tilde{n},j,\tilde{k}-\tilde{n},s}\frac{\partial^{\tilde{p}+\tilde{q}}}{\partial\zeta_1^{\tilde{p}}\partial\zeta_2^{\tilde{q}}}\zeta_1^{\tilde{k}}\zeta_2^s\right)\zeta_3^{\tilde{n}} \tag{5.17}$$

for $\tilde{p},\tilde{q}\in\mathbb{N}$ is absolutely convergent on the region given by $|\zeta_1|<\frac{1}{1+r}$, $\zeta_2\in\mathbb{C}$ and $|\zeta_3|<r$.

On the other hand, on the region $|\frac{z_2}{z_1}|\le\frac{r_2}{1+r_2}$ for positive $r_2<\frac{1}{r}$,

$$\left|\frac{z_2}{z_1-z_2}\right|\le\frac{|z_2|}{|z_1|-|z_2|}\le\frac{|z_2|}{\frac{1+r_2}{r_2}|z_2|-|z_2|}=r_2.$$

Then for $\frac{z_2}{z_1}\in\mathbb{C}$ satisfying $|\frac{z_2}{z_1}|\le\frac{r_2}{1+r_2}$ for positive $r_2<\frac{1}{r}$ and $\log z_1\in\mathbb{C}$, (5.15) is uniformly convergent with the inner sum also uniformly convergent on the same closed region. Thus (5.17) is uniformly convergent on the region $|\zeta_1|\le\frac{r_2}{1+r_2}$, $\zeta_2\in\mathbb{C}$ and $|\zeta_3|\le r$ with the inner sum also uniformly convergent on the same closed region. We have shown that the series (5.17) for $\tilde{p},\tilde{q}\in\mathbb{N}$ obtained by taking derivatives of each term in the series (5.17) with $\tilde{p},\tilde{q}=0$ with respect to ζ_1, ζ_2 and ζ_3 is uniformly convergent on the region $|\zeta_1|\le\frac{r_2}{1+r_2}$, $\zeta_2\in\mathbb{C}$ and $|\zeta_3|\le r$. In particular, the derivatives of (5.17) with $\tilde{p},\tilde{q}=0$ with respect to ζ_1, ζ_2 and ζ_3 exist and is equal to the sum of the series obtained by taking the corresponding derivatives of each term in (5.17) with $\tilde{p},\tilde{q}=0$ on the region $|\zeta_1|<\frac{r_2}{1+r_2}$, $\zeta_2\in\mathbb{C}$ and $|\zeta_3|<r$. Then the sum of (5.17) with $\tilde{p},\tilde{q}=0$ is an analytic function of ζ_1, ζ_2 and ζ_3 on the same open region.

For $r > 1$, let ζ_1, ζ_2, ζ_3 be complex numbers satisfying $|\zeta_1| < \frac{1}{(1+r)r}$, $\zeta_2 \in \mathbb{C}$ and $|\zeta_3| < r$. Then we have $\frac{|\zeta_1|}{1-|\zeta_1|} < (1+r)|\zeta_1| < \frac{1}{r}$. We choose r_2 such that $\frac{|\zeta_1|}{1-|\zeta_1|} < r_2 < (1+r)|\zeta_1|$. From $\frac{|\zeta_1|}{1-|\zeta_1|} < r_2$, we obtain $|\zeta_1| < \frac{r_2}{1+r_2}$. We also have $0 < r_2 < (1+r)|\zeta_1| < \frac{1}{r}$. Now ζ_1, ζ_3 satisfy $|\zeta_1| \le \frac{r_2}{1+r_2}$ and $|\zeta_3| < r$. This means that the sum of (5.17) with $\tilde{p}, \tilde{q} = 0$ is analytic and the derivatives can be calculated term by term at ζ_1, ζ_2, ζ_3. So the sum of (5.17) with $\tilde{p}, \tilde{q} = 0$ is analytic and the derivatives can be calculated term by term on the polydisc $|\zeta_1| < \frac{1}{(1+r)r}$, $\zeta_2 \in \mathbb{C}$ and $|\zeta_3| < r$ for any $r > 1$. Since analytic functions on polydiscs can be expanded as power series, the sum of (5.17) with $\tilde{p}, \tilde{q} = 0$ can be expanded as a power series in ζ_1, ζ_2 and ζ_3 and the coefficients of the power series expansion can be obtained using its derivatives with respect to ζ_1, ζ_2 and ζ_3 evaluated at $\zeta_1 = \zeta_2 = \zeta_3 = 0$. By taking the derivatives term by term, we see that the coefficients of the power series expansion of this analytic function are equal to the coefficients $(-1)^{\tilde{k}} a_{R_\mu - \tilde{n}, j, i} b_{R_\mu - \tilde{n}, j, \tilde{k} - \tilde{n}, s}$ of the iterated series (5.17) with $\tilde{p}, \tilde{q} = 0$. Since the power series expansion of an analytic function is an absolutely convergent multisum, we see that the triple series

$$\sum_{\tilde{n} \in \mathbb{N}} \sum_{\tilde{k} \in \mathbb{N}} \sum_{s=0}^{j} (-1)^{\tilde{k}} a_{R_\mu - \tilde{n}, j, i} b_{R_\mu - \tilde{n}, j, \tilde{k} - \tilde{n}, s} \zeta_1^{\tilde{k}} \zeta_2^{s} \zeta_3^{\tilde{n}} \tag{5.18}$$

is absolutely convergent on the region $|\zeta_1| < \frac{1}{(1+r)r}$, $\zeta_2 \in \mathbb{C}$ and $|\zeta_3| < r$.

In particular, on the region $|z_1| > (1+r)r|z_2|$, taking $\zeta_1 = \frac{z_2}{z_1}$, $\zeta_2 = \log z_1$ and $\zeta_3 = 1$ in (5.18), we see that the triple series

$$\sum_{\tilde{n} \in \mathbb{N}} \sum_{\tilde{k} \in \mathbb{N}} \sum_{s=0}^{j} (-1)^{\tilde{k}} a_{R_\mu - \tilde{n}, j, i} b_{R_\mu - \tilde{n}, j, \tilde{k} - \tilde{n}, s} \left(\frac{z_2}{z_1} \right)^{\tilde{k}} (\log z_1)^s \tag{5.19}$$

is absolutely convergent. Since r is an arbitrary real number satisfying $r > 1$, (5.19) is in fact absolutely convergent on the region $|z_1| > 2|z_2|$. Multiplying $\mathrm{e}^{(-\Delta + R_\mu + 1)\log z_1} \mathrm{e}^{(-R_\mu - 1)\log(-z_2)} \times (\log(-z_2))^i$ to (5.19) and summing over A, $j = 0, \dots, M$ and $i = 0, \dots, N$, we see that the multiseries

$$\begin{aligned}
&\sum_{\mu \in A} \sum_{\tilde{n} \in \mathbb{N}} \sum_{j=0}^{M} \sum_{i=0}^{N} \sum_{\tilde{k} \in \mathbb{N}} \sum_{s=0}^{j} (-1)^{\tilde{k}} a_{R_\mu - \tilde{n}, j, i} b_{R_\mu - \tilde{n}, j, \tilde{k} - \tilde{n}, s} \mathrm{e}^{(-\Delta + R_\mu + 1)\log z_1} \\
&\qquad \times \mathrm{e}^{(-R_\mu - 1)\log(-z_2)} (\log(-z_2))^i \left(\frac{z_2}{z_1} \right)^{\tilde{k}} (\log z_1)^s \\
&= \sum_{n \in D} \sum_{j=0}^{M} \sum_{i=0}^{N} \sum_{k \in \mathbb{N}} \sum_{s=0}^{j} a_{n,j,i} (-1)^k b_{n,j,k,s} \mathrm{e}^{(-\Delta + n + 1 - k)\log z_1} z_2^k (\log z_1)^s \\
&\qquad \times \mathrm{e}^{(-n-1)\log(-z_2)} (\log(-z_2))^i
\end{aligned} \tag{5.20}$$

is absolutely convergent on the region $|z_1| > 2|z_2| > 0$, proving the first part of the lemma.

In the case that the additional assumption holds, from the proof above and the additional assumption, the multisum (5.20), which is equal to the iterated series in the right-hand side of (5.7), is absolutely convergent on the region $|z_1| > 2|z_2| > 0$, $|\arg z_1 - \arg(z_1 - z_2)| < \frac{\pi}{2}$ to $f^e(z_1 - z_2, -z_2)$. In particular, (5.9) as an iterated sum of (5.20) is also absolutely convergent on the region $|z_1| > 2|z_2| > 0$, $|\arg z_1 - \arg(z_1 - z_2)| < \frac{\pi}{2}$ to $f^e(z_1 - z_2, -z_2)$. Since there is no singular point of $f^e(z_1 - z_2, -z_2)$ in the region $|z_1| > |z_2| > 0$, (5.9) must also be absolutely convergent when $|z_1| > |z_2| > 0$ and is thus absolutely convergent on the region $|z_1| > |z_2| > 0$, $|\arg z_1 - \arg(z_1 - z_2)| < \frac{\pi}{2}$ to $f^e(z_1 - z_2, -z_2)$. □

Remark 5.2. There is a subtlety about the convergence regions for (5.20) and (5.9). Note that in general the multisum (5.20) might not be absolutely convergent on the larger region $|z_1| > |z_2| > 0$. This is because (5.20) is not a series in powers of z_1 and z_2 and nonnegative integral powers of $\log z_1$ and $\log z_2$. Even if we restore the variable ζ_3 in the proof of the lemma above to obtain a series in powers of z_1, z_2, ζ_3 and nonnegative integral powers of $\log z_1$ and $\log z_2$, since we do not have the assumption that the sum of this series can be analytically extended to a region containing $|z_1| > |z_2| > 0$, this series might not be absolutely convergent on any region containing $|z_1| > |z_2| > 0$.

Remark 5.3. Lemma 5.1 can be generalized to the case of more than two variables. Consider the series

$$\sum_{\substack{n_1,\dots,n_k\in D\\ n_1+\cdots+n_k=\Delta-k}} \sum_{j_1,\dots,j_{k-1}=0}^{M} \sum_{i=0}^{N} a_{n_1,\dots,n_k,j_1,\dots,j_{k-1},i} \mathrm{e}^{(-n_1-1)\log z_1} (\log z_1)^{j_1} \cdots$$
$$\times \mathrm{e}^{(-n_{k-1}-1)\log z_l} (\log z_l)^{j_{k-1}} \mathrm{e}^{(-n_k-1)\log z_k} (\log z_k)^i \tag{5.21}$$

for $z_1, \dots, z_k \in \mathbb{C}^\times$. Replacing z_i for $i = 1, \dots, k-1$ and z_k by $z_i - z_k$ and $-z_k$, respectively, we obtain the following series:

$$\sum_{\substack{n_1,\dots,n_k\in D\\ n_1+\cdots+n_k=\Delta-k}} \sum_{j_1,\dots,j_{k-1}=0}^{M} \sum_{i=0}^{N} a_{n_1,\dots,n_k,j_1,\dots,j_{k-1},i} \mathrm{e}^{(-n_1-1)\log(z_1-z_k)} \log(z_1-z_k)^{j_1} \cdots$$
$$\times \mathrm{e}^{(-n_{k-1}-1)\log(z_{k-1}-z_k)} (\log(z_{k-1}-z_k))^{j_{k-1}} \mathrm{e}^{(-n_k-1)\log(-z_k)} (\log(-z_k))^i. \tag{5.22}$$

Expanding $\mathrm{e}^{(-n_1-1)\log(z_1-z_k)} \log(z_1-z_k)^{j_1}, \dots, \mathrm{e}^{(-n_{k-1}-1)\log(z_{k-1}-z_k)} (\log(z_{k-1}-z_k))^{j_l}$ in (5.22) using (5.5) with $-\Delta+n+1$ replaced by $-n_1-1, \dots, -n_l-1$, respectively, we obtain an iterated series. This iterated series also has the corresponding multisum and other iterated series. A generalization of Lemma 5.1 says that if (5.21) is absolutely convergent on the region $|z_1| > \cdots > |z_k| > 0$ and its sum is equal to a suitable single-valued branch of a multivalued analytic function on a suitable subregion, then the iterated series obtained by taking the sums in (5.22) first and then taking the sums given by the expansions of $\mathrm{e}^{(-n_1-1)\log(z_1-z_k)}(\log(z_1-z_k)^{j_1}, \dots, \mathrm{e}^{(-n_{k-1}-1)\log(z_{k-1}-z_k)}(\log(z_{k-1}-z_k))^{j_l}$ is absolutely convergent on the region $|z_1| > \cdots > |z_k| > 0$ and its sum is equal to a given single-valued branch on a suitable subregion. Here we omit the details of this generalization and its proof.

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